FINANCIAL ENGINEERING LAB

QUANTITATIVE FINANCE

From Mathematical Foundations to Empirical Research

Probability • Stochastic Calculus • Econometrics

Asset Pricing • Derivatives • Fixed Income

Portfolio and Risk • Computational Methods

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Research monograph and reproducible laboratory

Second expanded edition | 2026

Copyright and scope

Preface

Financial engineering is the disciplined conversion of economic questions into mathematical objects, statistical estimands, algorithms, and decisions. Its craft lies neither in formula collection nor in unconstrained prediction. It lies in knowing exactly what has been assumed, what has been identified by the data, what has merely been calibrated, and how an apparently precise answer can fail.

This book begins with proof and structure because quantitative work becomes fragile when notation hides gaps in reasoning. It then moves through macro-finance, econometrics, market institutions, asset pricing, derivatives, fixed income, portfolio construction, risk, and computation. Each part has four layers: *definitions*, which fix meaning; *theorems and proofs*, which establish what follows; *worked examples*, which turn abstractions into calculations; and *research protocols*, which connect the mathematics to reproducible evidence.

The intended progression is from a first serious encounter with the subject to the point at which the reader can read modern papers critically, implement models, and design an out-of-sample study. The text is deliberately explicit about conventions, regularity conditions, timestamps, and failure modes. Those details are not administrative: in quantitative finance they are part of the theorem.

How to study from this book

For a first pass, read the core definitions and worked examples, reproduce the calculations, and solve the exercises marked **Core**. For a second pass, write each proof without looking, implement the numerical algorithms, and challenge every empirical claim with a different sample. A research-level pass should include the advanced exercises, the projects in Part X, and a written model-risk memo.

Throughout, $\mathbb{P}$ denotes the physical or statistical probability measure and $\mathbb{Q}$ a pricing measure. Equality of random variables normally means almost-sure equality. Vectors are columns; $\mathbf{1}$ is a vector of ones; dates at which a decision is made are distinguished from dates at which data become observable. Annualization uses an explicitly stated clock and never repairs dependence by itself.

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Part I

Mathematical Language of Finance

Chapter 1

Analysis, Calculus, and Convexity

Learning objectives. Develop the language of sets, limits, continuity, differentiation, and convexity; derive approximation and optimality results used throughout finance; and learn to separate a formal calculation from a theorem whose assumptions have been verified.

1.1 Sets, functions, and proof

Mathematics starts by fixing the universe in which a claim is made. If A and B are sets, $A \subseteq B$ means every element of A belongs to B. The union, intersection, complement, and Cartesian product are denoted $A \cup B$, $A \cap B$, A^c, and $A \times B$. In probability an outcome lies in a sample space; in optimization a decision lies in a feasible set; in econometrics an estimand is a functional of a probability law. Confusing these objects produces errors that no amount of computation can repair.

Definition 1.1 (Function and inverse image)**.** A function $f : A \to B$ assigns exactly one $f(x) \in B$ to each $x \in A$. For $C \subseteq B$, its inverse image is

$$f^{-1}(C) = \{x \in A : f(x) \in C\}.$$

An inverse function exists on all of B only when f is bijective.

Inverse images, rather than ordinary inverses, are fundamental in probability: $X^{-1}((-\infty, x])$ is an event even when a random variable X is not one-to-one. A proof by contradiction assumes the negation of a statement and derives an impossibility. A proof by induction establishes a base case and shows that truth at n implies truth at $n+1$. A counterexample defeats a universal claim with one admissible case.

Worked example 1.2 (A quantifier matters)**.** The statement $\mathbb{E}[XY] = \mathbb{E}[X]\,\mathbb{E}[Y]$ for all integrable X, Y is false. Take $Y = X$ with $\mathrm{Var}(X) > 0$. The equality would imply $\mathbb{E}[X^2] = \mathbb{E}[X]^2$, hence zero variance. The equality does hold under independence. The missing hypothesis is the entire difference between a useful proposition and a false slogan.

1.2 Sequences, limits, and completeness

Definition 1.3 (Convergence)**.** A sequence (x_n) in $\mathbb{R}$ converges to x if for every $\varepsilon > 0$ there is an N such that $n \geq N$ implies $|x_n - x| < \varepsilon$. A sequence is Cauchy if for every $\varepsilon > 0$ there exists N such that $m, n \geq N$ imply $|x_m - x_n| < \varepsilon$.

The completeness of $\mathbb{R}$ says that every Cauchy sequence of real numbers converges to a real number. It supports existence arguments, numerical convergence, and the construction of integrals. A bounded monotone sequence converges. Compactness is the multidimensional workhorse: in $\mathbb{R}^n$, a set is compact exactly when it is closed and bounded.

Theorem 1.4 (Extreme-value theorem)**.** *If $K \subset \mathbb{R}^n$ is nonempty and compact and $f : K \to \mathbb{R}$ is continuous, then f attains both a minimum and a maximum on K.*

Proof. Let $m = \inf_{x \in K} f(x)$ and choose $x_k \in K$ with $f(x_k) \downarrow m$. Compactness gives a convergent subsequence $x_{k_j} \to x^\star \in K$. Continuity implies $f(x_{k_j}) \to f(x^\star)$, so $f(x^\star) = m$. Apply the same argument to $-f$ for the maximum. □

Existence is not uniqueness and uniqueness is not computational stability. A portfolio objective may attain a minimum while having many minimizers; a unique minimum may move dramatically under a small change in expected returns.

1.3 Continuity and differentiation

A function $f : \mathbb{R}^n \to \mathbb{R}^m$ is continuous at x when $x_k \to x$ implies $f(x_k) \to f(x)$. For scalar f, differentiability at x means there is a linear map represented by the gradient $\nabla f(x)$ such that

$$f(x+h) = f(x) + \nabla f(x)^\mathsf{T} h + o(\|h\|).$$

For a vector-valued function the derivative is its Jacobian. The Hessian $\nabla^2 f(x)$ collects second partial derivatives.

Theorem 1.5 (Multivariate Taylor formula)**.** *If f is three times continuously differentiable on a neighborhood of the segment between x and $x+h$, then*

$$f(x+h) = f(x) + \nabla f(x)^\mathsf{T} h + \frac{1}{2} h^\mathsf{T} \nabla^2 f(x) h + R_3, \qquad |R_3| \le C \|h\|^3$$

for a finite C determined by third derivatives on that segment.

Delta–gamma risk is this theorem applied to a derivative value $V(S)$:

$$\Delta V \approx \Delta^\mathsf{T} \Delta S + \frac{1}{2} \Delta S^\mathsf{T} \Gamma \Delta S.$$

The approximation is local. Large jumps, barrier crossings, volatility moves, and time decay can make the omitted terms economically decisive.

Worked example 1.6 (Duration and convexity)**.** For a bond price $P(y) = \sum_{k=1}^n C_k \mathrm{e}^{-y t_k}$,

$$\frac{\Delta P}{P} \approx -D_{\mathrm{mod}} \Delta y + \frac{1}{2} \mathcal{C} (\Delta y)^2, \quad D_{\mathrm{mod}} = -\frac{P'(y)}{P(y)}, \quad \mathcal{C} = \frac{P''(y)}{P(y)}.$$

The first-order term loses value when yields rise; positive convexity offsets part of the loss for moves of either sign.

1.4 Convex sets and convex functions

Definition 1.7 (Convexity)**.** A set $C \subseteq \mathbb{R}^n$ is convex if $\theta x + (1-\theta) y \in C$ for all $x, y \in C$ and $\theta \in [0,1]$. A function $f : C \to \mathbb{R}$ is convex when

$$f(\theta x + (1-\theta) y) \le \theta f(x) + (1-\theta) f(y).$$

It is strictly convex when the inequality is strict for distinct x, y and $\theta \in (0,1)$.

Theorem 1.8 (First-order characterization). *Let C be open and convex and let $f : C \to \mathbb{R}$ be differentiable. Then f is convex if and only if*

$$f(y) \geq f(x) + \nabla f(x)^\mathsf{T}(y - x) \quad \textit{for every } x, y \in C.$$

Proof. If f is convex, apply convexity to $x + t(y - x)$, rearrange, divide by $t > 0$, and let $t \downarrow 0$. Conversely, define $g(t) = f(x + t(y - x))$. The proposed inequality applied along the segment makes g' nondecreasing; integrating from 0 to θ and from θ to 1 gives the chord inequality. □

If f is twice continuously differentiable, convexity is equivalent to $\nabla^2 f(x) \succeq 0$ everywhere. Jensen's inequality is the probabilistic face of convexity.

Theorem 1.9 (Jensen's inequality). *For integrable X and convex f with $f(X)$ integrable,*

$$f(\mathbb{E}[X]) \leq \mathbb{E}[f(X)].$$

Proof. At an interior point $m = \mathbb{E}[X]$, a convex function has a supporting affine function: $f(x) \geq f(m) + a(x - m)$. Take expectations. Boundary cases follow by approximation. □

Jensen explains why volatility affects nonlinear payoffs. For the convex call payoff $f(S_T) = (S_T - K)^+$, dispersion can increase expected payoff even when the mean terminal price is fixed. It also explains why $\mathbb{E}[\log W] \leq \log \mathbb{E}[W]$: arithmetic average wealth and long-run geometric growth answer different questions.

1.5 Implicit functions and comparative statics

Suppose $F(x, \theta) = 0$ defines a calibrated parameter x as data or hyperparameter θ changes. If F is continuously differentiable and $F_x \neq 0$, the implicit function theorem gives

$$\frac{\mathrm{d}x}{\mathrm{d}\theta} = -\frac{F_\theta}{F_x}.$$

For vector equations, replace division by F_x^{-1}. Near singularity, comparative statics explode: this is a mathematical definition of weak identification or ill-conditioned calibration.

1.6 Problems

Exercise 1.10 (Core: Taylor risk)**.** For $V(S, \sigma, t)$, write the complete second-order expansion in ΔS and $\Delta\sigma$ and the first-order expansion in Δt. Identify delta, gamma, vega, vanna, and volga. Explain which cross term is omitted by a delta–gamma approximation.

Exercise 1.11 (Core: convexity)**.** Show that $x \mapsto -\log x$ is strictly convex on $\mathbb{R}_+$. Use Jensen's inequality to prove $\mathbb{E}[X]\ \mathbb{E}[X^{-1}] \geq 1$ for positive X with finite expectations.

Exercise 1.12 (Advanced: existence)**.** Let $C = \{w : \mathbf{1}^\mathsf{T} w = 1,\ \|w\| \leq M\}$. Prove that a continuous portfolio loss functional attains its minimum on C. Which hypothesis fails if the leverage bound is removed?

Chapter 2

Linear Algebra, Geometry, and Factor Structure

Learning objectives. Use vector spaces, projections, eigenvalues, singular values, and positive semidefinite matrices to understand portfolios, regressions, factors, and numerical conditioning.

2.1 Vector spaces and norms

A vector space is a set closed under vector addition and scalar multiplication. The span of columns of $A \in \mathbb{R}^{m\times n}$ is its column space; the null space is $\{x : Ax = 0\}$. The rank–nullity theorem gives

$$\operatorname{rank}(A) + \dim\ker(A) = n.$$

Thus a system $Ax = b$ is uniquely solvable for every b only when a square A has full rank.

A norm satisfies positivity, homogeneity, and the triangle inequality. Common norms are $\|x\|_1 = \sum_i |x_i|$, $\|x\|_2 = (x^\top x)^{1/2}$, and $\|x\|_\infty = \max_i |x_i|$. They encode different geometry. An ℓ_1 constraint encourages sparse exposures; an ℓ_2 constraint spreads shrinkage; an ℓ_∞ constraint caps each coordinate.

Theorem 2.1 (Cauchy–Schwarz)**.** *For an inner-product space,*

$$|\langle x, y\rangle| \le \|x\|\,\|y\|\,.$$

Equality holds exactly when x and y are linearly dependent.

Proof. For $y \neq 0$, nonnegativity of $\|x - ty\|^2$ at $t = \langle x, y\rangle \,/\, \|y\|^2$ yields $0 \leq \|x\|^2 - |\langle x, y\rangle|^2 \,/\, \|y\|^2$. □

Cauchy–Schwarz proves the upper bound $|\mathrm{Corr}(X, Y)| \leq 1$ and provides the geometry of the maximum Sharpe ratio.

2.2 Orthogonal projection and least squares

Let $X \in \mathbb{R}^{n\times k}$ have full column rank. The least-squares problem $\min_b \|y - Xb\|_2^2$ has normal equations $X^\top X b = X^\top y$. Hence

$$\widehat{b} = (X^\top X)^{-1} X^\top y, \qquad \widehat{y} = P_X y, \quad P_X = X(X^\top X)^{-1} X^\top.$$

The residual $M_X y = (I - P_X)y$ is orthogonal to every column of X.

Proposition 2.2 (Frisch–Waugh–Lovell)**.** *Partition $X = [X_1\ X_2]$. The coefficient on X_2 in the regression of y on X_1, X_2 equals the regression coefficient of $M_1 y$ on $M_1 X_2$, where $M_1 = I - P_{X_1}$.*

Proof. Minimize $\|y - X_1 b_1 - X_2 b_2\|^2$ first over b_1 for fixed b_2. The minimized residual is $M_1(y - X_2 b_2)$. Differentiating its squared norm in b_2 gives the partialled-out regression. □

The theorem clarifies what a regression coefficient means: it uses variation in a regressor left after its linear association with controls has been removed. It does not, by itself, make that residual variation exogenous.

2.3 Eigenvalues, covariance, and principal components

A symmetric matrix A admits an orthogonal eigendecomposition $A = Q\Lambda Q^\top$. It is positive semidefinite when $x^\top A x \geq 0$ for all x, equivalently when every eigenvalue is nonnegative. A covariance matrix is positive semidefinite because

$$a^\top \Sigma a = \mathrm{Var}(a^\top X) \geq 0.$$

Theorem 2.3 (Spectral theorem). *Every real symmetric A has an orthonormal basis of eigenvectors and real eigenvalues. Consequently $A = Q\Lambda Q^\top$ for orthogonal Q.*

For centered data matrix R, principal component analysis diagonalizes $S = R^\top R/(T-1)$. The first loading q_1 solves

$$\max_{\|q\|=1} q^\top S q$$

and equals the eigenvector associated with the largest eigenvalue.

Proof. Write $q = Qa$ in the eigenbasis. Then $q^\top S q = \sum_i \lambda_i a_i^2 \leq \lambda_1 \sum_i a_i^2 = \lambda_1$. Equality obtains at $q = q_1$. $\square$

In a yield curve, the first three empirical components are often interpreted as level, slope, and curvature. Those labels are sample-dependent summaries, not structural shocks. Sign is arbitrary, component order can change, and standard PCA ignores measurement error and dynamics.

2.4 Singular values and conditioning

Any $A \in \mathbb{R}^{m\times n}$ has a singular-value decomposition $A = U\Sigma V^\top$. The nonzero singular values are square roots of the positive eigenvalues of $A^\top A$. The two-norm condition number is $\kappa_2(A) = \sigma_{\max}/\sigma_{\min}$ for full-rank A.

If $\kappa_2(A)$ is large, small perturbations in inputs can produce large changes in solutions. Expected-return optimization, high-dimensional regression, and implied parameter inversion are commonly ill-conditioned. Regularization changes the question to stabilize the answer:

$$\widehat{\beta}_{\text{ridge}} = \arg\min_\beta \{\|y - X\beta\|^2 + \lambda\,\|\beta\|^2\} = (X^\top X + \lambda I)^{-1} X^\top y.$$

Model-risk warning 2.4. An invertible sample covariance matrix is not necessarily reliable. When the number of assets is close to the sample length, its smallest eigenvalues are estimated with enormous relative error. Optimization amplifies precisely those errors.

2.5 Factor models

A linear factor representation is

$$r_t = \alpha + Bf_t + \varepsilon_t, \qquad \mathbb{E}[\varepsilon_t \mid f_t] = 0, \qquad \mathrm{Cov}(\varepsilon_t, f_t) = 0.$$

It implies

$$\Sigma_r = B\Sigma_f B^\top + \Sigma_\varepsilon.$$

This low-rank plus idiosyncratic structure reduces estimation dimension and makes risk attribution possible. It does not decide whether factors are traded, causal, or merely statistical.

Worked example 2.5 (Marginal contribution to variance)**.** For portfolio variance $\sigma_p^2 = w^\top \Sigma w$,

$$\frac{\partial \sigma_p}{\partial w_i} = \frac{(\Sigma w)_i}{\sigma_p}, \qquad RC_i = w_i \frac{(\Sigma w)_i}{\sigma_p}.$$

Euler's theorem for homogeneous functions gives $\sum_i RC_i = \sigma_p$.

2.6 Problems

Exercise 2.6 (Core: projection)**.** Prove that $P_X = P_X^\top = P_X^2$. Show that $\mathrm{rank}(P_X) = k$ and its eigenvalues are 0 or 1.

Exercise 2.7 (Core: covariance)**.** For a two-asset covariance matrix, derive the condition under which an asset with positive standalone volatility reduces portfolio variance at the margin.

Exercise 2.8 (Advanced: SVD)**.** Use the SVD to derive the Moore–Penrose pseudoinverse and show that it gives the minimum-norm solution to $Ax = b$ when the system has infinitely many solutions.

Chapter 3

Optimization and Dynamic Decisions

Learning objectives. Derive first- and second-order conditions, understand constraint qualifications and duality, solve canonical portfolio programs, and formulate sequential decisions by dynamic programming.

3.1 Unconstrained and equality-constrained optimization

At an interior local minimum of differentiable f, $\nabla f(x^\star) = 0$. If f is twice differentiable, a positive definite Hessian at a stationary point is sufficient for a strict local minimum. Convexity upgrades local optimality to global optimality.

For

$$\min_x f(x) \quad \text{subject to} \quad h(x) = 0,$$

the Lagrangian is $\mathcal{L}(x, \lambda) = f(x) + \lambda^\mathsf{T} h(x)$. When gradients of active constraints are linearly independent, a local solution satisfies $\nabla_x \mathcal{L} = 0$ and $h(x) = 0$.

Worked example 3.1 (Global minimum-variance portfolio)**.** For positive definite Σ,

$$\min_w \frac{1}{2} w^\mathsf{T} \Sigma w \quad \text{s.t.} \quad \mathbf{1}^\mathsf{T} w = 1.$$

Stationarity gives $\Sigma w - \lambda \mathbf{1} = 0$. Enforcing the constraint yields

$$w_{\mathrm{GMV}} = \frac{\Sigma^{-1}\mathbf{1}}{\mathbf{1}^\mathsf{T}\Sigma^{-1}\mathbf{1}}, \qquad \sigma^2_{\mathrm{GMV}} = \frac{1}{\mathbf{1}^\mathsf{T}\Sigma^{-1}\mathbf{1}}.$$

For a target mean m, define

$$A = \mathbf{1}^\mathsf{T}\Sigma^{-1}\mathbf{1}, \quad B = \mathbf{1}^\mathsf{T}\Sigma^{-1}\mu, \quad C = \mu^\mathsf{T}\Sigma^{-1}\mu, \quad D = AC - B^2.$$

When $D > 0$, the efficient-frontier variance is $\sigma^2(m) = (Am^2 - 2Bm + C)/D$.

3.2 Inequality constraints and KKT conditions

Consider convex f, g_i and affine h:

$$\min_x f(x) \quad \text{s.t.} \quad g_i(x) \leq 0,\ h(x) = 0.$$

The Karush–Kuhn–Tucker conditions are primal feasibility, dual feasibility $\lambda_i \geq 0$, stationarity, and complementary slackness $\lambda_i g_i(x) = 0$.

Theorem 3.2 (KKT sufficiency)**.** *For a convex differentiable program, any primal–dual point satisfying the KKT conditions is globally optimal. Under Slater's condition, KKT conditions are also necessary and strong duality holds.*

Proof. Convexity gives $f(y) \geq f(x^\star) + \nabla f(x^\star)^\mathsf{T}(y - x^\star)$. Substitute stationarity. Convexity of active g_i, feasibility, dual feasibility, and complementary slackness make the remaining expression nonnegative for every feasible y. □

The multiplier is a shadow price: under regularity, it measures the marginal change in optimized value when a constraint is relaxed. In a risk limit, this quantity is economically informative only if units and sign conventions are explicit.

3.3 Duality and robust optimization

The Lagrange dual function $q(\lambda, \nu) = \inf_x \mathcal{L}(x, \lambda, \nu)$ provides a lower bound on a minimization problem. Maximizing that lower bound is the dual problem. The primal–dual gap is a numerical certificate when strong duality holds.

Robust optimization replaces a single estimated input with an uncertainty set. If $\mu \in \{\widehat{\mu} + u : u^\mathsf{T}\Omega^{-1}u \le \delta^2\}$, then

$$\min_{\mu} \; w^\mathsf{T}\mu = w^\mathsf{T}\widehat{\mu} - \delta\sqrt{w^\mathsf{T}\Omega w}.$$

Thus ellipsoidal mean uncertainty produces an explicit penalty for sensitivity to the weakest input.

3.4 Utility, certainty equivalents, and Kelly growth

Expected utility ranks uncertain terminal wealth by $\mathbb{E}[u(W)]$. For increasing, concave u, the Arrow–Pratt coefficient of absolute risk aversion is $A(W) = -u''(W)/u'(W)$ and relative risk aversion is $R(W) = WA(W)$. Exponential utility has constant absolute risk aversion; power utility has constant relative risk aversion.

For small zero-mean risk $\widetilde{X}$ added to wealth W, a second-order expansion gives the risk premium

$$\pi \approx \frac{1}{2}A(W)\,\mathrm{Var}(\widetilde{X}).$$

With normally distributed return and quadratic or exponential utility, maximizing a certainty equivalent leads to mean–variance form.

Log utility maximizes expected logarithmic growth. In a one-period approximation with risky excess return mean μ and variance σ^2, the growth-optimal fraction is $f^\star \approx \mu/\sigma^2$. Estimation error, fat tails, leverage constraints, and drawdown preferences make fractional Kelly policies more robust in practice.

3.5 Dynamic programming

A controlled Markov state evolves as $x_{t+1} = g(x_t, a_t, \varepsilon_{t+1})$. With reward $r(x,a)$ and discount β, the value function satisfies the Bellman equation

$$V_t(x) = \sup_{a\in\mathcal{A}(x)} \left\{ r_t(x,a) + \beta\,\mathbb{E}[V_{t+1}(x_{t+1}) \mid x_t = x, a_t = a] \right\}.$$

Theorem 3.3 (Principle of optimality)**.** *If a policy is optimal from an initial state, its continuation after any reachable state is optimal for the remaining subproblem.*

Proof. If a continuation were not optimal, replace it by a better continuation while keeping earlier actions fixed. This would improve the initial policy, a contradiction. □

Transaction costs make portfolio choice genuinely dynamic because trading today changes tomorrow's state. With proportional costs, an inaction region often emerges: rebalance only when the benefit exceeds the cost of moving toward the frictionless target.

3.6 Problems

Exercise 3.4 (Core: efficient frontier)**.** Derive the target-return efficient portfolio and verify the formula for its variance. What happens when expected returns are proportional to $\mathbf{1}$?

Exercise 3.5 (Core: KKT)**.** Solve a two-asset minimum-variance problem with $w_i \geq 0$ and characterize when each nonnegativity constraint binds.

Exercise 3.6 (Advanced: robust mean)**.** Derive the ellipsoidal worst-case mean by Cauchy–Schwarz after the substitution $u = \Omega^{1/2}z$. Interpret δ.

Chapter 4

Probability, Information, and Statistical Limits

Learning objectives. Construct probability models with sigma-algebras, work with conditional expectation and changes of variables, understand modes of convergence, and derive the laws that justify estimation.

4.1 Probability spaces and random variables

Definition 4.1 (Probability space)**.** A probability space is $(\Omega, \mathcal{F}, \mathbb{P})$, where Ω is the sample space, $\mathcal{F}$ is a sigma-algebra, and $\mathbb{P} : \mathcal{F} \to [0, 1]$ is countably additive with $\mathbb{P}(\Omega) = 1$.

A sigma-algebra contains Ω, complements, and countable unions. This closure ensures that limits of observable events remain observable. A random variable $X : (\Omega, \mathcal{F}) \to (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ is measurable: $X^{-1}(B) \in \mathcal{F}$ for every Borel set B. Its distribution is the pushforward measure $\mathbb{P}_X(B) = \mathbb{P}(X \in B)$.

For a nonnegative random variable, expectation is defined by approximation with simple functions; for general integrable X, write $X = X^+ - X^-$. This construction explains why expectation is linear and why limit-exchange theorems need hypotheses.

Theorem 4.2 (Monotone convergence)**.** *If* $0 \le X_n \uparrow X$ *almost surely, then* $\mathbb{E}[X_n] \uparrow \mathbb{E}[X]$*, possibly to infinity.*

Theorem 4.3 (Dominated convergence)**.** *If* $X_n \to X$ *almost surely and* $|X_n| \le Y$ *for an integrable* Y*, then* X *is integrable and* $\mathbb{E}[X_n] \to \mathbb{E}[X]$*.*

These theorems justify differentiation or limiting operations under an expectation. Without domination or uniform integrability, pointwise convergence need not preserve prices.

4.2 Distributions and transforms

The distribution function $F_X(x) = \mathbb{P}(X \le x)$ is nondecreasing, right-continuous, and has limits 0 and 1. If it is absolutely continuous, its derivative is a density and $\mathbb{E}[g(X)] = \int g(x) f_X(x)\,\mathrm{d}x$.

The moment-generating function $M_X(t) = \mathbb{E}[\mathrm{e}^{tX}]$, when finite near zero, determines all moments and the distribution. The characteristic function $\varphi_X(u) = \mathbb{E}[\mathrm{e}^{iuX}]$ always exists and uniquely determines the law. Fourier pricing methods exploit

$$\mathbb{E}[g(X)] = \frac{1}{2\pi} \int_{\mathbb{R}} \widehat{g}(-u)\varphi_X(u)\,\mathrm{d}u$$

when the transform conditions hold.

Key families include Bernoulli and binomial counts; Poisson arrivals; exponential waiting times; Gaussian sums; lognormal positive prices; Student distributions with heavy tails; and generalized extreme-value laws for maxima. A named distribution is a hypothesis, not a property of a ticker.

Worked example 4.4 (Lognormal moments)**.** If $X \sim N(m, s^2)$ and $S = \mathrm{e}^X$, then

$$\mathbb{E}[S^k] = \mathrm{e}^{km + \frac{1}{2}k^2 s^2}.$$

Thus $\mathbb{E}[S] = \mathrm{e}^{m+s^2/2}$ and $\mathrm{Var}(S) = \mathrm{e}^{2m+s^2}(\mathrm{e}^{s^2} - 1)$. The median e^m is below the mean.

4.3 Conditional expectation

Definition 4.5 (Conditional expectation)**.** For integrable X and sub-sigma-algebra $\mathcal{G} \subseteq \mathcal{F}$, $\mathbb{E}[X \mid \mathcal{G}]$ is the almost-surely unique $\mathcal{G}$-measurable random variable such that

$$\mathbb{E}[\mathbf{1}_A\, \mathbb{E}[X \mid \mathcal{G}]] = \mathbb{E}[\mathbf{1}_A X] \quad \text{for all } A \in \mathcal{G}.$$

It is a projection of X onto available information. It is linear, monotone, and respects known quantities: $\mathbb{E}[ZX \mid \mathcal{G}] = Z\,\mathbb{E}[X \mid \mathcal{G}]$ when Z is bounded and $\mathcal{G}$-measurable.

Theorem 4.6 (Tower property)**.** *If $\mathcal{H} \subseteq \mathcal{G}$, then*

$$\mathbb{E}[\mathbb{E}[X \mid \mathcal{G}] \mid \mathcal{H}] = \mathbb{E}[X \mid \mathcal{H}].$$

Proof. The left side is $\mathcal{H}$-measurable. For $A \in \mathcal{H} \subseteq \mathcal{G}$, apply the defining integral identity twice. Uniqueness completes the proof. □

Proposition 4.7 (Conditional Jensen)**.** *For convex ϕ and suitable integrability,*

$$\phi(\mathbb{E}[X \mid \mathcal{G}]) \le \mathbb{E}[\phi(X) \mid \mathcal{G}].$$

4.4 Dependence

Independence is stronger than zero covariance. Copulas separate marginal distributions from dependence: Sklar's theorem represents a joint distribution as

$$F(x_1, \dots, x_d) = C(F_1(x_1), \dots, F_d(x_d)).$$

Gaussian copulas have zero asymptotic tail dependence unless correlation is perfect; Student copulas can generate symmetric tail dependence. Rank correlations are invariant under strictly monotone marginal transformations but still do not fully describe joint tails.

Definition 4.8 (Tail dependence)**.** For continuous X, Y, upper-tail dependence is

$$\lambda_U = \lim_{u \uparrow 1} \mathbb{P}\Big(Y > F_Y^{-1}(u) \mid X > F_X^{-1}(u)\Big),$$

when the limit exists. Lower-tail dependence is defined analogously.

4.5 Convergence and limit theorems

Almost-sure convergence implies convergence in probability, which implies convergence in distribution. Convergence in L^p is convergence of $\mathbb{E}[|X_n - X|^p]$ to zero. Reverse implications require additional conditions.

Theorem 4.9 (Weak law of large numbers)**.** *If $X_1, X_2, \dots$ are iid with finite mean μ, then $\overline{X}_n \to \mu$ in probability.*

Proof. When variance is finite, Chebyshev's inequality yields

$$\mathbb{P}(\left|\overline{X}_n - \mu\right| > \varepsilon) \le \frac{\mathrm{Var}(X_1)}{n\varepsilon^2} \to 0.$$

The finite-mean result follows by truncation. □

Theorem 4.10 (Lindeberg–Lévy central limit theorem)**.** *For iid variables with mean μ and variance $0 < \sigma^2 < \infty$,*

$$\sqrt{n}\,\frac{\overline{X}_n - \mu}{\sigma} \Rightarrow N(0,1).$$

Financial returns are neither exactly iid nor always finite-variance in useful samples. Mixing, martingale-difference, and stable-law limit theories address different departures. Standard errors must match the dependence structure.

Proposition 4.11 (Markov and Chebyshev inequalities)**.** *For nonnegative X and $a > 0$, $\mathbb{P}(X \ge a) \le \mathbb{E}[X]/a$. Hence for finite-variance Y, $\mathbb{P}(|Y - \mathbb{E}\,Y| \ge a) \le \mathrm{Var}(Y)/a^2$.*

4.6 Bayesian updating

With prior $\pi(\theta)$ and likelihood $p(y \mid \theta)$,

$$\pi(\theta \mid y) = \frac{p(y \mid \theta)\pi(\theta)}{\int p(y \mid u)\pi(u)\,\mathrm{d}u}.$$

Posterior prediction integrates parameter uncertainty: $p(\tilde{y} \mid y) = \int p(\tilde{y} \mid \theta)\pi(\theta \mid y)\,\mathrm{d}\theta$. This differs from plugging in a point estimate.

Worked example 4.12 (Normal–normal learning)**.** If $Y_i \mid \mu \sim N(\mu, \sigma^2)$ and $\mu \sim N(m_0, s_0^2)$ with known σ^2, then the posterior is normal with

$$s_n^{-2} = s_0^{-2} + n\sigma^{-2}, \qquad m_n = s_n^2(s_0^{-2}m_0 + n\sigma^{-2}\overline{Y}).$$

The posterior mean is precision-weighted shrinkage toward the prior.

4.7 Problems

Exercise 4.13 (Core: conditioning)**.** For jointly Gaussian (X, Y), derive $\mathbb{E}[Y \mid X]$ and $\mathrm{Var}(Y \mid X)$ by completing the square.

Exercise 4.14 (Core: convergence)**.** Construct a sequence that converges in probability but not almost surely. State an extra summability condition under which the Borel–Cantelli lemma gives almost-sure convergence.

Exercise 4.15 (Advanced: change of variable)**.** Derive the density of $Y = g(X)$ for a continuously differentiable monotone g and then generalize to finitely many inverse branches.

Chapter 5

Stochastic Processes and Continuous-Time Finance

Learning objectives. Work with filtrations, martingales, Brownian motion, stochastic integrals, Itô's formula, stochastic differential equations, and equivalent changes of measure.

5.1 Time, information, and martingales

A stochastic process is a family $(X_t)_{t\in T}$ of random variables on a common probability space. A filtration $(\mathcal{F}_t)$ is increasing: $\mathcal{F}_s \subseteq \mathcal{F}_t$ for $s \leq t$. A process is adapted if X_t is $\mathcal{F}_t$-measurable. Predictability is a slightly stronger pre-trade measurability condition required of trading integrands.

Definition 5.1 (Martingale)**.** An adapted integrable process M is a martingale when

$$\mathbb{E}[M_t \mid \mathcal{F}_s] = M_s, \qquad s \leq t.$$

Replacing equality by $\leq$ gives a supermartingale and by $\geq$ a submartingale.

Martingale does not mean independent increments, constant paths, or absence of risk. It means that conditional on the specified information set, the next expected change is zero. A discounted traded price is a martingale only under a suitable pricing measure and market assumptions.

Theorem 5.2 (Optional stopping, bounded version)**.** *If M is a martingale and τ is a bounded stopping time, then $\mathbb{E}[M_\tau] = \mathbb{E}[M_0]$.*

Proof. For integer time with $\tau \le N$, $M_\tau = M_0 + \sum_{t=1}^{N}(M_t - M_{t-1})\mathbf{1}_{\{\tau \ge t\}}$. The indicator is $\mathcal{F}_{t-1}$-measurable, so every increment has zero expectation. Continuous time follows by approximation under the stated regularity. □

Unbounded stopping rules require uniform integrability or related hypotheses. Doubling strategies appear to beat fair games precisely by violating admissibility or optional-stopping conditions.

5.2 Brownian motion and quadratic variation

Definition 5.3 (Brownian motion)**.** A process W is standard Brownian motion if $W_0 = 0$, paths are continuous, increments are independent and stationary, and $W_t - W_s \sim N(0, t-s)$ for $t > s$.

Brownian paths are almost surely nowhere classically differentiable and have infinite total variation on every interval. Yet their quadratic variation is finite:

$$\sum_k (W_{t_{k+1}} - W_{t_k})^2 \to t$$

in probability as mesh tends to zero. In differential notation, $(\mathrm{d}W_t)^2 = \mathrm{d}t$, $\mathrm{d}W_t\,\mathrm{d}t = 0$, and $(\mathrm{d}t)^2 = 0$. These are mnemonic rules for quadratic covariation, not ordinary infinitesimals.

5.3 The Itô integral

For a simple predictable process $H_t = \sum_k H_k \mathbf{1}_{(t_k, t_{k+1}]}(t)$, define

$$\int_0^T H_t\,\mathrm{d}W_t = \sum_k H_k (W_{t_{k+1}} - W_{t_k}).$$

The construction extends by completion to square-integrable predictable processes.

Theorem 5.4 (Itô isometry)**.** *If* $\mathbb{E}\int_0^T H_t^2\,\mathrm{d}t < \infty$*, then*

$$\mathbb{E}\left[\left(\int_0^T H_t\,\mathrm{d}W_t\right)^2\right] = \mathbb{E}\int_0^T H_t^2\,\mathrm{d}t.$$

Proof. For a simple process, expand the squared sum. Cross terms vanish because later Brownian increments have conditional mean zero, and diagonal terms equal $\mathbb{E}[H_k^2](t_{k+1} - t_k)$. Pass to the L^2 limit. □

The stochastic integral is a martingale under square integrability. It is defined with left-endpoint information, encoding that a strategy cannot see the increment it is about to trade.

5.4 Itô's formula

Theorem 5.5 (Itô's formula)**.** *If* $\mathrm{d}X_t = b(t, X_t)\,\mathrm{d}t + a(t, X_t)\,\mathrm{d}W_t$ *and* $f \in C^{1,2}$*, then*

$$\mathrm{d}f(t, X_t) = \left(f_t + bf_x + \frac{1}{2}a^2 f_{xx}\right)\,\mathrm{d}t + af_x\,\mathrm{d}W_t.$$

For d *dimensions, the second-order term is* $\frac{1}{2}\operatorname{tr}(aa^\mathsf{T}\nabla^2 f)\,\mathrm{d}t$.

Proof idea. Apply a second-order Taylor expansion over a fine partition. Terms in Δt^2 and $\Delta t \Delta W$ vanish in aggregate; the sum of $(\Delta W)^2$ converges to elapsed time. Control of the Taylor remainder completes the limit. □

Worked example 5.6 (Geometric Brownian motion)**.** If $\mathrm{d}S_t = \mu S_t\,\mathrm{d}t + \sigma S_t\,\mathrm{d}W_t$, Itô's formula for $\log S$ gives

$$\mathrm{d}\log S_t = (\mu - \tfrac{1}{2}\sigma^2)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t.$$

Hence $S_t = S_0 \exp((\mu - \sigma^2/2)t + \sigma W_t)$, which is positive and satisfies $\mathbb{E}[S_t] = S_0 \mathrm{e}^{\mu t}$.

Worked example 5.7 (Ornstein–Uhlenbeck process)**.** For $\mathrm{d}X_t = \kappa(\theta - X_t)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t$,

$$X_t = \theta + (X_0 - \theta)\mathrm{e}^{-\kappa t} + \sigma \int_0^t \mathrm{e}^{-\kappa(t-s)}\,\mathrm{d}W_s.$$

It is Gaussian with conditional mean $\theta + (X_s - \theta)\mathrm{e}^{-\kappa(t-s)}$ and variance $\sigma^2(1 - \mathrm{e}^{-2\kappa(t-s)})/(2\kappa)$.

5.5 Existence, generators, and Feynman–Kac

Global Lipschitz and linear-growth conditions on b and a are sufficient for a unique strong SDE solution. Many finance diffusions, including square-root models, need more refined results at boundaries.

The infinitesimal generator is

$$\mathcal{L}f = b^{\mathsf{T}}\nabla f + \frac{1}{2}\operatorname{tr}(aa^{\mathsf{T}}\nabla^2 f).$$

If

$$u(t,x) = \mathbb{E}_{t,x}\left[\mathrm{e}^{-\int_t^T c(X_s)\,\mathrm{d}s} g(X_T) + \int_t^T \mathrm{e}^{-\int_t^s c(X_v)\,\mathrm{d}v} q(s,X_s)\,\mathrm{d}s\right],$$

then under regularity u solves $u_t + \mathcal{L}u - cu + q = 0$ with $u(T,x) = g(x)$. This Feynman–Kac bridge connects conditional expectations, pricing, and partial differential equations.

5.6 Change of measure

Let θ be adapted and satisfy an integrability condition such as Novikov's: $\mathbb{E}[\exp(\frac{1}{2}\int_0^T \theta_t^2\,\mathrm{d}t)] < \infty$. Define

$$Z_T = \exp\left(-\int_0^T \theta_t\,\mathrm{d}W_t - \frac{1}{2}\int_0^T \theta_t^2\,\mathrm{d}t\right), \qquad \frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}} = Z_T.$$

Theorem 5.8 (Girsanov)**.** *Under $\mathbb{Q}$, the process $W_t^{\mathbb{Q}} = W_t + \int_0^t \theta_s\,\mathrm{d}s$ is Brownian motion.*

If a stock obeys $\mathrm{d}S/S = \mu\,\mathrm{d}t + \sigma\,\mathrm{d}W^{\mathbb{P}}$ and the money-market rate is r, choose $\theta = (\mu - r)/\sigma$. Then under $\mathbb{Q}$, $\mathrm{d}S/S = r\,\mathrm{d}t + \sigma\,\mathrm{d}W^{\mathbb{Q}}$. The physical expected return disappears from the replication price, while volatility remains.

Model-risk warning 5.9*.* An equivalent pricing measure is not a forecast distribution. Risk-neutral probabilities incorporate risk prices and are designed to price traded payoffs. Scenario probabilities for risk management belong under a physical or stressed measure.

5.7 Jumps and semimartingales

Brownian paths cannot produce discontinuous prices. A jump diffusion may be written

$$\frac{\mathrm{d}S_t}{S_{t-}} = (\mu - \lambda\kappa_J)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t + (J-1)\,\mathrm{d}N_t,$$

where N is Poisson with intensity λ and $\kappa_J = \mathbb{E}[J-1]$. Jumps alter Itô's formula by adding the exact discontinuity $f(S_t) - f(S_{t-})$ rather than a local quadratic approximation.

The general no-arbitrage language uses semimartingales because they support a broad class of stochastic integrals while ruling out pathwise schemes that manufacture profit from infinite variation.

5.8 Problems

Exercise 5.10 (Core: stochastic exponential)**.** Use Itô's formula to show that $\mathcal{E}_t = \exp(\sigma W_t - \frac{1}{2}\sigma^2 t)$ is a martingale and solves $\mathrm{d}\mathcal{E}_t = \sigma\mathcal{E}_t\,\mathrm{d}W_t$.

Exercise 5.11 (Core: generator)**.** Find the generator of geometric Brownian motion and use it to verify the backward equation for a discounted European payoff.

Exercise 5.12 (Advanced: quadratic covariation)**.** For correlated Brownian motions with $\mathrm{d}\langle W^1, W^2\rangle_t = \rho\,\mathrm{d}t$, derive the two-dimensional Itô formula and apply it to X_tY_t.

Exercise 5.13 (Advanced: measure change)**.** Derive the market price of risk in a one-stock complete market. Explain why multiple pricing measures generally exist when Brownian risk has more dimensions than traded risky assets.

Part II

Macroeconomics and the Term Structure

Chapter 6

Intertemporal Macroeconomics

Learning objectives. Connect national accounts, consumption and saving, growth, and equilibrium to asset prices; derive Euler equations and understand where representative-agent models succeed and fail.

6.1 Stocks, flows, nominal quantities, and real quantities

Gross domestic product is the market value of final production within an economy during a period. The expenditure identity is

$$Y_t = C_t + I_t + G_t + NX_t.$$

It is an accounting identity, not a behavioral equation. Consumption, investment, government purchases, and net exports are flows. Wealth, debt, and capital are stocks measured at a point in time. A stock evolves through flows and valuation:

$$W_t - W_{t-1} = \text{saving}_t + \text{capital gain}_t + \text{transfers}_t.$$

Nominal GDP uses current prices. Real GDP applies a price index or chain-weighted volume measure. If P_t is a price index, inflation is $\pi_t = \log P_t - \log P_{t-1}$ and the approximate Fisher relation is $i_t \approx r_t + \mathbb{E}_t\,\pi_{t+1}$. Exact gross returns satisfy

$$1 + r_t = \frac{1 + i_t}{1 + \pi_{t+1}}.$$

Model-risk warning 6.1. Macro series are revised, seasonally adjusted, sampled at mixed frequencies, and released with lags. A backtest using the latest revised history has information that was unavailable in real time. Vintage data are required for a genuine forecasting experiment.

6.2 Two-period consumption

A household with initial wealth w, incomes y_0, y_1, and gross risk-free return R_f solves

$$\max_{c_0,c_1} u(c_0) + \beta u(c_1) \quad \text{s.t.} \quad c_0 + \frac{c_1}{R_f} = w + y_0 + \frac{y_1}{R_f}.$$

At an interior solution,

$$u'(c_0) = \beta R_f u'(c_1).$$

The relative price of future consumption is $1/R_f$. A higher rate has a substitution effect favoring future consumption and a wealth effect whose sign depends on whether the household is a borrower or lender.

6.3 Consumption under uncertainty

In many periods, the agent chooses adapted consumption and a portfolio subject to a budget constraint. The asset-specific Euler condition is

$$1 = \mathbb{E}_t[M_{t+1} R_{i,t+1}], \qquad M_{t+1} = \beta \frac{u'(C_{t+1})}{u'(C_t)}.$$

The random variable M is the stochastic discount factor. It is high in states where marginal utility is high, so an asset that pays in bad states is valuable.

For CRRA utility $u(C) = C^{1-\gamma}/(1-\gamma)$,

$$M_{t+1} = \beta \left(\frac{C_{t+1}}{C_t} \right)^{-\gamma}.$$

The risk-free rate satisfies $R_{f,t}^{-1} = \mathbb{E}_t[M_{t+1}]$, while the expected excess return obeys

$$\mathbb{E}_t[R_{i,t+1}] - R_{f,t} = -\frac{\mathrm{Cov}_t(M_{t+1}, R_{i,t+1})}{\mathbb{E}_t[M_{t+1}]}.$$

Proof. The Euler equation gives $1 = \mathbb{E}_t[M]\,\mathbb{E}_t[R_i] + \mathrm{Cov}_t(M, R_i)$. For the risk-free asset, $1 = R_f\,\mathbb{E}_t[M]$. Subtract and divide by $\mathbb{E}_t[M]$. □

This equation is central to macro-finance: expected returns compensate covariance with marginal utility, not variance in isolation.

6.4 Permanent income and precautionary saving

With quadratic utility, a constant rate, and certainty-equivalent income, consumption is proportional to annuitized total wealth. Under rational expectations, changes in consumption reflect news about permanent resources. With concave marginal utility, uncertainty adds precautionary saving when $u'''(c) > 0$.

Proposition 6.2 (Euler inequality under a borrowing constraint). *If the agent cannot borrow beyond a boundary, the consumption Euler condition becomes*

$$u'(C_t) \geq \beta R_f\,\mathbb{E}_t[u'(C_{t+1})],$$

with equality when the constraint does not bind.

Heterogeneous agents matter because constraints bind in bad states and marginal propensities to consume vary. A representative agent can match aggregate resource constraints while missing distributional channels important for monetary policy and risk premia.

6.5 Production and growth

Let output be $Y_t = F(K_t, A_t L_t)$ and capital evolve by $K_{t+1} = (1-\delta)K_t + I_t$. With constant returns to scale and Cobb–Douglas $F(K, AL) = K^{\alpha}(AL)^{1-\alpha}$, competitive factor prices satisfy

$$r_t^K = \alpha\frac{Y_t}{K_t}, \qquad w_t = (1-\alpha)\frac{Y_t}{L_t}.$$

In the Solow model per effective worker,

$$k_{t+1} = \frac{sf(k_t) + (1-\delta)k_t}{(1+n)(1+g)}.$$

The steady state equates investment per effective worker with break-even investment. The golden-rule stock maximizes steady-state consumption and satisfies $f'(k_{\mathrm{GR}}) = n + g + \delta$ in the continuous-time approximation.

Growth expectations affect long-duration cash flows, real rates, fiscal capacity, and default probabilities. But an observed equity index is not GDP: multinational revenue, valuation multiples, leverage, entry and exit, and profit shares separate market capitalization from domestic production.

6.6 Open-economy parity

Covered interest parity follows from no arbitrage:

$$\frac{F_{t,T}}{S_t} = \frac{B_t^d(T)^{-1}}{B_t^f(T)^{-1}}$$

under consistent quote conventions, where S_t is domestic currency per unit of foreign currency. In one-period notation, $F/S = (1+i_d)/(1+i_f)$. Uncovered interest parity replaces the forward rate with an expected future spot rate and is a risky equilibrium statement, not arbitrage. Cross-currency basis measures deviations after accounting for instruments, collateral, funding, and balance-sheet costs.

6.7 Problems

Exercise 6.3 (Core: Euler equation)**.** Derive the stochastic Euler equation by perturbing consumption and holdings of a traded asset. State the transversality condition needed in an infinite-horizon problem.

Exercise 6.4 (Core: Fisher relation)**.** Derive the exact real return and quantify the approximation error in $r \approx i - \pi$ to second order.

Exercise 6.5 (Advanced: habit)**.** Let utility depend on surplus consumption $C_t - hC_{t-1}$. Derive marginal utility and explain qualitatively how habit can amplify time-varying risk premia.

Chapter 7

Monetary Economics and Macro-Financial Dynamics

Learning objectives. Derive the canonical New Keynesian system, distinguish policy rules from shocks, understand identification around announcements, and connect financial conditions to the business cycle.

7.1 A three-equation model

A log-linearized New Keynesian system contains a dynamic IS curve,

$$x_t = \mathbb{E}_t\, x_{t+1} - \frac{1}{\sigma}\left(i_t - \mathbb{E}_t\, \pi_{t+1} - r_t^n\right),$$

a Phillips curve,

$$\pi_t = \beta\, \mathbb{E}_t\, \pi_{t+1} + \kappa x_t + u_t,$$

and a monetary-policy rule,

$$i_t = \rho i_{t-1} + (1-\rho)(\bar{i} + \phi_\pi \pi_t + \phi_x x_t) + \varepsilon_t^m.$$

Here x_t is the output gap, r_t^n the natural real rate, u_t a cost-push shock, and ε_t^m a policy innovation.

The IS equation is an Euler condition: demand falls when the ex ante real rate rises relative to its natural level. The Phillips curve comes from forward-looking price setting with nominal rigidity. The policy rule is a description of systematic behavior plus a residual; it is not the central bank's loss function.

Definition 7.1 (Taylor principle)**.** In a basic model, the policy rule satisfies the Taylor principle when the nominal rate responds more than one-for-one to persistent inflation, $\phi_\pi > 1$, so the real rate eventually rises.

Under familiar restrictions, the principle contributes to local determinacy. At the effective lower bound or with fiscal interactions, the result changes.

7.2 Present-value solutions and forward guidance

A stable scalar equation $x_t = a\,\mathbb{E}_t\,x_{t+1} + bz_t$ with $0 < a < 1$ has the forward solution

$$x_t = b\sum_{j=0}^{\infty} a^j\,\mathbb{E}_t\,z_{t+j},$$

provided the discounted terminal term vanishes. Asset prices and macro variables therefore react to the entire expected policy path, not only today's overnight rate.

Forward guidance can move long rates by changing expected future short rates and term premia. Its measured effect depends on whether an announcement changes the policy path, economic outlook, uncertainty, or balance-sheet policy.

7.3 Central-bank balance sheets and liquidity

An asset purchase exchanges one consolidated-government liability for another unless it changes duration, liquidity, risk bearing, or expected future policy. Candidate channels include duration extraction, scarcity, signaling, collateral value, and market-functioning support. The central bank balance sheet has assets, reserves, currency, capital, and income risk; accounting losses do not map mechanically into private-firm insolvency but can affect remittances and political constraints.

Money markets connect the policy rate to secured and unsecured funding. Repo haircuts and margins create nonlinear liquidity demand. A dealer financing a position with haircut h supplies equity hP and borrows $(1-h)P$;

a rising haircut forces deleveraging even if the asset's expected cash flow is unchanged.

7.4 Financial accelerator

Borrowers pay an external-finance premium that falls with net worth. A negative asset-price shock reduces collateral, tightens borrowing capacity, lowers investment, and may cause further price declines. In a reduced-form relation,

$$\text{spread}_t = \psi\left(\frac{\text{external finance}_t}{\text{net worth}_t}\right) + \eta_t, \qquad \psi' > 0.$$

This feedback makes state dependence essential. A linear model near normal times may understate crisis amplification.

7.5 Policy identification

Observed policy rates respond endogenously to inflation and growth. Regressing asset returns on rate changes mixes the policy response with underlying news. Identification strategies include:

- narrow intraday windows around scheduled announcements;
- high-frequency surprises from rate futures;
- sign or narrative restrictions in vector autoregressions;
- external instruments correlated with shocks but orthogonal to other news;
- local projections tracing dynamic responses.

Definition 7.2 (Instrument validity)**.** An instrument Z for endogenous treatment D must be relevant, $\text{Cov}(Z, D) \neq 0$, and excluded from the outcome except through D: $\mathbb{E}[Z\varepsilon] = 0$ in the structural equation.

Announcement surprises may combine a pure policy shock and central-bank information about the outlook. Joint moves in rates and equities can help separate these components, but the decomposition itself rests on sign and timing assumptions.

7.6 Inflation, discount rates, and asset classes

Unexpected inflation redistributes between nominal debtors and creditors. Nominal bond prices respond to expected future inflation, real-rate expectations, and risk premia. Equity response depends on pricing power, input costs, leverage, and the discount-rate channel. Commodities may respond to demand, supply, inventories, and currency simultaneously.

Break-even inflation is approximately

$$\begin{aligned} y_t^N(T) - y_t^R(T) = {} & \mathbb{E}_t^{\text{pricing}}[\bar{\pi}_{t,T}] \\ & + \text{inflation risk premium} + \text{liquidity differential}. \end{aligned}$$

It is not a pure survey expectation.

7.7 State-space models and the Kalman filter

Let

$$x_t = Ax_{t-1} + Bu_t + \eta_t, \qquad y_t = Cx_t + \varepsilon_t,$$

with Gaussian shocks of covariance Q and R. Given filtered mean $\widehat{x}_{t-1|t-1}$ and covariance $P_{t-1|t-1}$, prediction is

$$\widehat{x}_{t|t-1} = A\widehat{x}_{t-1|t-1} + Bu_t, \quad P_{t|t-1} = AP_{t-1|t-1}A^\mathsf{T} + Q.$$

The innovation and Kalman gain are

$$v_t = y_t - C\widehat{x}_{t|t-1}, \quad S_t = CP_{t|t-1}C^\mathsf{T} + R, \quad K_t = P_{t|t-1}C^\mathsf{T}S_t^{-1},$$

and updating gives

$$\widehat{x}_{t|t} = \widehat{x}_{t|t-1} + K_t v_t, \quad P_{t|t} = (I - K_t C)P_{t|t-1}.$$

Theorem 7.3 (Linear-Gaussian filtering). *Under the stated model, $\widehat{x}_{t|t} = \mathbb{E}[x_t \mid y_1, \ldots, y_t]$ and $P_{t|t}$ is its conditional covariance.*

Proof. Induct on t. Linear propagation preserves joint Gaussianity. Conditioning a joint Gaussian vector on the new observation gives exactly the gain update above. $\square$

A filtered estimate uses only information available by t; a smoothed estimate uses future observations. Using a smoother inside a historical trading rule creates look-ahead bias.

7.8 Problems

Exercise 7.4 (Core: policy rule)**.** In the static version of the three-equation model, solve for inflation and the output gap after a monetary shock. Track signs and state parameter restrictions.

Exercise 7.5 (Core: Kalman update)**.** Derive the scalar Kalman gain and show how it changes when measurement noise tends to zero or infinity.

Exercise 7.6 (Advanced: local projection)**.** Write a local-projection specification for the response of credit spreads to an instrumented policy surprise. State the timing, fixed effects, and standard-error choices.

Chapter 8

Yield Curves, Macro Factors, and Regimes

Learning objectives. Read a yield curve, decompose nominal rates, fit parsimonious curves, understand expectations and term premia, and build macro-financial regime models without look-ahead bias.

8.1 Discount factors, spot rates, and forwards

The time-t price of one unit at T is $P(t,T)$. With continuously compounded zero rate $y(t,T)$,

$$P(t,T) = \mathrm{e}^{-y(t,T)(T-t)}.$$

The instantaneous forward rate satisfies

$$f(t,T) = -\frac{\partial}{\partial T}\log P(t,T), \qquad P(t,T) = \exp\left(-\int_t^T f(t,u)\,\mathrm{d}u\right).$$

For discretely compounded market instruments, day count, settlement, calendars, and coupon conventions must be handled before comparing quoted yields.

8.2 Expectations hypothesis and term premia

An n-period yield can be decomposed schematically as

$$y_t^{(n)} = \frac{1}{n}\sum_{j=0}^{n-1} \mathbb{E}_t\, r_{t+j} + TP_t^{(n)}.$$

The first term is an expected path of short rates; the term premium compensates risk, duration supply, preferred habitat, and model-specific covariance. Neither component is observed directly. Different models can fit the same yields and disagree about the decomposition.

Under pure expectations plus constant premium, a steep curve predicts rising short rates. Empirically, forward rates also vary with risk premia. Regressions of excess bond returns on forward-rate combinations test this tension but are sensitive to persistence and small samples.

8.3 Nelson–Siegel representation

At maturity τ, the Nelson–Siegel yield is

$$y_t(\tau) = \beta_{0t} + \beta_{1t}\frac{1-\mathrm{e}^{-\lambda\tau}}{\lambda\tau} + \beta_{2t}\left(\frac{1-\mathrm{e}^{-\lambda\tau}}{\lambda\tau} - \mathrm{e}^{-\lambda\tau}\right).$$

The loadings motivate level, slope, and curvature interpretations. For fixed λ, factor estimates are linear least squares. Choosing λ by minimizing cross-sectional error is nonlinear and can be unstable on sparse curves.

Worked example 8.1 (Limiting loadings)**.** As $\tau \downarrow 0$, the slope loading tends to 1 and the curvature loading to 0. As $\tau \to \infty$, both tend to 0. Thus β_0 dominates long maturity, β_1 moves the short end, and β_2 produces a hump.

Dynamic Nelson–Siegel models the factors as a vector autoregression or state-space process. Measurement error at each maturity distinguishes curve-fitting noise from factor dynamics.

8.4 Affine term-structure models

Let state X_t follow an affine diffusion under $\mathbb{Q}$:

$$\mathrm{d}X_t = K^{\mathbb{Q}}(\theta^{\mathbb{Q}} - X_t)\,\mathrm{d}t + \Sigma(X_t)\,\mathrm{d}W_t^{\mathbb{Q}}, \qquad r_t = \delta_0 + \delta_1^{\top} X_t.$$

Under affine regularity, the zero-coupon price is exponential affine:

$$P(t,T) = \exp(A(\tau) - B(\tau)^{\top} X_t).$$

Substitution into the pricing PDE produces Riccati ordinary differential equations for A and B. A separate physical-measure drift determines forecasting dynamics and market prices of risk.

Worked example 8.2 (Vasicek bond coefficients)**.** For $\mathrm{d}r = \kappa(\theta - r)\,\mathrm{d}t + \sigma\,\mathrm{d}W^{\mathbb{Q}}$,

$$P(t,T) = A(\tau)\mathrm{e}^{-B(\tau)r_t}, \qquad B(\tau) = \frac{1 - \mathrm{e}^{-\kappa\tau}}{\kappa},$$

with

$$\log A(\tau) = \left(\theta - \frac{\sigma^2}{2\kappa^2}\right)(B(\tau) - \tau) - \frac{\sigma^2 B(\tau)^2}{4\kappa}.$$

Gaussian rates make the model tractable but allow negative rates.

8.5 Macro regimes

A discrete Markov regime $S_t \in \{1, \ldots, K\}$ has transition probabilities $p_{ij} = \mathbb{P}(S_t = j \mid S_{t-1} = i)$. Conditional observations can have regime-dependent mean and covariance:

$$y_t \mid S_t = k \sim N(\mu_k, \Sigma_k).$$

The Hamilton filter alternates prediction $\xi_{t|t-1} = P^{\top}\xi_{t-1|t-1}$ and Bayes updating using the conditional density. Smoothed probabilities use the entire sample and are inappropriate as real-time signals.

Definition 8.3 (Expected regime duration)**.** For a time-homogeneous Markov chain, the duration in regime i is geometrically distributed with mean $1/(1 - p_{ii})$.

Regime labels are unidentified up to permutation. More seriously, a finite mixture can use extra states to absorb outliers rather than discover economic regimes. Economic interpretation should follow estimation and be checked across vintages, initializations, and alternative emission distributions.

8.6 A disciplined macro-finance dashboard

A research dashboard can combine:

- growth: industrial production, payrolls, claims, and activity surveys;
- inflation: consumer prices, producer prices, wages, and expectations;
- policy: overnight rate, expected path, and balance-sheet measures;
- curve: short rate, two–ten slope, forward rates, and term-premium estimate;
- financial conditions: credit spreads, equity volatility, dollar, and liquidity;
- vintages: the actual release available on each decision date.

Transformations must match the question. Year-over-year growth is smooth but delayed; monthly annualized growth is timely but noisy. Standardizing over a full sample leaks future moments. A live procedure uses expanding or rolling estimates.

8.7 Problems

Exercise 8.4 (Core: forwards)**.** Starting from discount factors, derive a forward rate between T_1 and T_2 under continuous and annual compounding.

Exercise 8.5 (Core: Nelson–Siegel)**.** For fixed λ, express cross-sectional factor estimation in matrix form. Describe how you would weight maturities when quote noise differs.

Exercise 8.6 (Advanced: regime filtering)**.** Derive the two-state Hamilton filter and log-likelihood. Explain why filtered and smoothed performance must be reported separately.

Exercise 8.7 (Research). Build an expanding-window recession-probability model using only vintage-available predictors. Compare calibration, ROC area, and log score against a constant probability benchmark.

Part III

Econometrics and Causal Evidence

Chapter 9

Statistical Identification and Linear Models

Learning objectives. Distinguish an estimand from an estimator, derive least squares and its sampling properties, diagnose endogeneity, and use uncertainty estimates appropriate to financial data.

9.1 From a question to an estimand

An empirical project needs four separate objects:

1. a population or data-generating process;
2. an estimand, such as a conditional mean, causal effect, or risk functional;
3. an estimator computed from a sample;
4. a decision rule that uses the estimate.

Prediction asks for a function that performs well under a specified loss on new data. Structural estimation asks for parameters with economic meaning under a model. Causal inference asks how an outcome changes under an intervention. The same regression code can be used for all three, but its coefficient does not inherit an interpretation from the software.

Definition 9.1 (Identification)**.** A parameter θ is identified by the distribution of observables P if $P_{\theta_1} = P_{\theta_2}$ implies $\theta_1 = \theta_2$. It is set identified when the observables restrict it only to a non-singleton set.

Identification is a population property; precision is a sample property. Infinite data cannot identify a parameter that two observationally equivalent models map to different values.

9.2 Population regression

The best linear predictor of scalar Y from vector X solves

$$\beta^\star = \arg\min_b \mathbb{E}[(Y - X^\top b)^2].$$

If $\mathbb{E}[XX^\top]$ is nonsingular,

$$\beta^\star = \mathbb{E}[XX^\top]^{-1}\,\mathbb{E}[XY].$$

The projection error $u = Y - X^\top\beta^\star$ satisfies $\mathbb{E}[Xu] = 0$ by construction. This orthogonality does not imply $\mathbb{E}[u \mid X] = 0$ or causality.

9.3 Ordinary least squares

For $y = X\beta + \varepsilon$, OLS is

$$\widehat{\beta} = (X^\top X)^{-1}X^\top y = \beta + (X^\top X)^{-1}X^\top\varepsilon.$$

Assumption 9.2 (Classical conditional model)**.** X has full column rank, $\mathbb{E}[\varepsilon \mid X] = 0$, and $\mathrm{Var}(\varepsilon \mid X) = \sigma^2 I$.

Theorem 9.3 (Gauss–Markov)**.** *Under the classical conditional model, OLS is the best linear unbiased estimator: for any linear unbiased $\widetilde{\beta} = Ay$,*

$$\mathrm{Var}(\widetilde{\beta} \mid X) - \mathrm{Var}(\widehat{\beta} \mid X) \succeq 0.$$

Proof. Unbiasedness requires $AX = I$. Write $A = (X^\top X)^{-1}X^\top + D$, so $DX = 0$. Conditional variance is $\sigma^2 AA^\top = \sigma^2[(X^\top X)^{-1} + DD^\top]$ because cross terms vanish. □

Normality is unnecessary for Gauss–Markov. It gives exact finite-sample t and F distributions when combined with the other assumptions.

9.4 Large-sample inference

Under weak conditions,

$$\sqrt{n}(\widehat{\beta} - \beta^\star) \Rightarrow N(0, Q^{-1}\Omega Q^{-1}),$$

where $Q = \mathbb{E}[X_i X_i^\top]$ and $\Omega = \mathbb{E}[X_i X_i^\top u_i^2]$ for independent heteroskedastic observations. The sandwich estimator replaces expectations with sample analogues.

For time series, serial correlation adds long-run covariances:

$$\Omega = \sum_{h=-\infty}^{\infty} \mathbb{E}[X_t u_t u_{t-h} X_{t-h}^\top].$$

Newey–West HAC estimation truncates and weights sample autocovariances. Bandwidth choice trades bias against variance; HAC does not rescue a misspecified conditional mean or overlapping-label leakage.

Model-risk warning 9.4. Daily observations do not guarantee daily independent information. Persistent predictors, overlapping multi-day returns, common shocks, and clustered trades can reduce the effective sample far below the row count.

9.5 Omitted variables and measurement error

Suppose the true model is $Y = \beta X + \gamma Z + u$, but Z is omitted. In a scalar regression,

$$\operatorname{plim} \widehat{\beta}_{\text{short}} = \beta + \gamma \frac{\operatorname{Cov}(X, Z)}{\operatorname{Var}(X)}.$$

The sign depends on both the effect of Z and its association with X.

Classical measurement error $X^* = X + v$, with v independent, creates attenuation:

$$\operatorname{plim} \widehat{\beta} = \beta \frac{\operatorname{Var}(X)}{\operatorname{Var}(X) + \operatorname{Var}(v)}.$$

Nonclassical financial measurement error—stale prices, bid–ask bounce, imputed fundamentals—can produce any direction of bias.

9.6 Instrumental variables

For structural equation $Y = D\beta + X^\top\gamma + u$, an instrument Z supplies variation in D that is orthogonal to u. With one centered instrument and treatment,

$$\widehat{\beta}_{\mathrm{IV}} = \frac{\widehat{\mathrm{Cov}}(Z,Y)}{\widehat{\mathrm{Cov}}(Z,D)}.$$

Proposition 9.5 (Two-stage least squares)**.** *With instrument matrix Z including exogenous controls,*

$$\widehat{\beta}_{\mathrm{2SLS}} = (X^\top P_Z X)^{-1} X^\top P_Z y.$$

It equals OLS of y on the fitted values only with the correct second-stage standard errors based on the original structural residuals.

Weak instruments create biased, nonnormal finite-sample inference. Report first-stage strength, weak-IV robust confidence sets, and a substantive argument for exclusion.

9.7 Hypothesis testing and multiplicity

A p-value is the probability, under the null and the test's assumptions, of a statistic at least as extreme as observed. It is not the probability the null is true. A confidence interval is a procedure with repeated-sample coverage; it is not a posterior probability interval unless a Bayesian model supplies that meaning.

Testing hundreds of signals selects noise. Family-wise error control targets any false rejection; false discovery rate targets the expected proportion among discoveries. In research portfolios, dependence and adaptive specification search complicate both. A held-out test set and a complete research log are more credible than correcting only the final table.

9.8 Problems

Exercise 9.6 (Core: projection versus causality)**.** Construct a population in which the best linear predictor coefficient is nonzero but the causal effect of X on Y is zero.

Exercise 9.7 (Core: robust variance)**.** Derive the heteroskedastic sandwich covariance from a central limit theorem for $n^{-1/2} \sum_i X_i u_i$.

Exercise 9.8 (Advanced: weak instruments)**.** Analyze the IV ratio when $\mathrm{Cov}(Z, D)$ approaches zero at rate $n^{-1/2}$. Explain why ordinary normal approximations fail.

Chapter 10

Time-Series Econometrics

Learning objectives. Model stationarity, persistence, trends, cointegration, and multivariate dynamics; construct genuine forecasts; and understand the effect of parameter instability.

10.1 Stationarity, ergodicity, and autocovariance

A process is strictly stationary if every finite-dimensional distribution is invariant to a common time shift. It is covariance stationary if mean is constant, variance finite and constant, and covariance depends only on lag: $\gamma(h) = \mathrm{Cov}(X_t, X_{t-h})$. Ergodicity ensures time averages reveal population features.

The autocorrelation is $\rho(h) = \gamma(h)/\gamma(0)$. The spectral density

$$f(\omega) = \frac{1}{2\pi} \sum_{h=-\infty}^{\infty} \gamma(h) \mathrm{e}^{-i\omega h}$$

decomposes variance by frequency, with $\gamma(h) = \int_{-\pi}^{\pi} \mathrm{e}^{i\omega h} f(\omega)\, \mathrm{d}\omega$.

10.2 ARMA models

Using lag operator $LX_t = X_{t-1}$, an ARMA(p, q) is

$$\phi(L)X_t = c + \theta(L)\varepsilon_t.$$

Causality requires roots of $\phi(z) = 0$ outside the unit circle; invertibility requires the same for $\theta(z)$. These conditions yield stable infinite-order representations.

Worked example 10.1 (AR(1))**.** For $X_t = c + \phi X_{t-1} + \varepsilon_t$ with $|\phi| < 1$,

$$\mathbb{E}[X_t] = \frac{c}{1-\phi}, \quad \mathrm{Var}(X_t) = \frac{\sigma_\varepsilon^2}{1-\phi^2}, \quad \rho(h) = \phi^{|h|}.$$

The h-step forecast is

$$\widehat{X}_{t+h|t} = \bar{X} + \phi^h (X_t - \bar{X}),$$

and its error variance is $\sigma_\varepsilon^2 \sum_{j=0}^{h-1} \phi^{2j}$.

Information criteria balance fit and dimension: $\mathrm{AIC} = -2\ell + 2k$ and $\mathrm{BIC} = -2\ell + k \log n$. Selecting a model and reporting standard inference as if it were fixed understates uncertainty.

10.3 Unit roots and spurious regression

When $X_t = X_{t-1} + \varepsilon_t$, shocks have permanent effects and variance grows with t. Regressing one unrelated random walk on another can produce high R^2 and apparently significant coefficients.

The augmented Dickey–Fuller regression is

$$\Delta X_t = \alpha + \delta t + \gamma X_{t-1} + \sum_{j=1}^{p} \psi_j \Delta X_{t-j} + u_t.$$

The null $\gamma = 0$ has a nonstandard distribution. Failure to reject is not proof of a unit root, and deterministic terms materially change critical values.

Near-unit-root predictors cause severe finite-sample bias in return-predictability regressions, particularly when innovations to returns and predictors are correlated.

10.4 Cointegration and error correction

If nonstationary X_t and Y_t have a stationary linear combination $Y_t - \beta X_t$, they are cointegrated. The error-correction model

$$\Delta Y_t = \alpha(Y_{t-1} - \beta X_{t-1}) + \Gamma \Delta X_t + \varepsilon_t$$

combines long-run equilibrium and short-run adjustment. Cointegration is not a guarantee of profitable convergence trading; the relation can break, the error can be untradeable, and financing horizons can be shorter than mean reversion.

10.5 Vector autoregressions

A VAR(p) is

$$Y_t = c + A_1 Y_{t-1} + \cdots + A_p Y_{t-p} + u_t.$$

The reduced-form covariance Σ_u contains contemporaneously correlated shocks. Impulse responses require identifying structural shocks $u_t = B\varepsilon_t$. Since $BB^\top = \Sigma_u$ does not uniquely determine B, recursive ordering, long-run restrictions, signs, heteroskedasticity, or external instruments supply additional assumptions.

Local projections estimate each horizon separately:

$$Y_{t+h} - Y_{t-1} = \alpha_h + \beta_h \widehat{\varepsilon}_t + \Gamma_h^\top W_t + u_{t+h}.$$

They are flexible but can be noisy; VARs impose more dynamic structure.

10.6 Forecast evaluation

A valid forecast records information available at origin t, trains only through t, and predicts a target defined after t. For point forecast $\widehat{y}_t$, loss might be squared error, absolute error, quantile loss, or economic utility. Match the loss to the decision.

The Diebold–Mariano comparison uses loss differential $d_t = L(e_{1t}) - L(e_{2t})$ and tests $\mathbb{E}[d_t] = 0$ with a long-run variance estimate. Nested models and estimated parameters can require adjusted inference.

Directional accuracy is not enough if gains and losses are asymmetric. An economic backtest must define position timing, costs, leverage, financing, and benchmark. A forecast can improve mean squared error yet destroy utility, or vice versa.

10.7 Structural breaks and rolling models

Parameter instability can be abrupt or gradual. A Chow test assumes a known break; supremum tests search unknown break dates; CUSUM uses recursive residuals. Rolling windows adapt but discard information. Expanding windows stabilize estimates but adapt slowly.

Definition 10.2 (Pseudo-out-of-sample experiment)**.** Choose an initial training endpoint T_0. For each $t \geq T_0$, estimate the entire pipeline using data available at t, issue a forecast, advance time, and never revise the stored prediction after the outcome arrives.

10.8 Problems

Exercise 10.3 (Core: AR forecast)**.** Derive the AR(1) h-step forecast-error variance and its limit as $h \to \infty$.

Exercise 10.4 (Core: cointegration)**.** Show that if X_t is a random walk and $Y_t = \beta X_t + u_t$ with stationary u_t, then (Y_t, X_t) are cointegrated. Derive an error-correction representation.

Exercise 10.5 (Advanced: real-time design)**.** Write pseudocode for a pseudo-out-of-sample forecast with monthly predictors released on different dates and a daily trading target.

Chapter 11

Volatility, Dependence, and Extreme Risk

Learning objectives. Estimate conditional variance, distinguish realized from implied volatility, model heavy tails and multivariate dependence, and use extreme-value methods with honest uncertainty.

11.1 Volatility as a conditional object

For return $r_t = \mu_t + \varepsilon_t$, conditional variance is $h_t = \operatorname{Var}(r_t \mid \mathcal{F}_{t-1})$. It is latent. Squared return is a noisy ex post proxy; high-frequency realized variance estimates integrated variance under market microstructure conditions.

The ARCH(q) model is

$$h_t = \omega + \sum_{i=1}^{q} \alpha_i \varepsilon_{t-i}^2.$$

GARCH$(1,1)$ is

$$h_t = \omega + \alpha\varepsilon_{t-1}^2 + \beta h_{t-1}, \qquad \omega > 0,\ \alpha, \beta \geq 0.$$

If $\alpha + \beta < 1$, unconditional variance is $\omega/(1 - \alpha - \beta)$ under the standard innovation assumptions.

Proposition 11.1 (GARCH forecast)**.** *Let* $\bar{h} = \omega/(1 - \alpha - \beta)$. *Then*

$$\mathbb{E}_t[h_{t+k}] = \bar{h} + (\alpha + \beta)^{k-1}(\mathbb{E}_t[h_{t+1}] - \bar{h}).$$

Proof. Take conditional expectations of the recursion and use $\mathbb{E}_t[\varepsilon_{t+j}^2] = \mathbb{E}_t[h_{t+j}]$. Iterate the affine difference equation. □

When $\alpha + \beta$ is near one, volatility shocks decay slowly. Gaussian quasi-maximum likelihood can consistently estimate variance dynamics under certain non-Gaussian innovations, but robust standard errors are required.

11.2 Asymmetry and stochastic volatility

Equity volatility often rises more after negative returns than positive returns. GJR-GARCH adds $\gamma \mathbf{1}_{\{\varepsilon_{t-1}<0\}}\varepsilon_{t-1}^2$. EGARCH models log variance and removes positivity constraints.

A stochastic-volatility model is

$$r_t = \mu + \mathrm{e}^{h_t/2} z_t, \qquad h_t = \omega + \phi(h_{t-1} - \omega) + \sigma_h \eta_t,$$

possibly with $\mathrm{Corr}(z_t, \eta_t) = \rho$ for leverage. Here volatility has its own innovation and requires filtering or simulation-based inference.

11.3 Realized variation

For log price semimartingale $\mathrm{d}p_t = \mu_t \,\mathrm{d}t + \sigma_t \,\mathrm{d}W_t + \mathrm{d}J_t$, increasingly fine intraday returns satisfy

$$RV_t = \sum_{j=1}^{m} r_{t,j}^2 \longrightarrow \int_{t-1}^{t} \sigma_s^2 \,\mathrm{d}s + \sum_{t-1<s\le t} (\Delta p_s)^2.$$

At ultra-high frequency, bid–ask bounce, discreteness, and asynchronous timestamps create bias. Sampling, subsampling, realized kernels, and pre-averaging address different noise structures.

11.4 Implied volatility

Implied volatility is the value $\widehat{\sigma}$ that equates an option pricing formula to a market quote. It is a transformed price under a model, not a direct estimate of future physical volatility. Across strikes and maturities it forms a surface. Absence of static arbitrage imposes monotonicity and convexity in strike and calendar consistency.

The variance risk premium is often measured as a risk-neutral expected variance proxy minus subsequent realized variance. Interpretation depends on horizon, jump terms, discretization, and the physical forecast used.

11.5 Multivariate volatility

The exponentially weighted covariance recursion

$$\Sigma_t = \lambda \Sigma_{t-1} + (1-\lambda) r_{t-1} r_{t-1}^{\top}$$

is simple and remains positive semidefinite. It imposes one decay parameter.

Dynamic conditional correlation separates univariate conditional standard deviations D_t from correlation: $H_t = D_t R_t D_t$. Factor stochastic volatility reduces dimension. Any covariance forecast should be evaluated through portfolio-relevant losses, not only elementwise fit.

11.6 Extreme-value theory

For a high threshold u, peaks-over-threshold theory approximates excesses $Y = X - u \mid X > u$ by a generalized Pareto distribution

$$G_{\xi,\beta}(y) = 1 - \left(1 + \xi \frac{y}{\beta}\right)^{-1/\xi},$$

on the support $1 + \xi y/\beta > 0$. Shape $\xi > 0$ indicates heavy polynomial tails; $\xi = 0$ is the exponential limit.

If threshold exceedance probability is ζ_u, the high quantile estimate is

$$q_p = u + \frac{\beta}{\xi}\left[\left(\frac{1-p}{\zeta_u}\right)^{-\xi} - 1\right].$$

Threshold choice trades bias against variance. Dependence requires declustering or an extremal-index adjustment. Extrapolation far beyond the observed tail is mathematically possible and statistically fragile.

11.7 Copulas and stress dependence

For a Student copula with correlation ρ and degrees of freedom ν, symmetric tail dependence is

$$\lambda = 2t_{\nu+1}\left(-\sqrt{\frac{(\nu+1)(1-\rho)}{1+\rho}}\right).$$

Even a flexible copula assumes that fitted dependence persists into the stress being modeled. Scenario analysis should therefore supplement distributional estimation.

11.8 Backtesting risk forecasts

For VaR level α, exception indicator $I_t = \mathbf{1}_{\{L_t > \widehat{\mathrm{VaR}}_{\alpha,t}\}}$ should have conditional mean $1-\alpha$ under a correct model. Unconditional coverage checks frequency; independence tests clustering; conditional coverage combines them. Expected Shortfall backtests require joint identification with VaR or suitable scoring functions.

Model-risk warning 11.2. A model can pass a low-power backtest and still be dangerous. Passing means the sample did not reject a specified property, not that the full tail law is correct.

11.9 Problems

Exercise 11.3 (Core: GARCH half-life)**.** Show that the half-life of a variance shock is $\log(1/2)/\log(\alpha+\beta)$. Compute it for persistence 0.98.

Exercise 11.4 (Core: realized variance)**.** Explain why sampling more frequently can increase rather than decrease error in the presence of bid–ask bounce.

Exercise 11.5 (Advanced: EVT)**.** Derive Expected Shortfall for the generalized Pareto tail when $\xi < 1$. Identify the condition for a finite variance.

Chapter 12

Causal Inference, Panels, and Machine Learning

Learning objectives. Define causal estimands with potential outcomes, derive major identification strategies, use panel data correctly, and integrate machine learning without confusing predictive fit with causal evidence.

12.1 Potential outcomes

Let $Y_i(1)$ and $Y_i(0)$ denote outcomes under treatment and control. Observed outcome is

$$Y_i = D_iY_i(1) + (1 - D_i)Y_i(0).$$

The average treatment effect is $\tau = \mathbb{E}[Y(1) - Y(0)]$; the average effect on treated units conditions on $D = 1$. Only one potential outcome is observed for each unit, the fundamental missing-data problem.

Assumption 12.1 (Conditional ignorability and overlap)**.**

$$(Y(1), Y(0)) \perp D \mid X, \qquad 0 < e(X) = \mathbb{P}(D = 1 \mid X) < 1.$$

Then

$$\tau = \mathbb{E}[\mathbb{E}(Y \mid D = 1, X) - \mathbb{E}(Y \mid D = 0, X)]$$

and also admits inverse-propensity weighting.

Proposition 12.2 (Doubly robust moment)**.** *Let* $m_d(X) = \mathbb{E}[Y \mid D =$

$d, X]$. The score

$$\psi = m_1(X) - m_0(X) + \frac{D(Y - m_1(X))}{e(X)} - \frac{(1-D)(Y - m_0(X))}{1 - e(X)}$$

has expectation τ if either the outcome models are correct or the propensity model is correct.

Overlap is substantive. Extreme propensity weights signal that the data contain little comparable counterfactual information.

12.2 Difference in differences

With treated and control groups before and after an intervention, the estimator is

$$\widehat{\tau}_{\text{DiD}} = (\bar{Y}_{T,\text{post}} - \bar{Y}_{T,\text{pre}}) - (\bar{Y}_{C,\text{post}} - \bar{Y}_{C,\text{pre}}).$$

Identification requires parallel untreated trends, not merely similar pre-period levels. Event studies estimate dynamic leads and lags, but pretrend tests may have low power and treatment-effect heterogeneity complicates two-way fixed effects.

12.3 Regression discontinuity

If treatment changes discontinuously at running-variable cutoff c while untreated potential outcomes vary smoothly,

$$\tau_{\text{RD}} = \lim_{x \downarrow c} \mathbb{E}[Y \mid X = x] - \lim_{x \uparrow c} \mathbb{E}[Y \mid X = x].$$

Local polynomials, bandwidth sensitivity, density checks, and covariate balance are essential. The effect is local to units near the cutoff.

12.4 Panel data

The fixed-effects model

$$y_{it} = \alpha_i + \lambda_t + x_{it}^\mathsf{T}\beta + u_{it}$$

uses within-unit variation while absorbing time-invariant unit heterogeneity and common date shocks. It does not absorb time-varying confounders. Standard errors should cluster at the level of treatment assignment or dependence; two-way clustering may be necessary.

Dynamic panels with lagged outcomes and short time dimension suffer Nickell bias. Difference and system GMM use internal instruments, but instrument proliferation can overfit endogenous variables and weaken specification tests.

12.5 Prediction with regularization

Ridge regression minimizes

$$\frac{1}{n}\sum_i (y_i - x_i^\mathsf{T}\beta)^2 + \lambda \left\|\beta\right\|_2^2,$$

while lasso uses $\lambda \left\|\beta\right\|_1$. Lasso can set coefficients exactly to zero. Elastic net combines both. Standardization must be estimated within each training fold.

For a tree, recursive partitions approximate nonlinear interactions. Random forests average decorrelated trees; gradient boosting fits weak learners sequentially to loss gradients. These methods estimate predictive functions, not automatically structural mechanisms.

12.6 Cross-validation for dependent data

Random folds leak future regimes into the past. Financial validation should respect time. Walk-forward validation trains before validation. When labels overlap, purge training observations whose outcome windows intersect validation, and use an embargo around fold boundaries if feature construction transmits nearby information.

Hyperparameters are part of the fitted model. The final test set must be untouched by feature choice, tuning, threshold selection, and model comparison.

12.7 Double machine learning

For partially linear model

$$Y = \theta D + g(X) + U, \qquad D = m(X) + V,$$

an orthogonal moment is

$$\mathbb{E}[(D - m(X))(Y - g(X) - \theta(D - m(X)))] = 0.$$

Cross-fitting estimates nuisance functions on other folds. Orthogonality makes the target less sensitive, to first order, to nuisance estimation errors under rate and regularity conditions.

12.8 Model interpretation and uncertainty

Permutation importance can be distorted by correlated predictors. Partial dependence averages over possibly unrealistic feature combinations. Shapley-based explanations allocate a model's prediction under a specified background distribution; they are not causal effects.

Uncertainty includes sampling variation, tuning, feature discovery, nonstationarity, and decision costs. A single bootstrap around a fixed final model captures only part of this stack. Nested resampling or a transparent separation of confirmatory and exploratory results is preferable.

12.9 Problems

Exercise 12.3 (Core: IPW)**.** Prove identification of the ATE by inverse-propensity weighting under ignorability and overlap.

Exercise 12.4 (Core: fixed effects)**.** Derive the within transformation and explain why a time-invariant regressor cannot be separately estimated with unit fixed effects.

Exercise 12.5 (Advanced: orthogonality)**.** Differentiate the double-machine-learning moment with respect to perturbations of m and g at their true values and verify Neyman orthogonality.

Exercise 12.6 (Research)**.** Design a study of the effect of an index inclusion on liquidity. Specify the estimand, treatment timing, plausible control group, standard errors, and threats from anticipation.

Part IV

Markets, Instruments, and Institutions

Chapter 13

Securities, Trading, and Market Conventions

Learning objectives. Understand the contractual cash flows of major instruments, translate market quotes into economic exposures, and recognize how settlement, collateral, and conventions enter valuation.

13.1 The balance-sheet view

Every security is simultaneously an asset to one party and a liability or residual claim of another. A useful first analysis lists:

- contractual cash flows and their priority;
- states in which cash flows change;
- collateral, margin, and close-out rules;
- currency, settlement date, day count, and calendar;
- liquidity, financing, and counterparty exposure.

Prices are not cash flows. A model maps uncertain future cash flows and discounting to a price under explicit assumptions.

13.2 Equity

Common equity is a residual claim after contractual obligations. Corporate actions include dividends, splits, repurchases, rights issues, mergers, and spin-offs. Raw closing prices across these events do not produce total returns. If P_t is ex-dividend price and D_t cash distribution,

$$R_t = \frac{P_t + D_t}{P_{t-1}} - 1.$$

Adjusted-price vendors encode corporate actions with methodology that must be documented.

Market capitalization weights are proportional to float-adjusted value. Equal weighting, fundamental weighting, and price weighting create distinct rebalancing and factor exposures. Index membership is governed by rules and changes over time; using today's constituents historically creates survivorship bias.

13.3 Money-market instruments

Treasury bills, commercial paper, certificates of deposit, repo, and overnight indexed exposures use different quotation bases. A discount yield can be based on face value and a 360-day year; an investment yield uses purchase price. Comparing numbers without converting conventions creates false arbitrage.

In a repo, one party sells collateral and agrees to repurchase it. The repo rate is the financing rate; haircut protects the cash lender from collateral-value changes. Special collateral can finance below the general collateral rate because possession of a particular security has value.

13.4 Bonds

A fixed-rate bond pays coupons and principal subject to credit and contractual terms. Clean price excludes accrued interest; dirty price is the settlement amount:

$$P_{\text{dirty}} = P_{\text{clean}} + \text{accrued interest}.$$

Yield to maturity is the single internal rate equating discounted promised cash flows to price. It is not an expected return unless coupons reinvest at that yield, the bond is held to maturity, and promised cash flows occur.

Floating-rate notes reset coupons to a reference plus spread. Inflation-linked bonds index principal or coupons with publication lags and floors. Callable bonds embed an option held by the issuer; putable bonds embed one held by the investor.

13.5 Forwards and futures

A long forward with delivery price K has payoff $S_T - K$. With no income and deterministic rate, no arbitrage gives $K = S_0 e^{rT}$. For an asset with known continuous yield q,

$$F_{0,T} = S_0 e^{(r-q)T}.$$

Storage cost and convenience yield modify commodity carry.

Futures are marked to market daily. When rates are stochastic and correlated with the underlying, futures and forward prices can differ because gains and losses are realized at different times.

13.6 Swaps

An interest-rate swap exchanges fixed and floating cash flows on a notional. At inception, the par fixed rate makes value zero:

$$K_{\text{swap}} = \frac{P(0,T_0) - P(0,T_n)}{\sum_{i=1}^{n} \delta_i P(0,T_i)}$$

under a single-curve abstraction. Modern collateralized valuation often projects floating cash flows on a tenor curve and discounts on an overnight curve.

Currency swaps exchange principal and interest in different currencies. Total-return swaps transfer the price and income performance of an asset against financing. Contract details determine funding, collateral, and counterparty risk.

13.7 Options

A European call pays $(S_T - K)^+$ and a put $(K - S_T)^+$. American exercise is permitted before maturity; path-dependent options use the history of the underlying. At fixed maturity, call value is decreasing and convex in strike under no static arbitrage.

Proposition 13.1 (Put–call parity)**.** *For European options with the same K, T on a non-dividend-paying asset,*

$$C_0 - P_0 = S_0 - KP(0, T).$$

Proof. Both portfolios—long call plus the present value of K, and long put plus the asset—pay $\max(S_T, K)$ at T. No arbitrage equates their values. □

With known dividends, replace spot by prepaid forward value. Violations after bid, ask, funding, settlement, shorting, and exercise details are needed before claiming an executable arbitrage.

13.8 Problems

Exercise 13.2 (Core: carry)**.** Derive equity-index forward price with discrete known dividends and with a continuous dividend yield. State the assumptions.

Exercise 13.3 (Core: swap rate)**.** Replicate a floating-rate note to derive the par swap rate formula.

Exercise 13.4 (Advanced: futures convexity)**.** Use a two-period tree with stochastic rates to construct a case in which forward and futures prices differ.

Chapter 14

Market Microstructure and Execution

Learning objectives. Model the limit order book, spreads, information, impact, and execution; translate a statistical signal into a feasible trading rule with capacity and timestamp discipline.

14.1 Orders and the limit order book

A market order demands immediate execution at available quotes. A limit order sets a worst acceptable price and supplies liquidity while exposed to non-execution and adverse selection. The best bid b_t and ask a_t define quoted midpoint $m_t = (a_t + b_t)/2$ and spread $s_t = a_t - b_t$.

The transaction price can be written

$$p_t = m_t + q_t \frac{s_t}{2} + \eta_t,$$

where $q_t = +1$ for buyer initiation and -1 for seller initiation, with η_t capturing price improvement and classification error. Alternation between bid and ask creates negative first-order autocorrelation even when the efficient price is a martingale.

14.2 Why spreads exist

Spread compensates order-processing cost, inventory risk, and adverse selection. An informed trader preferentially buys before price rises and sells before it falls, so passive quotes lose conditionally to informed flow. Dealers widen or adjust quotes as the probability of information increases.

Effective spread for a buy at price p_t relative to pre-trade midpoint m_t is $2(p_t - m_t)$. Realized spread compares execution with a later midpoint and attempts to subtract information-driven price movement. Price impact is the complementary midpoint response.

14.3 Order flow and price discovery

Signed order flow can predict short-horizon price changes because trades reveal information and mechanically consume depth. Long memory in trade signs need not imply easily exploitable return predictability: impact can be transient or state-dependent and execution costs absorb it.

Hasbrouck-style vector autoregressions, Kyle's linear model, and structural microstructure models quantify different notions of information and impact. They rely on timestamp alignment and trade classification that vary across venues.

Worked example 14.1 (Kyle impact)**.** In a one-period Gaussian model, an informed trader submits $x = \beta(v - p_0)$ and noise traders submit u. A competitive market maker sets $p = \mathbb{E}[v \mid x + u] = p_0 + \lambda(x + u)$. Linear equilibrium chooses β and λ so the informed trader optimizes profit and the market maker breaks even. Impact λ rises with fundamental uncertainty and falls with noise-trading scale.

14.4 Transaction-cost models

A simple cost for trade Δw is

$$C(\Delta w) = \underbrace{\sum_i c_i \left|\Delta w_i\right|}_{\text{spread and fees}} + \underbrace{\sum_i k_i \left|\Delta w_i\right|^{3/2}}_{\text{nonlinear impact}}.$$

The square-root impact rule is an empirical approximation, not universal. Participation rate, volatility, volume, urgency, venue, and order type matter.

Portfolio turnover can be defined as $\frac{1}{2}\sum_i \left|w_{i,t}^{\text{target}} - w_{i,t}^{\text{pretrade}}\right|$. State which convention is used. Costs must apply to dollar trades after accounting for drift in pre-trade weights.

14.5 Optimal execution

In the Almgren–Chriss framework, a trader liquidates inventory $x(t)$ by balancing expected temporary impact against variance from holding risk. A continuous objective is

$$\min_{x(0)=X,\ x(T)=0} \int_0^T \left(\eta\dot{x}(t)^2 + \lambda\sigma^2 x(t)^2\right) \mathrm{d}t.$$

The Euler–Lagrange equation is $\eta\ddot{x} = \lambda\sigma^2 x$, producing a hyperbolic-sine schedule. Larger risk aversion or volatility accelerates trading; greater temporary impact slows it.

Real execution adds stochastic liquidity, price limits, queues, hidden orders, fill uncertainty, multiple venues, and feedback between child orders and signals.

14.6 Timestamps and backtests

A research row should distinguish:

- event time when the economic event occurs;
- publication time when a vendor makes data available;
- ingestion time when the strategy receives it;
- decision time when the signal is frozen;
- execution window and settlement.

Trading at the close with a signal computed from that same official close is usually impossible unless an executable auction mechanism and submission cutoff are modeled. Daily bar data hide the path and queue within the bar.

14.7 Capacity and crowding

Capacity is the capital at which expected marginal alpha equals marginal costs and constraints. Scaling leverage leaves returns unchanged only in a frictionless linear model. Impact, borrow availability, margin, concentration, and correlated liquidation make strategy P&L nonlinear in capital.

Crowding matters when many strategies share factors, signals, or risk controls. During deleveraging, normally diverse positions can become correlated through common flows. Stress tests should include spread widening, volume collapse, borrow recall, and delayed execution.

14.8 Market integrity

Research must distinguish legitimate liquidity provision from manipulative intent. Spoofing, layering, wash trading, misuse of confidential order information, and trading on material nonpublic information raise legal and ethical issues beyond model performance. Market-data licenses can restrict storage, redistribution, and publication.

14.9 Problems

Exercise 14.2 (Core: bid–ask bounce)**.** Assume a constant efficient price and independent trade signs. Compute the first-order autocovariance of transaction-price changes.

Exercise 14.3 (Core: optimal execution)**.** Solve the Euler–Lagrange equation for the stated boundary conditions and examine the risk-neutral limit $\lambda \downarrow 0$.

Exercise 14.4 (Research)**.** Design a transaction-cost calibration using order or bar data. Specify target, features, sample split, weighting, and how you would handle endogenous order size.

Chapter 15

Arbitrage, Collateral, and Financial Engineering Practice

Learning objectives. Use replication and dominance arguments, understand collateralized valuation, recognize counterparty and funding adjustments, and establish a professional model governance process.

15.1 Law of one price and arbitrage

If two portfolios deliver identical cash flows in every state and have identical institutional treatment, they must have the same price in an ideal market. An arbitrage is a self-financing strategy with zero or negative initial cost, nonnegative terminal value almost surely, and positive value with positive probability.

The phrase identical institutional treatment matters. Assets can differ in liquidity, financing eligibility, taxes, capital charges, haircuts, legal netting, or delivery optionality. A price basis can compensate these differences without being a free lunch.

15.2 State prices and complete markets

In a finite-state one-period economy, let payoff matrix $X \in \mathbb{R}^{S\times N}$ and price vector p. A positive state-price vector $q > 0$ prices assets when $p = X^{\mathsf{T}}q$. Absence of arbitrage is equivalent to existence of such a strictly positive linear pricing functional under standard finite-dimensional conditions.

The market is complete if $\operatorname{rank}(X) = S$: every state-contingent payoff is replicable and state prices are unique. In incomplete markets, arbitrage-free prices of an unspanned claim form an interval or depend on an added criterion.

Theorem 15.1 (Finite-state fundamental theorem). *In a finite one-period market, no arbitrage holds if and only if there exists $q \in \mathbb{R}^S$ with $q_s > 0$ and $p = X^\mathsf{T} q$.*

Proof sketch. If positive state prices exist, any nonnegative nonzero payoff has positive cost, so no arbitrage exists. Conversely, separation of the attainable-payoff cone from a strictly positive payoff produces a positive linear functional agreeing with market prices. □

15.3 Collateral and discounting

A collateral agreement specifies eligible assets, currency, threshold, minimum transfer amount, frequency, haircut, and remuneration. If variation margin is continuous, cash, and remunerated at overnight rate c_t, a stylized collateralized payoff is discounted at c:

$$V_t = \mathbb{E}_t^{\mathbb{Q}}\left[\mathrm{e}^{-\int_t^T c_s \,\mathrm{d}s} H_T\right].$$

Changing collateral currency can change value through funding and cross-currency basis. Optional collateral currency is itself an option.

15.4 Counterparty credit and valuation adjustments

Positive future exposure creates loss if the counterparty defaults. Expected positive exposure at t is $\mathbb{E}[V_t^+]$; potential future exposure is a quantile. A stylized unilateral CVA is

$$\mathrm{CVA} \approx (1-R)\int_0^T \mathbb{E}^{\mathbb{Q}}[D(0,t)V_t^+ \mid \tau = t] \,\mathrm{d}\mathbb{P}^{\mathbb{Q}}(\tau \le t).$$

Wrong-way risk occurs when exposure rises as counterparty credit quality deteriorates.

Modern valuation adjustment language also includes DVA for own default, FVA for funding, MVA for initial margin, and KVA for capital. Summing independently computed adjustments can double-count effects; the institution needs a consistent replication and accounting framework.

15.5 Hedging as a control problem

A model price is only one output. A desk holds a hedging policy under discrete trading, transaction costs, jumps, liquidity limits, and parameter uncertainty. Hedging P&L decomposes into model theta, realized gamma, volatility repricing, cross-Greeks, carry, funding, and residual.

For a delta-hedged option over a short interval under continuous diffusion,

$$\Delta\Pi \approx \frac{1}{2}\Gamma S^2(\sigma_{\mathrm{real}}^2 - \sigma_{\mathrm{imp}}^2)\Delta t$$

under a simplified convention. The formula motivates variance trading but omits jumps, surface dynamics, costs, and discrete hedge error.

15.6 Model lifecycle

A controlled model process contains:

1. purpose, owner, users, and prohibited uses;
2. mathematical specification and assumptions;
3. data lineage and implementation tests;
4. calibration and benchmark comparison;
5. sensitivity, stress, and limitation analysis;
6. independent validation and approval;
7. monitoring thresholds and change control;
8. retirement or remediation criteria.

Verification asks whether code correctly implements the specification. Validation asks whether the specification is adequate for its use. A correct implementation of an unsuitable model is still model risk.

15.7 Reproducibility and auditability

Every reported number should be traceable to immutable raw data or an identified vendor query, code version, configuration, dependency environment, and generation timestamp. Random seeds support debugging but do not create statistical validity. Figures should expose units, sampling frequency, and whether returns include distributions.

> **Research note.** A strong quant portfolio is not a gallery of final Sharpe ratios. It is a chain of questions, mathematical arguments, failed specifications, robustness checks, and reproducible decisions.

15.8 Problems

Exercise 15.2 (Core: state prices)**.** For two states and a bond plus stock, solve state prices and price a call. Determine when the market is complete.

Exercise 15.3 (Core: CVA)**.** Explain the difference between exposure under the physical and pricing measures. Which belongs in pricing and which in risk-management scenario design?

Exercise 15.4 (Advanced: governance)**.** Write a validation plan for an illiquid-option model. Include benchmarks, data limitations, hedging tests, uncertainty, and usage restrictions.

Part V

Asset Pricing

Chapter 16

Stochastic Discount Factors and Equilibrium Pricing

Learning objectives. Unify asset pricing through stochastic discount factors, derive risk premia and no-arbitrage bounds, and connect equilibrium preferences to empirical moments.

16.1 The fundamental pricing equation

For payoff X_{t+1} with price P_t, absence of arbitrage under suitable conditions implies a positive stochastic discount factor M_{t+1} such that

$$P_t = \mathbb{E}_t[M_{t+1}X_{t+1}].$$

For gross return $R_{i,t+1} = X_{i,t+1}/P_{i,t}$,

$$1 = \mathbb{E}_t[M_{t+1}R_{i,t+1}].$$

This equation encompasses consumption-based equilibrium, state-price valuation, and equivalent martingale measures.

The risk-free rate obeys $R_{f,t} = 1/\mathbb{E}_t[M]$. Therefore

$$\mathbb{E}_t[R_i] - R_f = -\frac{\mathrm{Cov}_t(M, R_i)}{\mathbb{E}_t[M]}.$$

Assets that pay when M is high hedge bad states and require lower average returns.

Theorem 16.1 (Hansen–Jagannathan bound)**.** *For any excess return R_i^e*

priced by M,

$$\frac{\sigma_t(M)}{\mathbb{E}_t[M]} \geq \frac{|\mathbb{E}_t[R_i^e]|}{\sigma_t(R_i^e)}.$$

For a vector of returns, the strongest bound is the maximum conditional Sharpe ratio.

Proof. Pricing gives $\mathbb{E}[MR_i^e] = 0$, so $\mathbb{E}[R_i^e] = -\operatorname{Cov}(M, R_i^e)/\mathbb{E}[M]$. Apply Cauchy–Schwarz and divide by $\sigma(R_i^e)$. □

The bound is diagnostic: a candidate consumption-based M must be sufficiently volatile to price observed Sharpe ratios. It does not identify the true SDF.

16.2 Projection and the minimum-variance SDF

Given gross returns R, any SDF can be decomposed into its projection on their span plus an orthogonal component. The minimum-second-moment SDF in the payoff span is

$$M^\star = a + b^\mathsf{T} R,$$

with coefficients satisfying pricing equations. Unspanned components do not affect prices of included assets but can matter for other claims.

In incomplete markets, many SDFs price the observed assets. Empirical asset pricing therefore tests a joint hypothesis about the factor representation, conditioning information, and set of test assets.

16.3 Representative-agent equilibrium

An agent maximizing $\mathbb{E}_t \sum_j \beta^j u(C_{t+j})$ has

$$M_{t+1} = \beta \frac{u'(C_{t+1})}{u'(C_t)}.$$

For lognormal consumption growth $g_{t+1} = \Delta \log C_{t+1}$ and CRRA utility,

$$\log M_{t+1} = \log \beta - \gamma g_{t+1}.$$

If (g, r_i) are conditionally jointly normal, expected log excess return includes consumption covariance and a Jensen term. Smooth aggregate consumption growth makes the observed equity premium and low risk-free rate difficult to match with moderate risk aversion, the equity-premium and risk-free-rate puzzles.

Extensions introduce habit, recursive utility, long-run risks, rare disasters, heterogeneous agents, incomplete markets, or intermediary constraints. Each adds state variables and identification demands; better fit alone does not establish the mechanism.

16.4 Conditional asset pricing

If instruments Z_t are known at t, conditional pricing implies unconditional moments

$$\mathbb{E}[(M_{t+1}R_{t+1} - \mathbf{1}) \otimes Z_t] = 0.$$

Generalized method of moments estimates parameters by minimizing $g_T(\theta)^\mathsf{T} W_T g_T(\theta)$. Efficient weighting uses the inverse long-run covariance of moments, but two-step estimation and weak identification affect finite-sample inference.

The overidentifying-restrictions statistic tests whether all sample moments can be close to zero. Rejection can reflect the SDF, return data, instruments, or maintained regularity. Non-rejection does not prove economic truth.

16.5 Entropy and good-deal restrictions

No-arbitrage price intervals can be wide in incomplete markets. Good-deal bounds restrict SDF volatility or admissible Sharpe ratios to exclude implausibly attractive unspanned opportunities. Entropy methods choose a pricing measure closest to a reference law subject to moment conditions:

$$\min_{\mathbb{Q}} \mathbb{E}^{\mathbb{P}} \left[\frac{d\mathbb{Q}}{d\mathbb{P}} \log \frac{d\mathbb{Q}}{d\mathbb{P}} \right].$$

This is an additional economic or statistical criterion, not a consequence of no-arbitrage.

16.6 Problems

Exercise 16.2 (Core: covariance pricing)**.** Derive expected excess return from the SDF equation and interpret the sign for an insurance-like payoff.

Exercise 16.3 (Core: HJ bound)**.** Derive the vector maximum-Sharpe form using the tangency portfolio.

Exercise 16.4 (Advanced: GMM)**.** Write moments for a consumption-based model with CRRA utility, derive the Jacobian, and describe HAC weighting with quarterly consumption and returns.

Chapter 17

CAPM, Multifactor Models, and Anomalies

Learning objectives. Derive the CAPM, understand beta and factor-mimicking portfolios, evaluate multifactor models cross-sectionally, and conduct anomaly research without common biases.

17.1 Mean–variance equilibrium

Suppose investors have homogeneous beliefs over one-period means and covariance, can borrow and lend at R_f, and choose portfolios by mean and variance. Every investor holds a combination of the risk-free asset and the same tangency portfolio. Market clearing makes that portfolio the market.

Theorem 17.1 (CAPM). *Under the mean–variance equilibrium assumptions,*

$$\mathbb{E}[R_i] - R_f = \beta_i(\mathbb{E}[R_M] - R_f), \qquad \beta_i = \frac{\operatorname{Cov}(R_i, R_M)}{\operatorname{Var}(R_M)}.$$

Proof. The tangency portfolio first-order condition gives $\mu - R_f\mathbf{1} = \lambda\Sigma w_M$. Component i is $\mu_i - R_f = \lambda \operatorname{Cov}(R_i, R_M)$. Multiply by w_M^T and sum to obtain $\mu_M - R_f = \lambda \operatorname{Var}(R_M)$; divide. □

The security market line is a statement about expected return and systematic covariance, not total volatility. The zero-beta CAPM adapts the result when risk-free borrowing is unavailable.

17.2 Time-series beta estimation

The market model

$$R^e_{i,t} = \alpha_i + \beta_i R^e_{M,t} + \varepsilon_{i,t}$$

estimates beta over a chosen window. Nonsynchronous trading, leverage change, regime-dependent covariance, errors in the market proxy, and conditional betas complicate interpretation. A statistically nonzero α is a joint rejection of model, benchmark, implementation, and sampling assumptions.

17.3 Arbitrage pricing theory

If returns have approximate factor structure $R_i = a_i + b_i^\mathsf{T} f + \varepsilon_i$ and idiosyncratic risk is diversifiable, absence of arbitrage implies expected returns are approximately linear in factor loadings:

$$\mathbb{E}[R_i] = \lambda_0 + b_i^\mathsf{T} \lambda.$$

APT needs fewer preference restrictions than CAPM but does not name the factors. Traded factors make λ directly tied to factor expected returns; nontraded macroeconomic factors need beta-pricing estimation.

17.4 Empirical factor models

Common factor families target size, value, profitability, investment, momentum, quality, low risk, carry, and liquidity. A characteristic is an asset attribute used to sort; a factor is a return. The distinction matters when a characteristic predicts returns without representing covariance risk.

Portfolio construction choices include:

- universe and delisting returns;
- breakpoint population and exchange;
- weighting and rebalancing;
- accounting-data publication lags;

- microcap and liquidity filters;
- neutralization and transaction costs.

Published factor data are excellent benchmarks but do not replace reconstruction when the research question concerns implementability.

17.5 Cross-sectional tests

The Fama–MacBeth procedure estimates cross-sectional regressions each period:

$$R_{i,t} = a_t + \beta_i^\top \lambda_t + u_{i,t},$$

then averages $\widehat{\lambda}_t$. Time-series standard errors of the premia must handle autocorrelation. Estimated betas introduce errors-in-variables; Shanken-type corrections address one component under strong assumptions.

Testing factors on portfolios sorted by the same characteristic can make power and interpretation circular. Include diverse test assets, report pricing errors in economic units, and compare against nested and nonnested benchmarks.

17.6 Anomalies and the research process

An anomaly may reflect compensation for risk, behavioral mispricing, institutional friction, data mining, or implementation error. A credible study:

1. states the economic mechanism before examining the test sample;
2. freezes feature definitions and availability dates;
3. accounts for delistings, survivorship, and restatements;
4. uses an out-of-sample or independent-market test;
5. reports turnover, spread, impact, borrow, and capacity;
6. compares to simple factor exposures;
7. publishes failures and sensitivity.

The probability of an extreme backtest rises with the number and dependence of tried strategies. Deflated Sharpe ratios and reality-check procedures attempt selection adjustments but require an honest count or model of the search process.

17.7 Problems

Exercise 17.2 (Core: CAPM)**.** Derive the zero-beta version when there is no risk-free asset but all investors share the same efficient frontier.

Exercise 17.3 (Core: factor neutrality)**.** Given loading matrix B, formulate the closest portfolio to target w_0 subject to $B^\top w = 0$ and full investment.

Exercise 17.4 (Advanced: two-pass bias)**.** Explain how noisy first-pass beta estimates bias the second-pass risk-premium regression. Propose portfolios or instruments that improve precision.

Chapter 18

Return Predictability and Present-Value Relations

Learning objectives. Derive present-value decompositions, connect valuation ratios to cash-flow and discount-rate news, and evaluate predictive regressions with appropriate skepticism.

18.1 Log-linear present value

Let p_t and d_t be log price and dividend. Log return $r_{t+1} = \log(P_{t+1} + D_{t+1}) - \log P_t$ can be approximated around the average dividend-price ratio:

$$r_{t+1} \approx k + \rho p_{t+1} + (1-\rho)d_{t+1} - p_t.$$

Rearranging and iterating under a terminal condition yields

$$d_t - p_t \approx -\frac{k}{1-\rho} + \sum_{j=0}^{\infty} \rho^j \left(\mathbb{E}_t\, r_{t+1+j} - \mathbb{E}_t\, \Delta d_{t+1+j}\right).$$

A high dividend-price ratio must forecast high future returns, low future dividend growth, or both within the approximation.

18.2 News decomposition

Unexpected return can be decomposed into cash-flow news and discount-rate news. Prices can fall on good cash-flow news if discount rates rise more. Interpreting a price response without separating the two is incomplete.

VAR-based decompositions estimate revisions to long-horizon expectations, so results depend on included state variables, persistence, sample, and terminal assumptions.

18.3 Predictive regressions

A standard regression is

$$r_{t+1} = \alpha + \beta x_t + \varepsilon_{t+1}.$$

When x_t is persistent and its innovation correlates with ε_{t+1}, finite-sample bias and nonstandard behavior arise. Overlapping long-horizon returns create moving-average residuals and mechanically smooth noise.

Out-of-sample R^2 relative to historical mean is

$$R^2_{\mathrm{OS}} = 1 - \frac{\sum_t (r_t - \widehat{r}_t)^2}{\sum_t (r_t - \bar{r}_{t-1})^2}.$$

Small positive values can be economically meaningful for volatile returns, but statistical comparison must reflect real-time estimation and model selection.

18.4 Economic value

A mean–variance investor might allocate

$$w_t = \frac{\widehat{\mu}_t}{\gamma \widehat{\sigma}_t^2},$$

subject to leverage and turnover constraints. Certainty-equivalent improvement evaluates forecasts through a decision. This exercise is sensitive to scaling, variance forecast, parameter caps, and costs.

Forecast combinations reduce model-selection risk. Simple equal weighting is a strong benchmark; optimal combination weights are themselves estimated and can be unstable.

18.5 Behavioral and institutional mechanisms

Underreaction can arise from slow information diffusion or anchoring; overreaction from extrapolation or constrained arbitrage. Limits to arbitrage include short-sale risk, funding, horizon mismatch, and noise-trader risk. A behavioral narrative must make distinct predictions about horizons, investor types, events, or limits; otherwise it is an after-the-fact label.

18.6 Problems

Exercise 18.1 (Core: present value)**.** Complete the iteration of the log-linear return identity and state the no-bubble terminal condition.

Exercise 18.2 (Core: overlapping returns)**.** Show that a k-period overlapping return induces an MA$(k-1)$ structure even when one-period returns are iid.

Exercise 18.3 (Research)**.** Compare valuation-ratio forecasts against the historical mean using recursive estimation. Report statistical loss, certainty equivalent, turnover, and performance across predeclared subperiods.

Part VI

Derivatives and Stochastic Models

Chapter 19

Discrete-Time Derivative Pricing

Learning objectives. Price derivatives by replication in one- and multi-period trees, understand completeness and early exercise, and derive convergence toward continuous-time models.

19.1 One-period binomial economy

Let a stock cost S_0 and move to $S_u = uS_0$ or $S_d = dS_0$, with $d < R < u$ where R is the risk-free gross return. A claim pays H_u, H_d. Choose Δ shares and cash B so

$$\Delta S_u + RB = H_u, \qquad \Delta S_d + RB = H_d.$$

Solving,

$$\Delta = \frac{H_u - H_d}{S_0(u-d)}, \qquad B = \frac{uH_d - dH_u}{R(u-d)}.$$

The no-arbitrage value is $V_0 = \Delta S_0 + B$.

Define risk-neutral probability

$$q = \frac{R-d}{u-d} \in (0,1).$$

Then

$$V_0 = \frac{1}{R}[qH_u + (1-q)H_d].$$

The physical up probability never entered replication. It matters for risk and forecasting, not the unique price in this complete tree.

Theorem 19.1 (No-arbitrage condition)**.** *The one-period stock–bond bino-*

mial market is arbitrage free if and only if $d < R < u$. It is complete when $u \neq d$.

Proof. If $R \leq d$, borrowing to buy stock has nonnegative payoff in both states and a strict gain in at least one; similarly $R \geq u$ supports the opposite trade. If $d < R < u$, $q \in (0,1)$ gives strictly positive state prices, excluding arbitrage. The two linearly independent securities span the two states. □

19.2 Multi-period trees

At each node, value satisfies backward recursion

$$V_t = \frac{1}{R}\left(qV_{t+1}^u + (1-q)V_{t+1}^d\right).$$

The replicating delta is recomputed at every node, making the hedge dynamic and self-financing. In a recombining n-step tree a European payoff has

$$V_0 = R^{-n}\sum_{j=0}^{n}\binom{n}{j}q^j(1-q)^{n-j}H(S_0u^jd^{n-j}).$$

The Cox–Ross–Rubinstein parameterization for step Δt uses $u = \mathrm{e}^{\sigma\sqrt{\Delta t}}$, $d = u^{-1}$, and $R = \mathrm{e}^{(r-q_d)\Delta t}$. As the grid is refined, the log-price random walk converges weakly to geometric Brownian motion under appropriate conditions.

19.3 American exercise

At an American-option node,

$$V_t = \max\left\{h(S_t), \frac{1}{R}\,\mathbb{E}_t^{\mathbb{Q}}[V_{t+1}]\right\}.$$

This is an optimal stopping problem. The value process is the smallest supermartingale dominating the discounted exercise payoff, the Snell envelope.

Proposition 19.2 (No early exercise for a non-dividend call)**.** *With non-*

negative rates and no dividends, it is never optimal to exercise an American call before maturity.

Proof. By put–call parity and $P_t \geq 0$, $C_t \geq S_t - KP(t,T) \geq S_t - K$. Continuation value is at least intrinsic value, and exercise destroys remaining optionality and the financing benefit of delaying K. □

American puts and dividend-paying calls can exercise early. A tree must align dividend dates, exercise opportunities, and contract conventions.

19.4 Incomplete trees and bounds

If there are more states than independent traded securities, replication may fail. Sub- and super-replication define no-arbitrage price bounds:

$$\pi_{\text{sub}}(H) \leq V_0 \leq \pi_{\text{super}}(H).$$

A utility indifference price, minimal entropy measure, or calibrated model selects a point only by adding criteria beyond no arbitrage.

19.5 Problems

Exercise 19.3 (Core: two-step call)**.** Price and replicate a two-step European call. Record delta and cash at every node and verify terminal payoffs.

Exercise 19.4 (Core: American put)**.** Construct a two-step example where early put exercise is optimal and explain the roles of rate and moneyness.

Exercise 19.5 (Advanced: convergence)**.** Match the first two conditional moments of the CRR log return and show their limits as $\Delta t \downarrow 0$.

Chapter 20

Black–Scholes–Merton Theory and Greeks

Learning objectives. Derive the pricing PDE and closed form, understand hedging and Greeks, and identify exactly which conclusions rely on which assumptions.

20.1 Market assumptions

In the benchmark model,

$$\mathrm{d}S_t = (\mu - q)S_t\,\mathrm{d}t + \sigma S_t\,\mathrm{d}W_t^{\mathbb{P}}, \qquad \mathrm{d}B_t = rB_t\,\mathrm{d}t,$$

with constant $\sigma > 0$, deterministic r, q, continuous trading, no jumps, frictionless markets, unrestricted admissible trading, and a European payoff. The stock including dividends earns drift μ.

Under the pricing measure,

$$\mathrm{d}S_t = (r - q)S_t\,\mathrm{d}t + \sigma S_t\,\mathrm{d}W_t^{\mathbb{Q}}.$$

20.2 Delta-hedging derivation

For value $V(t, S)$, Itô's formula gives

$$\mathrm{d}V = \left(V_t + (\mu - q)SV_S + \frac{1}{2}\sigma^2 S^2 V_{SS}\right)\mathrm{d}t + \sigma SV_S\,\mathrm{d}W.$$

Hold one derivative and short $\Delta = V_S$ shares. After incorporating dividend on the stock hedge, stochastic risk cancels. No arbitrage makes the locally riskless portfolio earn r, yielding

$$V_t + (r-q)SV_S + \frac{1}{2}\sigma^2 S^2 V_{SS} - rV = 0.$$

The physical drift μ cancels because the source of diffusion risk is spanned.

20.3 Risk-neutral valuation

Feynman–Kac gives

$$V(t,S) = \mathrm{e}^{-r\tau}\,\mathbb{E}^{\mathbb{Q}}[h(S_T) \mid S_t = S], \qquad \tau = T - t,$$

with

$$\log S_T \mid S_t \sim N\left(\log S + (r - q - \tfrac{1}{2}\sigma^2)\tau, \sigma^2\tau\right).$$

Theorem 20.1 (European call and put)**.**

$$C = S\mathrm{e}^{-q\tau}\Phi(d_1) - K\mathrm{e}^{-r\tau}\Phi(d_2),$$

$$P = K\mathrm{e}^{-r\tau}\Phi(-d_2) - S\mathrm{e}^{-q\tau}\Phi(-d_1),$$

where

$$d_1 = \frac{\log(S/K) + (r - q + \frac{1}{2}\sigma^2)\tau}{\sigma\sqrt{\tau}}, \qquad d_2 = d_1 - \sigma\sqrt{\tau}.$$

Proof. Write the call as $\mathrm{e}^{-r\tau}(\mathbb{E}^{\mathbb{Q}}[S_T\mathbf{1}_{S_T>K}] - K\mathbb{P}^{\mathbb{Q}}(S_T > K))$. Standardizing the lognormal threshold gives $\mathbb{P}^{\mathbb{Q}}(S_T > K) = \Phi(d_2)$. Exponential tilting of the normal density gives $\mathbb{E}^{\mathbb{Q}}[S_T\mathbf{1}_{S_T>K}] = S\mathrm{e}^{(r-q)\tau}\Phi(d_1)$. The put follows from parity. $\square$

20.4 Greeks

For a call,

$$\Delta_C = \mathrm{e}^{-q\tau}\Phi(d_1), \qquad \Gamma = \frac{\mathrm{e}^{-q\tau}\phi(d_1)}{S\sigma\sqrt{\tau}},$$

$$\mathcal{V} = \frac{\partial C}{\partial \sigma} = S\mathrm{e}^{-q\tau}\phi(d_1)\sqrt{\tau},$$
$$\Theta_C = -\frac{S\mathrm{e}^{-q\tau}\phi(d_1)\sigma}{2\sqrt{\tau}} - rK\mathrm{e}^{-r\tau}\Phi(d_2) + qS\mathrm{e}^{-q\tau}\Phi(d_1),$$
$$\rho_C = K\tau\mathrm{e}^{-r\tau}\Phi(d_2).$$

Greek units matter: vega above is per absolute volatility unit, so divide by 100 for a one-volatility-point move.

The PDE implies a theta–gamma–carry relation. For a delta-hedged option, time decay compensates for expected gamma gains under the model. Large gamma near expiry makes discrete hedging difficult exactly when theta is largest.

20.5 Discrete hedging error

With true realized quadratic variation different from model variance, a continuous delta-hedge approximation gives

$$\Pi_T - \Pi_0\mathrm{e}^{rT} \approx \frac{1}{2}\int_0^T \mathrm{e}^{r(T-t)}\Gamma_t S_t^2(\,\mathrm{d}\langle\log S\rangle_t - \sigma^2_{\mathrm{model}}\,\mathrm{d}t)$$

before costs and jumps. Discrete rebalancing adds error of order depending on mesh, payoff regularity, and strategy. Transaction costs grow with trading frequency, so continuous hedging is not an operational limit without a cost model.

20.6 Digital options and density

A cash-or-nothing call pays $\mathbf{1}_{\{S_T>K\}}$ and is worth $\mathrm{e}^{-r\tau}\Phi(d_2)$. Under regularity,

$$-\frac{\partial C(K,T)}{\partial K} = P(t,T)\mathbb{Q}(S_T > K), \qquad \frac{\partial^2 C(K,T)}{\partial K^2} = P(t,T)f^{\mathbb{Q}}_{S_T}(K).$$

Thus a continuum of option prices reveals the risk-neutral terminal distribution. Noise and discrete strikes make differentiation an ill-posed estimation problem requiring arbitrage-aware smoothing.

20.7 Problems

Exercise 20.2 (Core: Greeks)**.** Differentiate the call formula to obtain delta, gamma, and vega. Use the identity $S\mathrm{e}^{-q\tau}\phi(d_1) = K\mathrm{e}^{-r\tau}\phi(d_2)$.

Exercise 20.3 (Core: PDE)**.** Verify that the call formula satisfies the PDE, terminal condition, and boundary behavior.

Exercise 20.4 (Advanced: digital replication)**.** Show that a tight call spread converges to a digital payoff and quantify the finite strike-spacing approximation.

Chapter 21

Volatility Surfaces, Local and Stochastic Volatility

Learning objectives. Construct arbitrage-consistent volatility surfaces, derive local volatility, study stochastic-volatility dynamics, and distinguish calibration fit from hedging validity.

21.1 Why volatility is a surface

Market implied volatility depends on strike and maturity. Equity index smiles often show expensive downside protection; term structure changes with event and regime. Quoting implied volatility normalizes option prices but preserves model dependence.

Useful coordinates include log forward moneyness $k = \log(K/F_{t,T})$, delta, and total variance $w(k,T) = \sigma^2_{\text{imp}}(k,T)T$. Interpolating raw volatility independently can create arbitrage even when input quotes are valid.

21.2 Static-arbitrage restrictions

For fixed T, undiscounted call price is decreasing and convex in K:

$$\partial_K C \leq 0, \qquad \partial_{KK} C \geq 0.$$

It lies between intrinsic lower bounds and the prepaid forward. Across maturities, calendar consistency is naturally expressed for option prices under aligned forwards and discounting; total variance should not decrease under simplified settings.

Butterfly arbitrage corresponds to negative risk-neutral density. Calendar arbitrage corresponds to a more valuable shorter-dated claim under conditions where the longer-dated claim contains it.

21.3 Dupire local volatility

Assume under $\mathbb{Q}$

$$\frac{\mathrm{d}S_t}{S_t} = (r_t - q_t)\,\mathrm{d}t + \sigma_{\mathrm{loc}}(t, S_t)\,\mathrm{d}W_t.$$

Given a sufficiently smooth arbitrage-free call surface,

$$\sigma^2_{\mathrm{loc}}(T, K) = \frac{\partial_T C + (r-q)K\partial_K C + qC}{\frac{1}{2}K^2\partial_{KK}C}$$

for deterministic rates and yields under the stated convention.

Local volatility exactly matches the one-time marginal distributions encoded by the surface. It does not necessarily reproduce joint dynamics, forward smiles, or realistic spot–volatility co-movement. Numerical differentiation amplifies quote noise, especially through $\partial_{KK}C$.

21.4 Heston stochastic volatility

The Heston model under $\mathbb{Q}$ is

$$\frac{\mathrm{d}S_t}{S_t} = (r-q)\,\mathrm{d}t + \sqrt{v_t}\,\mathrm{d}W_t^S,$$

$$\mathrm{d}v_t = \kappa(\theta - v_t)\,\mathrm{d}t + \xi\sqrt{v_t}\,\mathrm{d}W_t^v, \qquad \mathrm{d}\langle W^S, W^v\rangle_t = \rho\,\mathrm{d}t.$$

Mean reversion is controlled by κ, long-run variance by θ, volatility of variance by ξ, and skew by ρ. The Feller condition $2\kappa\theta \geq \xi^2$ keeps zero unattainable under one common classification but is not required for every numerical scheme or admissible calibration.

The characteristic function is affine and supports Fourier pricing. Parameters can be poorly identified from a narrow option panel: κ and θ, for example, may trade off while producing similar maturities.

21.5 SABR

For forward F,

$$\mathrm{d}F_t = \alpha_t F_t^{\beta}\,\mathrm{d}W_t^1, \qquad \mathrm{d}\alpha_t = \nu\alpha_t\,\mathrm{d}W_t^2, \qquad \mathrm{d}\langle W^1, W^2\rangle_t = \rho\,\mathrm{d}t.$$

SABR is widely used to interpolate smiles, especially in rates. β controls backbone, ν smile curvature, and ρ skew. Common asymptotic implied-volatility formulas can be inaccurate at long maturity, extreme strikes, or near boundaries.

21.6 Jumps and Lévy models

Merton's jump diffusion adds compound Poisson lognormal jumps. Variance gamma and CGMY use infinite-activity Lévy processes. Jumps create short-maturity skew and kurtosis that continuous diffusions struggle to match.

Risk-neutral jump intensity and distribution are not physical jump forecasts. Hedging jump risk is incomplete when no traded asset spans it; calibration chooses prices consistent with selected instruments rather than manufacturing completeness.

21.7 Calibration

For quotes P_i^{mkt} and weights w_i,

$$\widehat{\theta} = \arg\min_{\theta\in\Theta} \sum_i w_i(P_i^{\mathrm{model}}(\theta) - P_i^{\mathrm{mkt}})^2 + \lambda\mathcal{R}(\theta).$$

Price-error weighting, vega weighting, relative errors, and bid–ask normalization answer different questions. A robust workflow checks:

- no-arbitrage cleaning and quote timestamps;
- multiple starting values and parameter bounds;
- residuals by strike and maturity;
- stability across days;
- prices of withheld instruments;

- hedge performance and scenario behavior;
- parameter uncertainty and numerical tolerance.

Model-risk warning 21.1. Six decimal places in a calibrated parameter are usually optimizer output, not six decimal places of economic information.

21.8 Problems

Exercise 21.2 (Core: convexity)**.** Use a butterfly portfolio to prove convexity of call price in strike.

Exercise 21.3 (Core: Heston moments)**.** Derive $\mathbb{E}[v_t \mid v_0]$ and the long-run mean. Discuss what changes under the physical measure.

Exercise 21.4 (Advanced: Dupire)**.** Derive the Dupire equation from the forward equation for the transition density and integration by parts.

Chapter 22

Exotic Derivatives and Incomplete-Market Hedging

Learning objectives. Analyze path-dependent payoffs, apply change-of-numeraire methods, value options with exercise features, and measure hedging risk in incomplete markets.

22.1 Change of numeraire

Let $N_t > 0$ be a traded numeraire. There exists an associated measure $\mathbb{Q}^N$ under which every traded price divided by N is a martingale:

$$\frac{V_t}{N_t} = \mathbb{E}_t^{\mathbb{Q}^N}\left[\frac{V_T}{N_T}\right].$$

Choosing a zero-coupon bond as numeraire gives a forward measure, under which the corresponding forward price is a martingale. This simplifies caplet, swaption, and quanto calculations by matching measure to payoff.

22.2 Asian options

An arithmetic-average call pays

$$\left(\frac{1}{n}\sum_{j=1}^{n} S_{t_j} - K\right)^+.$$

The arithmetic average of lognormal variables is not lognormal, so no Black–Scholes closed form exists. Geometric-average options are tractable and provide control variates. Bounds follow from Jensen and comonotonic constructions.

22.3 Barrier options

A down-and-out call ceases to exist if spot reaches barrier $H < S_0$. Continuous monitoring differs from daily monitoring because crossings can occur between observations. Reflection-principle formulas exist in Black–Scholes for standard contracts. Greeks become sharp near the barrier and discrete hedge error grows.

In–out parity gives

$$V_{\text{vanilla}} = V_{\text{knock-in}} + V_{\text{knock-out}}$$

when rebates and monitoring conventions align.

22.4 Lookbacks and cliquets

Lookback payoffs depend on running maximum or minimum. Cliquets lock periodic returns subject to local and global caps or floors. Their values depend strongly on the joint forward volatility surface, not only today's marginal smiles. A model calibrated to vanillas can therefore disagree materially on exotics.

22.5 Bermudan options and least-squares Monte Carlo

At exercise dates t_m, a Bermudan value is

$$V_{t_m} = \max\{h(X_{t_m}), \mathbb{E}^{\mathbb{Q}}[D_{m,m+1}V_{t_{m+1}} \mid X_{t_m}]\}.$$

Least-squares Monte Carlo regresses discounted continuation cash flows on basis functions of state among relevant paths, then compares fitted continuation with exercise value.

Using the same paths to fit and value creates in-sample bias. Independent valuation paths, cross-fitting, basis sensitivity, and dual upper bounds improve credibility. The exercise rule should respect all state variables, not spot alone when rates or volatility are stochastic.

22.6 Variance and volatility derivatives

A variance swap pays realized variance minus fixed strike. Under ideal continuous sampling and diffusion,

$$\int_0^T \sigma_t^2 \,\mathrm{d}t = 2\int_0^T \frac{\mathrm{d}S_t}{S_t} - 2\log\frac{S_T}{S_0}$$

after appropriate drift terms. A log contract can be replicated by a continuum of out-of-the-money options, giving a model-light fair variance strike under carefully stated jump and discretization conventions.

Volatility swap payoff uses square root of variance, so Jensen creates a convexity adjustment; it cannot be priced by simply taking the square root of variance-swap strike.

22.7 Quanto and correlation risk

A quanto pays an asset return in a different currency at a fixed exchange rate. Under the domestic pricing measure, covariance between the foreign asset and FX changes its drift. This quanto adjustment makes correlation a first-order input. Cross-asset derivatives reveal why specifying only marginal models is insufficient.

22.8 Hedging in incomplete markets

When risk is unspanned, exact replication is impossible. A minimum-variance hedge chooses strategy ϕ to minimize

$$\mathbb{E}[(H - V_0 - G_T(\phi))^2].$$

The Galtchouk–Kunita–Watanabe decomposition writes payoff into initial value, stochastic integral with traded assets, and an orthogonal residual.

Utility indifference and risk-measure hedges choose different objectives and therefore different prices.

22.9 Problems

Exercise 22.1 (Core: geometric Asian)**.** Derive the distribution of a discretely sampled geometric average under Black–Scholes and write its call value.

Exercise 22.2 (Core: variance replication)**.** Show how a twice differentiable payoff can be statically represented using cash, forward, and a continuum of calls and puts. Apply it to $-\log S_T$.

Exercise 22.3 (Advanced: LSM)**.** Implement least-squares Monte Carlo for an American put and report lower-bound stability across bases, paths, and time steps.

Part VII

Fixed Income and Credit

Chapter 23

Bond Mathematics and Curve Construction

Learning objectives. Price fixed-income cash flows under market conventions, construct discount curves, measure rate sensitivity, and diagnose interpolation and bootstrapping risk.

23.1 Present value and compounding

For deterministic cash flows C_i at dates T_i,

$$P_t = \sum_i C_i P(t, T_i).$$

If a flat continuously compounded yield y is used, $P(y) = \sum_i C_i \mathrm{e}^{-y(T_i - t)}$. Market instruments instead use instrument-specific compounding, day counts, settlement lags, calendars, and accrued-interest rules.

An annualized rate is incomplete without basis. ACT/360 and ACT/365 change accrual; 30/360 changes date counting; business-day adjustment changes payment date. Production curve code treats conventions as data.

23.2 Bootstrapping

Bootstrapping solves discount factors sequentially from liquid instruments. A par coupon bond priced at par satisfies

$$1 = c_n \sum_{i=1}^{n} \delta_i P(0, T_i) + P(0, T_n).$$

If earlier discounts are known,

$$P(0,T_n) = \frac{1 - c_n \sum_{i=1}^{n-1} \delta_i P(0,T_i)}{1 + c_n \delta_n}.$$

Deposits, futures or forward-rate agreements, overnight swaps, term swaps, and bonds can occupy different curve segments. Multi-curve construction discounts collateralized cash flows on an overnight curve while projecting each floating tenor on its own forward curve.

23.3 Interpolation

Quotes exist at discrete maturities while pricing needs all dates. Interpolating zero rates, log discounts, forwards, or par rates imposes different shapes. Linear interpolation of log discount factors gives piecewise constant forwards. Cubic splines are smooth but can oscillate and create negative forward rates. Monotone-convex methods aim to preserve economically sensible local behavior.

Curve risk includes quote noise, interpolation choice, instrument selection, and extrapolation. A perfect repricing of bootstrap instruments does not make the between-knot curve observed.

23.4 Duration and convexity

For cash-flow weights $w_i = C_i \mathrm{e}^{-y t_i}/P$, Macaulay duration is $D_M = \sum_i w_i t_i$. With continuous compounding, modified duration equals D_M:

$$-\frac{1}{P}\frac{\partial P}{\partial y} = D_M.$$

Convexity is $\mathcal{C} = P^{-1}\partial^2 P/\partial y^2 = \sum_i w_i t_i^2$.

Yield duration assumes a parallel flat-yield move. Key-rate duration shocks one curve node with an interpolation rule. Partial derivatives to market quotes provide more realistic curve risk but depend on the bootstrap Jacobian.

Proposition 23.1 (Positive convexity of an option-free bond)**.** *For fixed* *positive cash flows under a parallel continuously compounded yield,* $P''(y) > 0$*.*

Proof. $P''(y) = \sum_i t_i^2 C_i e^{-yt_i} > 0$ when at least one positive cash flow occurs after time zero. □

23.5 Immunization

A liability is locally immunized by matching present value and duration with an asset portfolio. Convexity improves protection for parallel moves. Multiple key-rate constraints immunize selected curve deformations:

$$\min_w w^\top \Sigma w \quad \text{s.t.} \quad A^\top w = b,$$

where rows of A contain value and key-rate exposures. Rebalancing, transaction costs, nonparallel moves, spread risk, and optionality limit immunization.

23.6 Problems

Exercise 23.2 (Core: bootstrap)**.** Bootstrap annual discount factors from three par rates and compute one-year forward rates. Check discount monotonicity.

Exercise 23.3 (Core: duration)**.** Derive duration and convexity for a zero-coupon bond and compare the approximation with the exact price change.

Exercise 23.4 (Advanced: curve Jacobian)**.** Use the implicit function theorem to derive sensitivities of bootstrapped discount factors to input par swap rates.

Chapter 24

Interest-Rate Models and Rate Derivatives

Learning objectives. Model short rates and forward rates under risk-neutral and physical measures, price caps and swaptions, and understand calibration and measure consistency.

24.1 Short-rate models

In a short-rate model,

$$P(t,T) = \mathbb{E}_t^{\mathbb{Q}}\left[\exp\left(-\int_t^T r_s \,\mathrm{d}s\right)\right].$$

Vasicek uses Gaussian mean reversion. Cox–Ingersoll–Ross uses

$$\mathrm{d}r_t = \kappa(\theta - r_t)\,\mathrm{d}t + \sigma\sqrt{r_t}\,\mathrm{d}W_t^{\mathbb{Q}},$$

which is affine and nonnegative under boundary conditions. Hull–White extends Vasicek with time-dependent drift:

$$\mathrm{d}r_t = (\theta(t) - ar_t)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t^{\mathbb{Q}},$$

allowing exact fit to today's initial curve.

Exact initial fit does not identify future dynamics. Mean reversion and volatility are calibrated to options or estimated under a measure; the physical drift requires risk-premium specification.

24.2 Heath–Jarrow–Morton

Instead of a short rate, HJM models the entire instantaneous forward curve:

$$\mathrm{d}f(t,T) = \alpha(t,T)\,\mathrm{d}t + \sigma(t,T)^\top\,\mathrm{d}W_t^{\mathbb{Q}}.$$

No arbitrage restricts drift:

$$\alpha(t,T) = \sigma(t,T)^\top \int_t^T \sigma(t,u)\,\mathrm{d}u$$

under the basic money-market measure convention. Thus volatility specification determines risk-neutral drift.

Finite-dimensional Markov representations require special volatility structures. Direct curve models make it clear that today's curve is a state, not a scalar rate.

24.3 Forward measures and caplets

The simply compounded forward rate for $[T_1, T_2]$ is

$$L_t(T_1,T_2) = \frac{1}{\delta}\left(\frac{P(t,T_1)}{P(t,T_2)} - 1\right).$$

Under the T_2-forward measure it is a martingale in a standard setting. A caplet payoff at T_2 is $N\delta(L_{T_1} - K)^+$ and Black's formula is

$$V_t = NP(t,T_2)\delta[F\Phi(d_1) - K\Phi(d_2)]$$

for lognormal forward $F = L_t(T_1,T_2)$.

Negative rates motivate shifted lognormal or normal Bachelier quoting. A normal volatility has rate units; a lognormal volatility has percentage units. Conversion depends on forward, strike, and maturity.

24.4 Swaptions

A payer swaption grants the right to pay fixed K and receive floating. Under the annuity numeraire

$$A_t = \sum_i \delta_i P(t, T_i),$$

the forward swap rate

$$S_t = \frac{P(t, T_0) - P(t, T_n)}{A_t}$$

is a martingale under its swap measure. Black-style pricing treats it as lognormal:

$$V_t = A_t[S_t \Phi(d_1) - K\Phi(d_2)].$$

Smile, cash settlement, day count, collateral, and exercise conventions enter real contracts. Bermudan swaptions depend on the joint curve evolution and optimal exercise.

24.5 Market models

The LIBOR market model specifies dynamics of a family of forward rates under their natural measures. Changing to one common simulation measure introduces drift terms depending on correlations and other forwards. Calibration matches cap and swaption volatilities but correlation is weakly identified by vanilla prices and critical for exotics.

Post-benchmark-reform markets use overnight and term reference rates with fallback and compounding conventions. Mathematical notation should name the actual rate and accrual mechanism instead of retaining obsolete labels by habit.

24.6 Problems

Exercise 24.1 (Core: CIR moments)**.** Derive conditional mean of the CIR rate. Research the noncentral chi-square transition law and explain its exact-simulation benefit.

Exercise 24.2 (Core: caplet measure)**.** Derive caplet pricing by choosing the T_2 bond as numeraire.

Exercise 24.3 (Advanced: HJM drift)**.** Derive the HJM drift restriction by requiring discounted bond prices to be martingales.

Chapter 25

Credit Risk

Learning objectives. Model default with structural and intensity approaches, bootstrap survival curves, price credit protection, and separate credit, liquidity, and recovery assumptions.

25.1 Credit events and loss

Credit risk includes default, downgrade, spread widening, restructuring, and counterparty deterioration. If exposure at default is EAD, loss fraction is $LGD = 1 - R$, and default indicator is D, loss is

$$L = \mathrm{EAD} \times LGD \times D.$$

Expected loss is not unexpected loss. Dependence among EAD, recovery, and default is important in stress.

25.2 Structural models

In Merton's model, firm asset value follows a diffusion and equity is a call on assets with strike equal to promised debt:

$$E_0 = V_0\Phi(d_1) - D\mathrm{e}^{-rT}\Phi(d_2).$$

Default at maturity occurs when $V_T < D$. Risky debt equals risk-free debt minus a put on firm assets.

The model connects leverage and asset volatility to credit spreads, but firm asset value and volatility are latent, default can occur only at specified barriers, and short-maturity spreads are often too low without jumps or liquidity.

First-passage models default when asset value crosses a barrier. Barrier design and capital structure become central.

25.3 Reduced-form intensity models

Let default time τ have intensity λ_t so, informally,

$$\mathbb{P}(t < \tau \le t + \mathrm{d}t \mid \tau > t, \mathcal{F}_t) = \lambda_t \, \mathrm{d}t.$$

Conditional survival is

$$\mathbb{P}(\tau > T \mid \mathcal{F}_t) = \mathbb{E}_t \left[\exp \left(- \int_t^T \lambda_s \, \mathrm{d}s \right) \right]$$

under appropriate filtration and measure conditions.

With deterministic intensity and recovery paid at default, a risky zero-coupon value is the sum of survival payment and expected recovery payment. Under zero recovery, $P^{\text{risky}}(0,T) = P(0,T)S(0,T)$ when rates and default separate appropriately.

25.4 Credit default swaps

A CDS protection buyer pays spread s until maturity or default and receives $1 - R$ upon covered credit event. The par spread equates premium and protection legs:

$$s \sum_i \delta_i P(0,T_i) S(0,T_i) \approx (1-R) \sum_i P(0,T_i)[S(0,T_{i-1}) - S(0,T_i)],$$

with production valuation adding accrual on default and exact timing.

Given recovery and discount curve, quoted spreads bootstrap survival probabilities or piecewise intensities. Recovery and intensity are not separately well identified by CDS spreads alone.

25.5 Credit portfolio dependence

In a one-factor Gaussian latent-variable model,

$$Z_i = \sqrt{\rho_i}Y + \sqrt{1-\rho_i}\varepsilon_i,$$

and obligor i defaults if Z_i is below a threshold matching marginal default probability. Conditional on Y, defaults are independent. The common factor creates default clustering.

Tail loss is extraordinarily sensitive to asset correlation, exposure concentration, recovery dependence, and model form. Gaussian dependence can understate joint extremes. Stress testing should include sector and macro scenarios rather than rely only on a fitted copula.

25.6 Migration matrices

A rating transition matrix P has rows summing to one. Multi-year transition under time homogeneity is P^n. A continuous-time generator Q has nonnegative off-diagonals, rows summing to zero, and transition matrix $\exp(Qt)$. Not every empirical discrete matrix is embeddable in a valid generator.

Historical transition probabilities are physical. Pricing spreads reflects risk-neutral probabilities, recovery, liquidity, capital, and risk premia.

25.7 Counterparty exposure

Netting sets combine positive and negative trade values under enforceable legal agreements. Variation margin reduces current exposure; initial margin targets future exposure over margin period of risk. Rehypothecation, thresholds, disputes, and close-out delay affect residual exposure.

Potential future exposure, expected shortfall of exposure, CVA, and regulatory exposure are different metrics. A simulation needs joint market factors, collateral rules, default dependence, and pathwise netting.

25.8 Problems

Exercise 25.1 (Core: Merton)**.** Show that risky debt equals a risk-free bond minus a put. Derive the risk-neutral default probability and distinguish it from expected loss.

Exercise 25.2 (Core: CDS bootstrap)**.** Bootstrap piecewise constant survival probabilities from two annual CDS spreads under a simplified payment convention.

Exercise 25.3 (Advanced: wrong-way risk)**.** Construct a joint model in which exposure and intensity share a factor. Compare CVA with an independence approximation.

Chapter 26

Securitization and Structured Credit

Learning objectives. Understand cash-flow waterfalls, tranche loss, prepayment and extension risk, and why structured-credit models are highly sensitive to dependence and legal detail.

26.1 Pooling and tranching

A securitization pools asset cash flows and allocates them through a contractual waterfall. A tranche with attachment A and detachment D has loss at portfolio loss fraction L:

$$L_{A,D}(L) = \min\{(L - A)^+, D - A\}.$$

Equity absorbs first loss; mezzanine absorbs intermediate loss; senior tranches lose only after subordination is exhausted.

Expected tranche loss is nonlinear in the full portfolio-loss distribution. Two portfolios with the same expected total loss can have very different senior-tranche risk.

26.2 Dependence and base correlation

Increasing default correlation moves mass toward very low and very high portfolio loss. This can reduce equity expected loss at some points while increasing senior tail loss. Base correlation fits each equity tranche $[0, D]$

with a separate correlation, creating an interpolation convention rather than one internally consistent joint distribution.

26.3 Mortgage cash flows

Mortgage-backed securities contain borrower prepayment options. Falling rates can accelerate refinancing, shortening duration when price would otherwise rise; rising rates slow prepayment, extending duration when price falls. This negative convexity drives dynamic hedging demand.

Prepayment depends on incentive, seasoning, burnout, housing turnover, credit, seasonality, and operational friction. A model needs loan-level or pool variables available at the valuation date.

26.4 Waterfall simulation

A pathwise engine applies:

1. scheduled interest and principal;
2. prepayment and default;
3. recovery lag and severity;
4. fees and servicing;
5. interest and principal priority;
6. triggers, reserve accounts, and diversion;
7. cleanup calls and termination.

The legal prospectus is part of the model specification. A mathematically elegant loss model cannot compensate for an incorrect waterfall.

26.5 Stress and model risk

Structured products combine long horizons, nonlinear allocation, uncertain behavior, and sparse tail data. Stress unemployment, home prices, rates, refinancing capacity, recovery, correlation, and servicer performance.

Report scenario cash flows and breakeven assumptions rather than only a single spread or rating.

26.6 Problems

Exercise 26.1 (Core: tranche loss)**.** Plot tranche loss as a function of portfolio loss for equity, mezzanine, and senior tranches. Identify delta with respect to total loss away from attachment points.

Exercise 26.2 (Advanced: large homogeneous pool)**.** Conditional on a Gaussian common factor, derive the limiting default fraction in a homogeneous portfolio and integrate to obtain expected tranche loss.

Part VIII

Portfolio Construction and Risk

Chapter 27

Portfolio Theory and Estimation

Learning objectives. Derive efficient portfolios, understand estimation error, apply shrinkage and robust construction, and evaluate portfolios through implementable out-of-sample decisions.

27.1 Returns and wealth

For simple return $R_t = (P_t + D_t)/P_{t-1} - 1$, wealth compounds as $W_T = W_0 \prod_t (1 + R_t)$. Log return $r_t = \log(1 + R_t)$ adds through time. Cross-sectionally, portfolio simple return is $R_{p,t} = w_{t-1}^\top R_t$ for pre-return weights. Portfolio log return is not the weighted average of asset log returns.

Annual arithmetic mean is often scaled by periods; volatility by square root of periods under stationarity and suitable dependence. Long-horizon quantiles and drawdowns do not obey a universal square-root rule.

27.2 Mean–variance optimization

For expected return μ and covariance Σ,

$$\max_w \left\{ w^\top \mu - \frac{\gamma}{2} w^\top \Sigma w \right\} \quad \text{s.t.} \quad \mathbf{1}^\top w = 1.$$

With a risk-free asset and unconstrained risky weights, the tangency direction is $\Sigma^{-1}(\mu - r_f \mathbf{1})$. Its maximum squared Sharpe ratio is

$$SR^2_{\max} = (\mu - r_f \mathbf{1})^\top \Sigma^{-1} (\mu - r_f \mathbf{1}).$$

This is an ex ante population statement. Replacing inputs by noisy estimates can produce extreme weights and poor realized performance.

27.3 Estimation error

Expected returns are much harder to estimate than covariance at ordinary horizons. Optimization maps small errors through Σ^{-1}, emphasizing directions of apparently low risk that often correspond to noise.

Sampling uncertainty is only one issue. Structural change, survivorship, asynchronous markets, stale holdings, and benchmark revisions create model and data uncertainty. Report weights and risk contributions, not just the objective value.

27.4 Covariance shrinkage

A linear shrinkage estimator is

$$\widehat{\Sigma}_\delta = \delta F + (1-\delta)S, \qquad \delta \in [0,1],$$

where S is sample covariance and F a structured target. Shrinkage trades some bias for lower variance and improved conditioning.

Factor covariance uses $B\Sigma_f B^\mathsf{T} + D$. Robust covariance estimators reduce outlier sensitivity. Exponentially weighted estimates adapt to changing volatility but reduce effective sample size.

27.5 Expected-return shrinkage and views

Bayesian or empirical-Bayes estimation shrinks noisy asset means toward a grand mean or factor-implied prior. Black–Litterman starts from equilibrium-implied excess returns

$$\pi = \delta \Sigma w_{\mathrm{mkt}}$$

and combines views $P\mu = q + \text{noise}$ with a prior. Posterior mean depends on view uncertainty and prior scaling; these are economic inputs, not tuning after seeing performance.

27.6 Constraints and regularization

Long-only, leverage, box, sector, factor, turnover, and tracking-error constraints stabilize portfolios while encoding mandate. A generic program is

$$\min_{w} \frac{1}{2} w^{\mathsf{T}} \widehat{\Sigma} w - \lambda w^{\mathsf{T}} \widehat{\mu} + \eta \left\| w - w_{\mathrm{prev}} \right\|_1$$

subject to exposure constraints. The turnover penalty approximates linear costs. Constraints create shadow prices that quantify the value of relaxing a mandate under the model.

27.7 Resampling and robust portfolios

Resampled efficiency averages optimized weights across simulated inputs. Distributionally robust optimization protects against laws in an ambiguity set. Worst-case methods can be conservative; uncertainty sets must be calibrated before observing desired weights.

The one-over-N portfolio is a serious benchmark. A complex method must earn its estimation and trading costs relative to equal weight, global minimum variance, and a simple factor-balanced allocation.

27.8 Problems

Exercise 27.1 (Core: tangency)**.** Derive the tangency direction and maximum Sharpe ratio using Cauchy–Schwarz.

Exercise 27.2 (Core: shrinkage)**.** Show that a convex combination of positive semidefinite covariance matrices remains positive semidefinite. When is it positive definite?

Exercise 27.3 (Advanced: sensitivity)**.** Differentiate unconstrained mean–variance weights with respect to μ and Σ. Identify why small eigenvalues amplify error.

Chapter 28

Risk Measures, Backtesting, and Stress

Learning objectives. Define coherent and convex risk measures, estimate VaR and Expected Shortfall, backtest forecasts, and design scenarios that expose rather than conceal model limitations.

28.1 Loss conventions

Let $L = -\Delta V$ be positive when money is lost. A risk number without horizon, confidence, currency, valuation time, data window, and treatment of positions is ambiguous.

Definition 28.1 (Value at Risk)**.**

$$\mathrm{VaR}_\alpha(L) = \inf\{\ell : \mathbb{P}(L \le \ell) \ge \alpha\}.$$

Definition 28.2 (Expected Shortfall)**.** A general definition is

$$\mathrm{ES}_\alpha(L) = \frac{1}{1-\alpha}\int_\alpha^1 \mathrm{VaR}_u(L)\,\mathrm{d}u.$$

For a continuous distribution, this equals $\mathbb{E}[L \mid L \ge \mathrm{VaR}_\alpha(L)]$.

28.2 Coherent risk

A risk measure ρ is coherent when it is monotone, translation invariant, positively homogeneous, and subadditive. Convex risk measures replace homogeneity and subadditivity by convexity, allowing liquidity or concentration penalties.

Theorem 28.3 (Expected Shortfall is coherent)**.** *For integrable losses,* ES_α *satisfies the four coherence axioms.*

Proof. Use the representation

$$\mathrm{ES}_\alpha(L) = \inf_{z\in\mathbb{R}} \left\{ z + \frac{1}{1-\alpha}\,\mathbb{E}[(L-z)^+] \right\}.$$

Monotonicity and translation follow directly. Positive homogeneity follows by scaling z. Subadditivity uses $(X+Y-z_X-z_Y)^+ \le (X-z_X)^+ + (Y-z_Y)^+$ and takes infima. □

VaR is not generally subadditive for discontinuous or heavy-tailed losses. Under elliptical distributions it may behave well, but coherence is a property over a declared domain.

28.3 Parametric and historical estimation

Under normal $L \sim N(\mu_L, \sigma_L^2)$,

$$\mathrm{VaR}_\alpha = \mu_L + \sigma_L z_\alpha, \qquad \mathrm{ES}_\alpha = \mu_L + \sigma_L \frac{\phi(z_\alpha)}{1-\alpha}.$$

Student tails, mixtures, filtered historical simulation, Monte Carlo, and EVT change tail assumptions. Historical simulation places zero probability beyond observed losses and assumes the selected past is representative.

Filtered historical simulation standardizes returns by estimated conditional volatility, resamples residuals, then applies current volatility. This separates scale dynamics from residual shape but inherits both model and resampling assumptions.

28.4 Marginal and component risk

For differentiable homogeneous $\rho(w)$, Euler's theorem gives

$$\rho(w) = \sum_i w_i \frac{\partial \rho}{\partial w_i}.$$

Component risk is $CR_i = w_i \partial\rho/\partial w_i$. It allocates the current portfolio's risk but is not the same as incremental risk from deleting a position, which is a finite change.

28.5 VaR and ES backtesting

For forecast $\widehat{q}_t$, exception $I_t = \mathbf{1}_{\{L_t > \widehat{q}_t\}}$ should have rate $1-\alpha$. Kupiec tests unconditional rate; Christoffersen tests transition dependence. Quantile loss

$$\ell_\alpha(q, L) = (\alpha - \mathbf{1}_{\{L \le q\}})(L - q)$$

is strictly consistent for the quantile.

ES alone is not elicitable by a scalar loss, but $(\mathrm{VaR}, \mathrm{ES})$ is jointly elicitable under conditions. Compare models through proper scoring, calibration, and economic consequences, not only exception counts.

28.6 Stress testing

Historical stress replays observed shocks. Hypothetical stress imposes coherent factor moves. Reverse stress asks which scenario breaches a solvency or liquidity threshold:

$$\min_{\Delta x} \; \Delta x^\mathsf{T} \Omega^{-1} \Delta x \quad \text{s.t.} \quad \Delta V(\Delta x) \le -L^\star.$$

This finds a most plausible damaging shock under metric Ω, not the only dangerous scenario.

Scenarios should propagate rates, volatility surfaces, spreads, FX, liquidity, defaults, collateral, and management actions consistently. Holding correlations fixed in stress can be the main error.

28.7 Liquidity risk

Market liquidity concerns executable price and depth; funding liquidity concerns ability to finance positions and margin. They interact through margin spirals. Liquidity-adjusted loss includes liquidation horizon and impact:

$$L_{\text{liq}} = L_{\text{market}} + \sum_i \left(\frac{s_i}{2} Q_i + \eta_i Q_i^{3/2} \right).$$

The horizon is position- and state-dependent. A ten-day scalar multiplier is not a substitute for a liquidation schedule.

28.8 Problems

Exercise 28.4 (Core: ES)**.** Derive normal Expected Shortfall by integrating the normal tail.

Exercise 28.5 (Core: coherence)**.** Give a two-position discrete example where VaR violates subadditivity.

Exercise 28.6 (Advanced: reverse stress)**.** Linearize portfolio loss and solve the Mahalanobis reverse-stress problem in closed form. Then discuss gamma effects.

Chapter 29

Risk Allocation, Drawdowns, and Alternative Objectives

Learning objectives. Construct risk-parity and drawdown-aware portfolios, distinguish volatility from loss, and connect portfolio objectives to investor liabilities and horizons.

29.1 Risk parity

For volatility $\sigma(w) = \sqrt{w^\top \Sigma w}$, component contribution is

$$RC_i = w_i \frac{(\Sigma w)_i}{\sigma(w)}.$$

Equal-risk-contribution portfolios solve $RC_i = \sigma/n$ under positive weights. One convex formulation minimizes

$$\frac{1}{2} w^\top \Sigma w - \sum_i b_i \log w_i$$

and rescales the solution, where budgets $b_i > 0$.

Risk parity does not make assets equally risky in all states. Covariance estimation, volatility scaling, leverage, and concentration in a common macro factor determine economic exposure. Levered bonds can dominate liquidity and inflation stress.

29.2 Volatility targeting

A strategy with raw return r_t and forecast volatility $\widehat{\sigma}_t$ uses exposure

$$\ell_t = \min\left(\ell_{\max}, \frac{\sigma^\star}{\widehat{\sigma}_t}\right).$$

It reduces exposure after volatility rises. Performance can improve through risk stabilization and a negative return–volatility relation, but rapid deleveraging can sell into stress and contribute to crowding.

The volatility estimate must use data through the decision time. Lag, rebalance frequency, cap, financing, and turnover define the strategy.

29.3 Drawdown

For wealth W_t, running peak $M_t = \max_{s \le t} W_s$ and drawdown $D_t = 1 - W_t/M_t$. Maximum drawdown is path-dependent and its sample distribution depends strongly on horizon. It is not a coherent one-period risk measure.

Conditional drawdown at risk and average drawdown measures focus on the distribution of drawdowns. Drawdown constraints create dynamic policies because the state includes the running maximum.

29.4 Kelly and risk of ruin

Log-optimal growth maximizes $\mathbb{E}[\log(W_{t+1}/W_t)]$. For repeated independent bets, it maximizes asymptotic growth under its model. It can accept deep interim drawdowns and is extremely sensitive to probability error and omitted tail outcomes.

Fractional Kelly multiplies estimated optimal exposure by $0 < f < 1$. This sacrifices some modeled growth for reduced variance and parameter sensitivity. Hard ruin, margin, or drawdown constraints should be modeled explicitly rather than expected to emerge from log utility.

29.5 Liability-driven investment

An investor with liabilities cares about surplus $S_t = A_t - L_t$. Asset-only volatility can be misleading: a long-duration asset may be risky standalone but hedge a long-duration liability. Objectives include surplus variance, funding-ratio shortfall, and expected utility of contributions.

Liability discounting itself can be regulatory, accounting, market-consistent, or actuarial. The chosen curve changes both measured funding and hedge.

29.6 Tracking error and active risk

For benchmark w_b, active weights $a = w - w_b$ have ex ante tracking error

$$TE = \sqrt{a^\top \Sigma a}.$$

Information ratio is expected active return divided by tracking error. Fundamental law approximations relate it to skill, breadth, and transfer coefficient under strong independence and stationarity assumptions.

Constraints reduce the transfer from signal to portfolio. Active share measures weight deviation but ignores covariance. Report both holdings-based and risk-based measures.

29.7 Problems

Exercise 29.1 (Core: ERC)**.** Derive first-order conditions of the log-barrier risk-budgeting objective and show that risk contributions are proportional to b_i after scaling.

Exercise 29.2 (Core: surplus)**.** Derive the minimum-variance asset hedge of a random liability using covariance projection.

Exercise 29.3 (Advanced: drawdown state)**.** Formulate a dynamic program for terminal utility with a maximum-drawdown constraint. Specify all state variables.

Chapter 30

Backtesting and Strategy Research

Learning objectives. Build a point-in-time backtest, avoid data leakage, quantify costs and uncertainty, and judge a strategy as a research claim rather than a chart.

30.1 The event-driven timeline

At each decision time:

1. ingest only records whose publication timestamp has passed;
2. update features using training-only transformations;
3. estimate or update the model;
4. freeze the target portfolio;
5. simulate orders in a stated execution window;
6. apply fills, costs, financing, borrow, and corporate actions;
7. mark positions and store an immutable audit row.

Vectorized backtests are fast but can conceal timing. An event-driven ledger makes cash, holdings, orders, fills, and value reconcile.

30.2 Biases

Look-ahead bias uses future information directly or through centered filters, revised data, global standardization, or same-bar execution. Survivorship bias excludes failed securities. Selection bias chooses favorable markets or periods. Data-snooping bias chooses a model after examining outcomes. Publication bias hides failures.

Point-in-time universes, delisting returns, publication lags, and a research registry are defenses. No statistical correction fully reconstructs undocumented exploration.

30.3 Performance measurement

For periodic excess returns r_t^e, estimated Sharpe ratio is

$$\widehat{SR} = \frac{\bar{r}^e}{s_r}$$

times an annualization factor under a stated frequency. Serial correlation changes its standard error and invalidates naive square-root scaling of information.

Other metrics include Sortino ratio, maximum drawdown, Calmar ratio, hit rate, skewness, tail ratio, turnover, capacity, beta, factor alpha, and certainty equivalent. Choose predeclared metrics aligned with the strategy.

30.4 Sharpe uncertainty

Under iid normal returns, approximate standard error depends on sample size and Sharpe itself. Non-normality and serial dependence require robust methods. A probabilistic Sharpe ratio compares an observed estimate to a benchmark using estimated skew and kurtosis; a deflated Sharpe also adjusts for selection among trials.

Block bootstrap resamples consecutive observations to preserve local dependence. Stationary bootstrap uses random block lengths. Strategy re-estimation should occur inside each bootstrap if model-fitting uncertainty is part of the question.

30.5 Costs and financing

Net return includes gross asset P&L, spread, commission, impact, borrow fees, dividends on shorts, futures variation margin, funding, FX conversion, and tax if in scope. Apply costs to trades, not average holdings. Use bid and ask where available.

Sensitivity should vary cost multipliers, execution delay, participation caps, and assets under management. A strategy whose conclusion reverses under a small plausible cost change is not robust.

30.6 Benchmarking and attribution

Benchmarks include cash, passive market, equal weight, factor-matched portfolios, and the previous production model. Attribution can separate allocation, selection, timing, factors, carry, convexity, and residual. Attribution is model-dependent and must reconcile algebraically to total return.

30.7 Decision protocol

Before looking at the final test:

- state hypotheses and directional predictions;
- freeze universe, features, estimator, tuning protocol, and costs;
- define rejection and economic materiality thresholds;
- name stress tests and subgroup analyses;
- preserve code and configuration hash.

After the test, report every predeclared result and label new investigations exploratory. A negative result with a clear mechanism test is valuable research.

30.8 Problems

Exercise 30.1 (Core: timing audit)**.** Draw the complete timeline for a monthly accounting signal traded at the next month's open. Include fiscal period end, filing, vendor ingestion, signal, and fill.

Exercise 30.2 (Core: ledger)**.** Write identities reconciling starting cash, trades, fees, distributions, financing, and ending portfolio value.

Exercise 30.3 (Research)**.** Reproduce a published factor using point-in-time inputs. Compare gross, quoted-spread, and impact-adjusted results, then test on an untouched geography.

Part IX

Numerical and Statistical Computing

Chapter 31

Numerical Accuracy and Optimization Algorithms

Learning objectives. Control floating-point and discretization error, solve nonlinear equations and optimization problems, and report numerical uncertainty alongside statistical uncertainty.

31.1 Error taxonomy

Numerical work contains:

- model error from an imperfect representation of the world;
- statistical error from finite data;
- discretization error from time, space, or quadrature grids;
- iteration error from stopping an algorithm;
- floating-point error from finite precision;
- implementation error from incorrect code.

Increasing Monte Carlo paths reduces sampling error but not model or time-step bias. An optimizer tolerance smaller than uncertainty in input quotes adds digits, not information.

31.2 Floating-point arithmetic

Machine numbers have finite relative precision. Subtracting nearly equal quantities causes catastrophic cancellation. Summing values with different magnitudes loses small terms; compensated summation helps. Exponentials overflow and underflow; log-sum-exp stabilizes

$$\log \sum_i e^{x_i} = m + \log \sum_i e^{x_i - m}, \quad m = \max_i x_i.$$

Comparing floats for exact equality is usually inappropriate. A tolerance should reflect scale and the numerical problem, not a universal constant.

31.3 Root finding

Bisection on a continuous f with opposite signs at a, b converges robustly and halves interval each iteration. Newton's method

$$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}$$

has local quadratic convergence near a simple root but can diverge or leave the domain. Brent's method combines bracketing reliability and interpolation speed.

Implied volatility inversion is monotone for ordinary European options with positive vega. First enforce no-arbitrage price bounds, bracket volatility, and handle near intrinsic options where vega is tiny.

31.4 Gradient-based optimization

Gradient descent is

$$x_{k+1} = x_k - \eta_k \nabla f(x_k).$$

Newton uses Hessian curvature:

$$x_{k+1} = x_k - \nabla^2 f(x_k)^{-1} \nabla f(x_k).$$

Quasi-Newton methods approximate the inverse Hessian from gradients. Constrained solvers use active-set, interior-point, or sequential quadratic programming ideas.

A convex problem has powerful global guarantees; a calibrated stochastic-volatility objective is often nonconvex. Multiple starts, parameter transformations, gradient checks, and profile plots are necessary.

31.5 Automatic differentiation and checks

Automatic differentiation applies the chain rule to program operations and can produce exact derivatives up to floating point. It differentiates the implemented program, including any mistake. Compare analytic, automatic, finite-difference, and complex-step derivatives on small cases.

Central finite difference has truncation error $O(h^2)$ but rounding error grows as h shrinks. A convergence plot across h reveals the useful region.

31.6 Linear systems

Never form a matrix inverse merely to solve $Ax = b$. Use a factorization: Cholesky for positive definite matrices, QR for least squares, SVD for rank deficiency, and sparse methods for large structured systems. Report condition numbers or residuals.

31.7 Problems

Exercise 31.1 (Core: bisection)**.** Prove the interval-error bound after n bisection steps and determine iterations needed for a desired implied-volatility tolerance.

Exercise 31.2 (Core: cancellation)**.** Derive a numerically stable formula for the smaller root of a quadratic when $b^2 \gg 4ac$.

Exercise 31.3 (Advanced: gradient audit)**.** Design a derivative test for a constrained portfolio objective and distinguish absolute from relative tolerance near zero.

Chapter 32

Monte Carlo and Variance Reduction

Learning objectives. Simulate stochastic models, quantify standard errors and time-step bias, use variance reduction, and price path-dependent claims with reproducible diagnostics.

32.1 Plain Monte Carlo

For $V = \mathbb{E}[g(X)]$, iid samples give

$$\widehat{V}_N = \frac{1}{N}\sum_{i=1}^{N} g(X_i), \qquad \widehat{\mathrm{se}} = \frac{s_g}{\sqrt{N}}.$$

The central limit theorem yields an approximate interval $\widehat{V}_N \pm z_{1-\alpha/2}\widehat{\mathrm{se}}$. Halving standard error requires roughly four times as many independent paths.

Pseudorandom seeds make a run repeatable. Separate random streams prevent small code changes from silently changing every comparison. Common random numbers improve precision of differences between scenarios.

32.2 Simulating diffusions

Euler–Maruyama for $\mathrm{d}X_t = b(X_t)\,\mathrm{d}t + a(X_t)\,\mathrm{d}W_t$ is

$$X_{n+1} = X_n + b(X_n)\Delta t + a(X_n)\sqrt{\Delta t}Z_{n+1}.$$

Under regularity, strong convergence order is typically $1/2$ and weak order 1. Milstein adds derivative terms and can achieve strong order 1 in scalar settings.

For geometric Brownian motion use its exact transition. Naive Euler can produce negative prices. For CIR variance, full truncation, quadratic-exponential, or exact transition schemes handle boundaries more reliably.

Definition 32.1 (Strong and weak error)**.** Strong error measures pathwise approximation such as $\mathbb{E}[\left|X_T - \widehat{X}_T\right|]$. Weak error measures expectation error $\left|\mathbb{E}[g(X_T)] - \mathbb{E}[g(\widehat{X}_T)]\right|$.

Pricing a European payoff needs weak accuracy; hedging, barriers, and exposure paths can require stronger path fidelity.

32.3 Antithetic variables

If Z is standard normal, simulate $g(Z)$ and $g(-Z)$ and average. Variance is

$$\frac{1}{2}\operatorname{Var}(g(Z)) + \frac{1}{2}\operatorname{Cov}(g(Z), g(-Z))$$

relative to a pair-based convention. Negative covariance reduces error. Antithetics are especially effective for monotone payoffs under symmetric shocks.

32.4 Control variates

Let Y have known mean μ_Y. Use

$$\widehat{V}_c = \overline{X} - c(\overline{Y} - \mu_Y).$$

The variance-minimizing coefficient is $c^\star = \operatorname{Cov}(X,Y)/\operatorname{Var}(Y)$ and reduction factor is $1 - \rho_{XY}^2$. Estimate c on an independent pilot or account for its estimation.

For an arithmetic Asian call, a geometric Asian or discounted terminal stock is a natural control.

32.5 Importance sampling

If rare loss A is seldom observed under p, sample from q and weight:

$$\mathbb{P}_p(A) = \mathbb{E}_q\left[\mathbf{1}_A(X)\frac{p(X)}{q(X)}\right].$$

The support of q must cover relevant p states. Poor likelihood ratios can have infinite variance even when the estimator is unbiased. Exponential tilting chooses a proposal that makes the rare event less rare.

32.6 Quasi-Monte Carlo

Low-discrepancy sequences fill the unit cube more uniformly than random points. Randomized Sobol sequences allow error estimation. Effective dimension matters: Brownian bridge or principal-component construction assigns early coordinates to important variation.

Classical N^{-1}-like convergence requires smoothness and dimension conditions; discontinuous barriers and exercise rules reduce gains. Scrambling and replicate randomizations provide honest empirical errors.

32.7 Nested simulation

CVA, capital, and risk of future value may require outer market scenarios and inner conditional valuations. Cost grows multiplicatively. Regression proxies, least-squares Monte Carlo, multilevel Monte Carlo, and allocation of inner paths reduce cost. Proxy validation must focus on tails and boundary states, not only average error.

32.8 Problems

Exercise 32.2 (Core: control variate)**.** Derive $c^\star$ and the variance-reduction factor.

Exercise 32.3 (Core: discretization study)**.** Design a grid refinement study that separately estimates Monte Carlo error and weak time-step bias.

Exercise 32.4 (Advanced: importance sampling)**.** For a Gaussian tail probability, derive exponential mean shifting and the likelihood ratio. Explore variance as the shift changes.

Chapter 33

Finite Differences, Trees, and Transform Methods

Learning objectives. Discretize pricing PDEs, analyze stability and convergence, use Fourier methods, and choose an algorithm that respects payoff and model structure.

33.1 Finite-difference grids

For Black–Scholes PDE, transform or discretize time t_n and price S_j. Central spatial differences approximate

$$V_S \approx \frac{V_{j+1} - V_{j-1}}{2\Delta S}, \qquad V_{SS} \approx \frac{V_{j+1} - 2V_j + V_{j-1}}{\Delta S^2}.$$

Explicit time stepping computes the next layer directly but has a stability restriction. Implicit stepping solves a tridiagonal system and is unconditionally stable in a norm sense. Crank–Nicolson averages explicit and implicit operators, giving second-order time accuracy for smooth solutions.

Nonsmooth terminal payoffs can make Crank–Nicolson oscillate. Rannacher smoothing uses initial implicit half steps. Grid boundaries should follow asymptotic payoff behavior and be stress-tested.

33.2 Consistency, stability, convergence

Theorem 33.1 (Lax equivalence, informal). *For a well-posed linear initial-value problem and a consistent finite-difference scheme, stability is equiva-*

lent to convergence.

Consistency means local truncation error vanishes. Stability controls amplification of perturbations. Convergence means numerical solution approaches the true solution. Checking only a visually smooth surface proves none of them.

33.3 American complementarity

An American option solves a linear complementarity problem:

$$\max\{h - V,\ V_t + \mathcal{L}V - rV\} = 0$$

under a sign convention. Projected SOR, penalty methods, and policy iteration enforce the obstacle. Verify value exceeds intrinsic, exercise boundary behaves sensibly, and European limits are recovered.

33.4 Lattices

Binomial and trinomial lattices discretize state transitions while preserving recombination. Trinomial trees can match drift and variance with greater flexibility. For multiple factors, tree size grows quickly and correlation matching can create negative transition probabilities.

Trees naturally handle early exercise and discrete dividends. They converge nonmonotonically for some strikes; Richardson extrapolation and smoothing help.

33.5 Fourier pricing

If a model has known characteristic function of log price, option values can be computed by Fourier inversion. Damping makes a call transform integrable. The Carr–Madan style transform applies FFT across a strike grid:

$$C(k) = \frac{\mathrm{e}^{-\alpha k}}{\pi} \int_0^\infty \mathrm{Re}\left[\mathrm{e}^{-iuk}\Psi(u;\alpha)\right] \, \mathrm{d}u.$$

FFT is fast for many evenly spaced log strikes but couples strike spacing and frequency cutoff. Aliasing, truncation, damping, and interpolation must be checked. Direct quadrature can be preferable for a few options.

33.6 Numerical quadrature

Trapezoidal and Simpson rules approximate regular finite integrals. Gaussian quadrature chooses nodes and weights exact for polynomial families under a weight. Adaptive quadrature allocates evaluations where estimated error is high.

Singular integrands, oscillation, and infinite domains require transformations or specialized schemes. Report absolute error against a closed form when one exists.

33.7 Algorithm selection

Method	Strength	Main limitation
Closed form	speed, transparency, Greeks	narrow assumptions
Tree	early exercise, discrete events	dimension and convergence
Finite difference	PDEs, free boundaries	state dimension, grids
Monte Carlo	high dimension, paths	early exercise, noisy Greeks
Fourier	many strikes, affine models	transform and grid errors
Quadrature	low-dimensional accuracy	dimension and singularities

A production library prices benchmarks by two independent methods. Agreement within tolerance is a control, not redundant work.

33.8 Problems

Exercise 33.2 (Core: explicit stability)**.** Derive coefficients of an explicit Black–Scholes scheme and a sufficient condition for nonnegative weights.

Exercise 33.3 (Core: boundary)**.** Specify lower and upper boundaries for European calls and puts with dividends.

Exercise 33.4 (Advanced: FFT)**.** Derive the damped call transform and analyze how log-strike spacing relates to the frequency grid.

Chapter 34

Statistical Computing and Reproducible Software

Learning objectives. Design auditable quantitative software, test mathematical properties, preserve data lineage, and make research reproducible across machines and time.

34.1 Architecture

Separate:

- immutable source-data ingestion;
- validation and canonical schemas;
- pure mathematical functions;
- stateful simulation or backtest engines;
- experiment configuration;
- reports and generated artifacts.

Pure functions are easy to test. Domain objects should carry units, currency, calendar, and convention rather than relying on hidden globals. A pricing function should not silently download data.

34.2 Testing layers

Unit tests cover small functions and edge cases. Property tests verify invariants: call price increases with spot, discount factors are positive, a covariance estimate is positive semidefinite, and portfolio weights reconcile. Regression tests freeze trusted examples. Integration tests exercise complete data-to-report flows.

Numerical tests use tolerances tied to a benchmark and grid. Tests should include zero time, zero volatility limits, deep moneyness, negative rates when supported, missing dates, and invalid inputs.

34.3 Data contracts

A canonical dataset records:

- provider and exact endpoint;
- instrument identifier and mapping history;
- event, publication, and ingestion timestamps with timezone;
- units, currency, frequency, adjustment, and missing-value codes;
- download time, query, checksum, and license note;
- transformations from raw to analytical columns.

Never overwrite raw files. A transformation manifest makes derived data reproducible. CSV is portable but loses types; Parquet preserves schemas and is efficient. Database constraints can enforce uniqueness and temporal validity.

34.4 Environments and randomness

Pin direct and transitive dependencies for a released result. Record interpreter, operating system, BLAS, and compiler where numerical differences matter. Containers improve portability but do not guarantee permanent access to external data.

A random seed initializes a generator. Parallel simulations need independent substreams, not repeated identical seeds. Store seed and generator family in experiment metadata.

34.5 Logging and experiment manifests

An experiment manifest includes configuration, Git commit, input checksums, start time, code version, environment, and output checksums. Logs should report decisions and anomalies without leaking credentials or licensed data.

Generated tables should come directly from saved results. Manual transcription breaks lineage. LaTeX can input CSV-derived tables or generated fragments, but the generation step must be deterministic and documented.

34.6 Code review for quantitative models

Review mathematical specification separately from software behavior. Ask:

1. Do equations, conventions, and units match the design?
2. Are array shapes, labels, and time axes explicit?
3. Are invalid domains rejected rather than silently coerced?
4. Are calibration residuals and solver status checked?
5. Are data and model timestamps point-in-time?
6. Are independent benchmarks and sensitivities present?
7. Are performance optimizations behavior-preserving?

34.7 Problems

Exercise 34.1 (Core: property tests)**.** Specify property-based tests for Black–Scholes call price, historical Expected Shortfall, and a covariance shrinkage estimator.

Exercise 34.2 (Core: manifest)**.** Design a JSON experiment manifest for a yield-curve study with raw-source checksums and point-in-time settings.

Exercise 34.3 (Advanced: reproducibility)**.** Explain why an unpinned vendor endpoint can make a computationally deterministic script scientifically irreproducible.

Chapter 35

Modern Machine Learning for Quantitative Finance

Learning objectives. Formulate machine learning as statistical decision theory, understand kernels and neural networks, build probabilistically calibrated sequence models, use reinforcement learning cautiously, and evaluate adaptive systems under financial nonstationarity.

35.1 Learning as risk minimization

Let $Z = (X, Y) \sim P$ and prediction rule f lie in class $\mathcal{F}$. Population risk is

$$\mathcal{R}(f) = \mathbb{E}_P[\ell(Y, f(X))].$$

Empirical risk minimization chooses

$$\widehat{f} \in \arg\min_{f\in\mathcal{F}} \left\{ \frac{1}{n} \sum_{i=1}^{n} \ell(Y_i, f(X_i)) + \lambda \mathcal{J}(f) \right\}.$$

The loss defines the statistical target. Squared loss targets conditional mean; absolute loss targets median; pinball loss targets a conditional quantile; log loss targets a conditional distribution. A trading objective may instead depend on cost-aware utility, but directly optimizing a noisy backtest makes selection bias especially severe.

Expected generalization error decomposes conceptually into irreducible noise, approximation error, estimation error, and distribution shift. Increasing model capacity can reduce approximation error and increase estimation error. Regularization, data scale, and stable inductive structure govern the balance.

35.2 Features and invariances

Prices in currency units are usually nonstationary and arbitrary under stock splits. Returns, spreads, ratios, ranks, and curve coordinates often encode more stable objects, but no transformation guarantees stationarity. A feature definition must include lookback window, missingness rule, publication lag, cross-sectional universe, and training-only normalization.

Useful inductive biases include:

- permutation invariance across an unordered asset set;
- translation invariance across time in local convolutions;
- monotonicity where economic dominance requires it;
- no-arbitrage constraints on price surfaces;
- equivariance to currency or notional scaling;
- sparsity or low rank for high-dimensional panels.

Encoding a known invariance reduces sample burden and prevents economically impossible extrapolation.

35.3 Kernel methods

A positive definite kernel $k(x, x')$ represents an inner product $\langle \phi(x), \phi(x') \rangle$ in a possibly infinite-dimensional feature space. Kernel ridge regression solves

$$\min_{f \in \mathcal{H}_k} \sum_{i=1}^{n} (y_i - f(x_i))^2 + \lambda \|f\|_{\mathcal{H}_k}^2 .$$

Theorem 35.1 (Representer theorem)**.** *For a broad class of losses depending on training evaluations and a strictly increasing penalty in RKHS norm, a minimizer has form*

$$\widehat{f}(x) = \sum_{i=1}^{n} \alpha_i k(x_i, x).$$

Proof idea. Decompose any f into a component in the span of $\{k(x_i,\cdot)\}_{i=1}^n$ and an orthogonal component. The latter is zero at every training point by the reproducing property, so it cannot improve fit and can only increase the norm penalty. □

For squared loss, $\hat{\alpha} = (K + \lambda I)^{-1}y$. Kernel methods are strong for structured small-to-medium data but scale cubically without approximation. Kernel and hyperparameter selection must occur within time-respecting validation.

35.4 Neural networks

A feed-forward network composes affine maps and nonlinear activations:

$$h^{(0)} = x, \qquad h^{(\ell)} = \sigma(W_\ell h^{(\ell-1)} + b_\ell), \qquad \hat{y} = W_L h^{(L-1)} + b_L.$$

Backpropagation applies reverse-mode automatic differentiation to compute the gradient of loss with respect to every parameter. Stochastic gradient methods update using minibatches; momentum and adaptive scaling change optimization dynamics.

Universal approximation results state that sufficiently wide networks can approximate broad classes of continuous functions on compact sets. They do not say finite financial samples identify the correct function, that gradient descent finds it, or that it extrapolates across regimes.

Weight decay, dropout, early stopping, augmentation, and architectural constraints regularize differently. Batch normalization can leak cross-sectional or temporal information if a batch mixes future observations; inference statistics and panel construction require explicit design.

35.5 Sequence models

A recurrent state model updates $h_t = g_\theta(h_{t-1}, x_t)$ and emits a forecast. Gated recurrent units address vanishing gradients. Temporal convolutions process fixed receptive fields in parallel.

Self-attention maps queries, keys, and values:

$$\mathrm{Attention}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^\top}{\sqrt{d}}\right)V.$$

Causal masks prevent a position from attending to future tokens. Positional or time encodings distinguish order and irregular spacing.

Financial panels add complications: changing universes, mixed frequencies, release lags, missing-not-at-random data, and corporate events. A transformer with a causal mask can still leak if input features were standardized globally or fundamentals were aligned to fiscal period rather than publication.

35.6 Cross-sectional learning

At each date, a model can rank assets by a score $s_{i,t}$. Cross-sectional losses include pairwise ranking and listwise objectives. Portfolio construction converts scores into weights under neutrality, risk, turnover, and capacity:

$$w_t = \arg\max_w \left\{ s_t^\top w - \frac{\gamma}{2} w^\top \widehat{\Sigma}_t w - C(w - w_{t-1}) \right\}.$$

This separation permits evaluation of forecast quality and portfolio translation. End-to-end optimization can be useful, but it entwines model learning with an estimated risk and cost layer.

Panel validation blocks by time, not individual rows. Otherwise the model sees the same market shock in training assets and validation assets. A geography or asset-class holdout tests cross-sectional transport but does not replace a future-time test.

35.7 Probabilistic forecasts and calibration

A probabilistic model should be sharp and calibrated. For event probability p_t, calibration means $\mathbb{P}(Y_t = 1 \mid p_t = p) = p$ in an ideal population sense. Reliability diagrams and Brier decomposition diagnose calibration; log loss heavily penalizes confident errors.

Quantile forecasts should attain coverage and respond to state. A collection of independently estimated quantiles can cross; monotone parameterizations or rearrangement fix ordering. Distributional networks can output location, scale, mixture weights, or a normalizing flow.

Conformal prediction constructs finite-sample marginal coverage under exchangeability. Time dependence and distribution shift break the basic guarantee; weighted, blockwise, or online variants add assumptions rather than magically restoring it.

35.8 Uncertainty

Aleatoric uncertainty describes outcome randomness conditional on inputs. Epistemic uncertainty describes limited knowledge of the predictive rule. Ensembles, Bayesian neural approximations, bootstrap, and last-layer methods estimate parts of epistemic uncertainty, but common model misspecification can make every ensemble member agree and be wrong.

For a portfolio decision, propagate predictive scenarios through constraints and costs. Confidence in a score is not the same as confidence in net P&L because covariance, impact, and crowding are additional uncertain layers.

35.9 Online learning and drift

Online learning receives data sequentially, predicts, observes loss, and updates. Regret relative to comparator f is

$$\mathrm{Regret}_T(f) = \sum_{t=1}^{T} \ell_t(f_t) - \sum_{t=1}^{T} \ell_t(f).$$

Sublinear regret means average loss approaches the best fixed comparator in hindsight under the theorem's assumptions. Markets may change so a dynamic comparator or variation budget is more relevant.

Drift monitoring includes feature distributions, residuals, calibration, turnover, and economic exposures. A detected change triggers a predeclared action: investigate, reduce risk, retrain, or retire. Repeatedly changing the system after losses without an audit creates an adaptive overfit.

35.10 Reinforcement learning

A Markov decision process has state s_t, action a_t, transition law, reward r_t, and discount γ. For policy π,

$$V^\pi(s) = \mathbb{E}^\pi\left[\sum_{k=0}^{\infty} \gamma^k r_{t+k} \mid s_t = s\right].$$

The Bellman optimality equation is

$$V^\star(s) = \max_a \mathbb{E}[r(s, a, S') + \gamma V^\star(S') \mid s, a].$$

In trading, the true state is partially observed, actions affect costs and perhaps prices, rewards have long horizons, and historical data were generated by another policy. Offline RL cannot freely test counterfactual actions absent support. Importance sampling can have explosive variance; model-based simulation inherits simulator error.

A safe design begins with a transparent stochastic-control benchmark, hard risk constraints, conservative off-policy evaluation, and stress outside the training distribution. Reward must include costs, financing, inventory, and risk; a raw P&L reward invites leverage.

35.11 Generative models and synthetic scenarios

Variational autoencoders, adversarial networks, diffusion models, and autoregressive models can generate return paths or order-book states. Fidelity should be evaluated on marginal tails, serial dependence, cross-asset dependence, conditional dynamics, and decision-relevant stress.

Synthetic data are useful for privacy, augmentation, and controlled experiments. They are not observed evidence and cannot validate the assumptions used to generate them. Rare crises are precisely where flexible generators have least information.

35.12 Evaluation protocol

A modern ML study should report:

1. a simple linear and economic benchmark;
2. model capacity, all tuned ranges, and number of research trials;
3. time and cross-sectional split with purge and embargo if needed;
4. training curves and seed variability;
5. proper statistical loss and calibration;
6. net decision performance with costs and constraints;
7. factor, regime, and subgroup attribution;
8. perturbation, missingness, delay, and cost stress;
9. inference-time latency, stability, and monitoring plan.

Complexity is justified by stable incremental evidence, not by the sophistication of the architecture.

35.13 Problems

Exercise 35.2 (Core: representer theorem)**.** Complete the orthogonal-decomposition proof of the representer theorem and derive kernel-ridge coefficients.

Exercise 35.3 (Core: attention leakage)**.** List every way a causally masked sequence model can still use future financial information through preprocessing, data alignment, and universe construction.

Exercise 35.4 (Advanced: probabilistic portfolio)**.** Design a model that forecasts a joint return distribution and a policy that optimizes Expected Shortfall subject to turnover. Specify a proper forecast score and a separate decision score.

Exercise 35.5 (Advanced: offline control)**.** Formulate inventory liquidation as an MDP. State what historical-policy support is needed for off-policy evaluation and how a model error could create a fictitious profitable action.

Exercise 35.6 (Research)**.** Compare ridge, gradient-boosted trees, and a small causal transformer on one predeclared target. Use identical point-in-time features, nested walk-forward tuning, multiple seeds, cost-aware portfolio mapping, and a locked final period.

Part X

Reproducible Quantitative Research

Chapter 36

Research Design and Data Provenance

Learning objectives. Turn an economic idea into a falsifiable, reproducible quantitative study; choose authoritative data; and preserve point-in-time lineage from source to conclusion.

36.1 The research question

A useful question names a population, horizon, information set, target, and loss. For example: *Does an expanding-window term-structure model improve the one-month log score of recession probabilities relative to a constant-probability benchmark, using only releases available at each month end?* This is testable. *Can AI predict markets?* is not yet a research design.

Before coding, write:

- economic mechanism and competing explanation;
- estimand or forecast target;
- observation and decision unit;
- sample, universe, and inclusion rules;
- information-availability and execution timeline;
- benchmark and loss;
- planned robustness and failure criteria.

36.2 A hierarchy of evidence

Descriptive evidence summarizes a defined sample. Predictive evidence demonstrates performance on genuinely new observations. Causal evidence requires an intervention or identification strategy. Structural evidence estimates an economic model under maintained restrictions. A backtest is not automatically any one of these; its claim must be named.

Theory disciplines which variables and signs matter. Exploratory visualization finds errors and hypotheses. A validation set selects models. A locked test estimates generalization once. Reusing the test turns it into training data.

36.3 Primary and secondary data

Prefer a primary producer for definitions and release metadata:

Domain	Source	Research use and caution
U.S. macro	FRED and ALFRED, `https://fred.stlouisfed.org/docs/api/fred/`	API access to observations and vintages; inspect underlying agency, units, seasonal adjustment, revisions, and release time.
U.S. yields	U.S. Treasury, `https://home.treasury.gov/resource-center/data-chart-center/interest-rates`	Official daily par curves and methodology; par yields are interpolated curve values, not transaction yields or zero rates.
Monetary policy	Federal Reserve and New York Fed, `https://www.newyorkfed.org/markets`	Rates, operations, reference rates, and SOMA material; align business dates and publication timestamps.

U.S. filings	SEC EDGAR APIs, https://www.sec.gov/search-filings/edgar-application-programming-interfaces	Company facts and filings; comply with fair-access policy, map identifiers, preserve filing and acceptance times, and handle amendments.
Factors	Kenneth French Data Library, https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html	Research factor benchmarks with construction notes; methodology can change and published portfolios are not a substitute for point-in-time reconstruction.
Options	Cboe market statistics, https://www.cboe.com/data/market_statistics/	Official exchange statistics and selected downloads; detailed historical quotes and trades may require licensed products.
Europe	ECB Data Portal, https://data.ecb.europa.eu/	Monetary, financial, yield, and macro series with structured metadata; inspect frequency, seasonal adjustment, and area composition.
Global	BIS Data Portal, https://data.bis.org/	Credit, banking, liquidity, debt, FX, and property series; cross-country definitions and breaks require care.

Convenience services are useful for prototypes but may change schemas, adjustments, symbols, and terms. The repository's market-price pipeline uses Yahoo Finance through `yfinance` as a convenient adjusted-price feed and labels it as such. Claims requiring production-quality price history should be repeated with licensed, point-in-time data.

36.4 Data due diligence

For every field, answer:

1. Who creates the underlying observation?

2. What exactly is measured and in what units?
3. Is it a level, rate, index, yield, return, or annualized transformation?
4. When did the event occur, when was it published, and when was it revised?
5. What do missing and preliminary values mean?
6. Did methodology, coverage, identifier, or calendar change?
7. May the data be stored and redistributed?

Automated checks include uniqueness, monotonic timestamps, valid domains, missingness, outliers, cross-field identities, and checksums. A large move is quarantined for review, not automatically deleted.

36.5 Revisions and vintages

Macro observations often evolve from advance to revised estimates. Let $x_{t|v}$ be the estimate for economic period t as recorded at vintage v. Forecasts made at v may use only $x_{t|v}$ for releases available then. A revision study can treat $x_{t|v_{\text{final}}} - x_{t|v_{\text{initial}}}$ as an outcome and examine whether revisions are news or noise.

Cross-sectional fundamentals also have point-in-time versions. Filing date, vendor processing, and restatement history determine availability. Lagging every field by an arbitrary fixed number of months is a crude defense, not full lineage.

36.6 Reproducible outputs

Each project should emit:

- a machine-readable configuration;
- raw or cached data with allowed provenance;
- validation report;
- tidy analytical data;

- model estimates and predictions;
- tables and figures generated from saved results;
- an interpretation and model-risk memo;
- a command that rebuilds everything.

36.7 Problems

Exercise 36.1 (Core: provenance)**.** Choose one FRED series and write a complete data contract including underlying producer, units, frequency, seasonal adjustment, vintage, missing codes, and release availability.

Exercise 36.2 (Core: test protocol)**.** Pre-register a market forecast including all choices that could otherwise be tuned after observing test returns.

Exercise 36.3 (Advanced: revision triangle)**.** Design a database table for real-time macro vintages and a query that reconstructs the information set at any historical date.

Chapter 37

Empirical Laboratories

Learning objectives. Integrate the theory in end-to-end projects spanning regime risk, the yield curve, factors, options, and causal event studies.

37.1 Laboratory I: cross-asset regime risk

Question

How do equity, duration, gold, and volatility exposures behave across latent or observable volatility regimes, and does volatility targeting improve the full distribution after costs?

Mathematics

Let return vector r_t have conditional model

$$r_t \mid S_t = k \sim t_{\nu_k}(\mu_k, \Sigma_k), \qquad \mathbb{P}(S_t = j \mid S_{t-1} = i) = p_{ij}.$$

Estimate filtered regime probabilities, not smoothed probabilities, for decisions. Compare with a transparent observable regime based on lagged market volatility.

Portfolio exposure is

$$w_t = \min\left(w_{\max}, \frac{\sigma^\star}{\widehat{\sigma}_{t|t-1}}\right) w_0.$$

Test whether scaling changes volatility, Expected Shortfall, drawdown, turnover, and factor exposure. A lower drawdown is not evidence of alpha if average exposure fell.

Protocol

Use adjusted total-return proxies with explicitly documented limitations. Split by time. Fit all regime thresholds and covariance models in the expanding training sample. Charge spread and nonlinear stress costs. Bootstrap blocks of strategy returns and repeat under alternative start dates.

Failure modes

Smoothed-state leakage, VIX treated as an investable total-return asset, constant transaction costs in crises, and synthetic fallback data presented as empirical evidence are prohibited. The repository labels fallback output in provenance.

37.2 Laboratory II: Treasury curve factors and forecasting

Question

How much yield-curve variation is described by level, slope, and curvature, and do Nelson–Siegel dynamics forecast yields or bond excess returns out of sample?

Mathematics

At each date, fit

$$y_t(\tau) = \beta_t^{\mathsf{T}} \ell(\tau; \lambda) + \varepsilon_t(\tau)$$

by weighted least squares. Compare the fitted factors with PCA. Model factors by VAR(1), random walk, and shrinkage VAR. Map forecast yields into zero-coupon price changes and duration-scaled returns using an explicit par-to-zero approximation or a proper bootstrap.

Protocol

Use official Treasury daily par yields or documented FRED constant-maturity series. Do not call them zero rates. Freeze λ in training or tune it within nested validation. Forecast recursively. Evaluate yield RMSE by maturity, directional accuracy, and a bond-pricing loss. Account for the Treasury methodology break in December 2021.

Extensions

Estimate a state-space dynamic Nelson–Siegel model; include macro factors; decompose forecast errors by level, slope, and curvature; test whether accuracy survives post-2020 and methodology-change subsamples.

37.3 Laboratory III: factor replication and robust allocation

Question

Can public research factors explain cross-asset portfolio returns, and does factor-aware covariance shrinkage improve an implementable allocation?

Mathematics

Estimate

$$r_t - r_{f,t}\mathbf{1} = \alpha + Bf_t + \varepsilon_t$$

with HAC inference. Construct $\widehat{\Sigma} = B\widehat{\Sigma}_f B^\top + \widehat{D}$, shrink it toward a diagonal or constant-correlation target, and solve constrained minimum variance.

Protocol

Download factor returns with their current construction notes and archive the file checksum. Align monthly periods, percentages, and risk-free convention. Use only past data for factor selection and covariance. Compare equal weight, sample covariance, factor covariance, and shrinkage after turnover.

Interpretation

A factor alpha is not skill unless benchmark choice was predeclared and holdings are implementable. Factor exposure can change across regimes. Report active risk and drawdown alongside Sharpe.

37.4 Laboratory IV: arbitrage-aware option surface

Question

Can a cleaned option chain produce a smooth risk-neutral density and stable local volatility without static arbitrage?

Mathematics

Infer forwards from put–call parity, filter quotes with executable bounds, and fit call prices subject to decreasing and convex strike constraints. Recover

$$f_T^{\mathbb{Q}}(K) = P(0,T)^{-1}\partial_{KK}C(K,T).$$

Compare spline, parametric total-variance, and constrained quadratic fits.

Protocol

Use synchronized bid and ask, not last prices. Record underlying, rates, dividends, exercise style, multiplier, and timestamps. Fit mid prices while penalizing errors outside bid–ask. Withhold strikes and maturities. Stress numerical derivatives.

Extensions

Calibrate Heston; compare price residuals with hedge residuals; study the risk-neutral tail over time; never interpret it as a physical crash probability without an explicit risk-premium model.

37.5 Laboratory V: causal policy event study

Question

What is the response of rates, equities, FX, and credit to an identified monetary policy surprise rather than to the endogenous policy decision?

Design

Use a narrow window around scheduled announcements and a surprise measured from an appropriate high-frequency futures instrument. Separate target-rate and path components by principal components of the futures curve. Regress intraday asset changes on surprises, then use local projections for lower-frequency responses.

Threats

Central-bank information effects, overlapping news, time-zone mismatch, contract roll, anticipated announcements, heteroskedasticity, and small event count all matter. Report event-level scatterplots and leave-one-event-out sensitivity.

37.6 Problems

Exercise 37.1 (Core: project map)**.** For each laboratory, list the theorem, estimator, numerical method, validation design, and decision metric it uses.

Exercise 37.2 (Advanced: independent replication)**.** Choose one laboratory and specify an independent data source or market that would provide the strongest falsification test.

Chapter 38

From Student Project to Research Portfolio

Learning objectives. Communicate advanced quantitative work with clarity, build a coherent portfolio of evidence, and identify a path from foundations to independent research.

38.1 What a strong repository demonstrates

A serious financial-engineering portfolio demonstrates:

- mathematical fluency through derivations and proofs;
- statistical discipline through identification and uncertainty;
- market knowledge through conventions and execution;
- software quality through tests and reproducibility;
- judgment through limitations and negative results;
- communication through readable reports and figures.

A large number of shallow notebooks is weaker than a small number of complete research pipelines. Each flagship project should connect a theory chapter to a data contract, code, tests, generated results, and a model-risk memo.

38.2 Suggested sequence

1. **Foundation.** Implement returns, discounting, Black–Scholes, bond analytics, optimization, and statistical tests with unit and property tests.
2. **Replication.** Reproduce a textbook result and one published empirical result, including differences.
3. **Extension.** Change an assumption, data source, market, or validation period for a falsification test.
4. **Synthesis.** Combine forecasting, portfolio construction, costs, and risk in an end-to-end decision.
5. **Original question.** Pre-register a mechanism and test it on point-in-time data.

38.3 Writing a research report

The report structure is:

1. abstract with question, method, result, and limitation;
2. economic motivation and falsifiable hypotheses;
3. data provenance and sample construction;
4. mathematical model and identification assumptions;
5. estimator and numerical implementation;
6. predeclared validation and benchmarks;
7. results with uncertainty and economic magnitude;
8. robustness, stress, and negative findings;
9. limitations and next experiment;
10. reproducibility appendix.

Put derivations where a reader can audit them. Put large robustness tables in an appendix. A figure title states the finding; axes state units; caption states sample and construction.

38.4 Reading research

For every paper, identify:

- the claim and counterfactual;
- the economic mechanism;
- the key equation and its assumptions;
- data availability and selection;
- variation that identifies the parameter;
- uncertainty and multiple testing;
- whether implementation resembles a tradable decision;
- the strongest alternative explanation.

Reproduce a central table before extending the paper. Disagreement is useful when it can be localized to data, timing, estimator, or assumption.

38.5 Ethics and epistemic standards

Do not present synthetic data as observed evidence, revised macro history as a real-time forecast, midquote returns as executable fills, or a selected backtest as an untouched test. Do not distribute data beyond its license. Protect credentials and confidential records.

State uncertainty in language. A model estimate is conditional; a p-value is not truth; a scenario is not a probability forecast; a risk-neutral density is not a physical belief. Intellectual honesty is a technical skill.

38.6 Mastery checklist

The reader is ready for independent work when they can:

- prove core pricing and probability results and state hypotheses;
- derive an estimator and its limiting uncertainty;
- implement two independent numerical methods for a benchmark;

- construct point-in-time datasets and audit timestamps;
- design a walk-forward or causal validation;
- decompose portfolio risk and trading costs;
- explain where a model will fail before seeing the result;
- write a reproducible report that another researcher can challenge.

38.7 Capstone

Exercise 38.1 (Capstone research dossier)**.** Complete one flagship laboratory. Deliver a preregistration, data dictionary, mathematical appendix, tested code, immutable predictions, cost-aware evaluation, stress results, model card, and ten-page research paper. Include at least one deliberate falsification test and one negative finding.

Part XI

Reference Material

Appendix A

Notation and Mathematical Reference

A.1 Core notation

Symbol	Meaning
$(\Omega, \mathcal{F}, \mathbb{P})$	physical probability space
$(\mathcal{F}_t)$	information filtration
$\mathbb{Q}$	equivalent pricing measure, with numeraire understood
$\mathbb{E}_t[\cdot]$	conditional expectation given $\mathcal{F}_t$
$X^\top$	transpose of vector or matrix
$A \succeq 0$	positive semidefinite matrix
$\lVert x \rVert_p$	ℓ_p norm
$\mathbf{1}_A$	indicator of event A
$P(t, T)$	time-t price of a zero-coupon payment of one at T
R_t, r_t	simple gross/net return as defined locally; log return is stated
M_{t+1}	stochastic discount factor
$W^{\mathbb{P}}, W^{\mathbb{Q}}$	Brownian motion under physical and pricing measures
$\mathrm{VaR}_\alpha, \mathrm{ES}_\alpha$	loss quantile and tail average at confidence α
$\widehat{\theta}$	estimate; population parameter has no hat
$\Rightarrow, \operatorname{plim}$	convergence in distribution and probability limit

A.2 Matrix identities

For conformable matrices:

$$(AB)^\top = B^\top A^\top, \qquad (AB)^{-1} = B^{-1}A^{-1},$$

$$\frac{\partial}{\partial x}(a^\top x) = a, \qquad \frac{\partial}{\partial x}(x^\top Ax) = (A + A^\top)x.$$

If $A = A^\top$, the last gradient is $2Ax$.

The Sherman–Morrison formula is

$$(A + uv^\top)^{-1} = A^{-1} - \frac{A^{-1}uv^\top A^{-1}}{1 + v^\top A^{-1}u},$$

when denominator is nonzero. Woodbury generalizes:

$$(A + UCV)^{-1} = A^{-1} - A^{-1}U(C^{-1} + VA^{-1}U)^{-1}VA^{-1}.$$

These identities accelerate factor-covariance inversions and clarify shrinkage.

For a block Gaussian covariance, the Schur complement appears in conditional variance:

$$\mathrm{Var}(Y \mid X) = \Sigma_{YY} - \Sigma_{YX}\Sigma_{XX}^{-1}\Sigma_{XY}.$$

A.3 Calculus identities

Chain rule: $D(f \circ g)(x) = Df(g(x))Dg(x)$. Product Itô rule:

$$\mathrm{d}(X_tY_t) = X_t\,\mathrm{d}Y_t + Y_t\,\mathrm{d}X_t + \mathrm{d}\langle X, Y\rangle_t.$$

Leibniz integral differentiation, under domination:

$$\frac{\mathrm{d}}{\mathrm{d}\theta}\int_{a(\theta)}^{b(\theta)} f(x,\theta)\,\mathrm{d}x = f(b,\theta)b' - f(a,\theta)a' + \int_a^b \partial_\theta f(x,\theta)\,\mathrm{d}x.$$

Euler's theorem: if differentiable $f(cx) = c^k f(x)$ for $c > 0$, then $x^\top \nabla f(x) = kf(x)$.

A.4 Convex-analysis identities

The convex conjugate is

$$f^*(y) = \sup_x \{y^\top x - f(x)\}.$$

Fenchel–Young inequality: $f(x) + f^*(y) \geq x^\top y$, with equality when $y \in \partial f(x)$. For $f(x) = \frac{1}{2}x^\top Ax$ with $A \succ 0$, $f^*(y) = \frac{1}{2}y^\top A^{-1}y$.

The subdifferential of $|x|$ is $\{-1\}$ for $x < 0$, $[-1, 1]$ for $x = 0$, and $\{1\}$ for $x > 0$. This kink permits lasso coefficients to be exactly zero.

Appendix B

Distribution and Statistical Formula Sheet

B.1 Selected univariate distributions

Law	Support/density or mass	Mean	Variance
Bernoulli	$p;\ x \in \{0,1\}$	p	$p(1-p)$
Poisson	$\lambda;\ k \in \mathbb{N}_0$	λ	λ
Exponential	$\lambda;\ x \geq 0$	$1/\lambda$	$1/\lambda^2$
Normal	$\mu, \sigma^2;\ x \in \mathbb{R}$	μ	σ^2
Lognormal	$\mu, \sigma^2;\ x > 0$	$\mathrm{e}^{\mu+\sigma^2/2}$	$(\mathrm{e}^{\sigma^2}-1)\mathrm{e}^{2\mu+\sigma^2}$
Student t_ν	$\mathbb{R}$, polynomial tail	0 if $\nu > 1$	$\nu/(\nu-2)$ if $\nu > 2$
Gamma	$a, b;\ x > 0$, rate b	a/b	a/b^2
Beta	$a, b;\ 0 < x < 1$	$a/(a+b)$	$ab/[(a+b)^2(a+b+1)]$

B.2 Normal identities

If $X \sim N(\mu, \sigma^2)$,

$$\mathbb{E}[\mathrm{e}^{tX}] = \mathrm{e}^{t\mu + \frac{1}{2}t^2\sigma^2}.$$

For $Z \sim N(0,1)$,

$$\mathbb{E}[Z\mathbf{1}_{\{Z>a\}}] = \phi(a), \qquad \mathbb{E}[Z \mid Z > a] = \frac{\phi(a)}{1-\Phi(a)}.$$

For jointly Gaussian (X, Y),

$$\mathbb{E}[Y \mid X = x] = \mu_Y + \Sigma_{YX}\Sigma_{XX}^{-1}(x - \mu_X),$$

$$\mathrm{Var}(Y \mid X) = \Sigma_{YY} - \Sigma_{YX}\Sigma_{XX}^{-1}\Sigma_{XY}.$$

B.3 Likelihood reference

For log likelihood $\ell_n(\theta) = \sum_i \log p(Y_i; \theta)$, score is $s_n(\theta) = \partial\ell_n/\partial\theta$ and observed information is $-\partial^2\ell_n/\partial\theta\partial\theta^\mathsf{T}$. Under correct regularity,

$$\sqrt{n}(\widehat{\theta} - \theta_0) \Rightarrow N(0, I(\theta_0)^{-1}), \quad I(\theta_0) = \mathbb{E}[s_i(\theta_0)s_i(\theta_0)^\mathsf{T}].$$

Under misspecification, quasi-MLE targets the Kullback–Leibler projection and uses sandwich covariance $H^{-1}JH^{-1}$.

B.4 Information criteria and scoring rules

$$\mathrm{AIC} = -2\ell(\widehat{\theta}) + 2k, \qquad \mathrm{BIC} = -2\ell(\widehat{\theta}) + k\log n.$$

For density forecast p_t, log score is $-\log p_t(y_t)$. For quantile q, pinball loss is $(\alpha - \mathbf{1}_{\{y\le q\}})(y - q)$. For probability p and binary outcome y, Brier score is $(p - y)^2$. Proper scores encourage honest probabilistic forecasts under their target functional.

B.5 Bootstrap choices

Method	Preserves	Main caution
iid bootstrap	empirical marginal law	destroys serial dependence
moving block	local serial dependence	block-boundary and length choice
stationary bootstrap	random-length dependence	tuning expected block length

residual bootstrap	fitted model structure	model must capture dependence
wild bootstrap	heteroskedastic regression	cluster/time variants differ
cross-sectional cluster	within-cluster dependence	needs enough clusters

Appendix C

Finance Formula Sheet

C.1 Returns and compounding

$$R_t^{\text{simple}} = \frac{P_t + D_t}{P_{t-1}} - 1, \qquad r_t = \log(1 + R_t).$$

$$1 + R_{0:T} = \prod_{t=1}^{T}(1 + R_t), \qquad r_{0:T} = \sum_{t=1}^{T} r_t.$$

For small returns, $r \approx R - \frac{1}{2}R^2$. Arithmetic mean exceeds geometric mean by approximately half the variance under a small-return lognormal approximation.

C.2 Portfolio

$$\mu_p = w^\top \mu, \qquad \sigma_p^2 = w^\top \Sigma w.$$

$$w_{\text{GMV}} = \frac{\Sigma^{-1}\mathbf{1}}{\mathbf{1}^\top \Sigma^{-1}\mathbf{1}}, \qquad w_{\tan} \propto \Sigma^{-1}(\mu - r_f \mathbf{1}).$$

$$\beta_i = \frac{\mathrm{Cov}(R_i, R_M)}{\mathrm{Var}(R_M)}, \qquad \alpha_i = \mathbb{E}[R_i - r_f] - \beta_i \, \mathbb{E}[R_M - r_f].$$

C.3 Fixed income

$$P(t,T) = \mathrm{e}^{-y(t,T)(T-t)}, \quad f(t,T) = -\partial_T \log P(t,T).$$

$$\frac{\Delta P}{P} \approx -D\Delta y + \frac{1}{2}\mathcal{C}(\Delta y)^2.$$

$$K_{\text{swap}} = \frac{P(0,T_0) - P(0,T_n)}{\sum_i \delta_i P(0,T_i)}.$$

C.4 Options

$$C - P = S\mathrm{e}^{-qT} - K\mathrm{e}^{-rT}.$$

$$C = S\mathrm{e}^{-qT}\Phi(d_1) - K\mathrm{e}^{-rT}\Phi(d_2), \quad d_{1,2} = \frac{\log(S/K) + (r - q \pm \sigma^2/2)T}{\sigma\sqrt{T}}.$$

$$\Delta_C = \mathrm{e}^{-qT}\Phi(d_1), \quad \Gamma = \frac{\mathrm{e}^{-qT}\phi(d_1)}{S\sigma\sqrt{T}}, \quad \mathcal{V} = S\mathrm{e}^{-qT}\phi(d_1)\sqrt{T}.$$

C.5 Risk

$$\mathrm{VaR}_\alpha(N(\mu,\sigma^2)) = \mu + \sigma z_\alpha,$$

$$\mathrm{ES}_\alpha(N(\mu,\sigma^2)) = \mu + \sigma\frac{\phi(z_\alpha)}{1-\alpha}.$$

$$\mathrm{ES}_\alpha(L) = \inf_z \left[z + \frac{1}{1-\alpha}\,\mathbb{E}(L-z)^+\right].$$

C.6 Continuous time

$$\mathrm{d}f(t, X_t) = \left(f_t + bf_x + \frac{1}{2}a^2 f_{xx}\right)\mathrm{d}t + af_x\,\mathrm{d}W_t.$$

$$V_t + (r-q)SV_S + \frac{1}{2}\sigma^2 S^2 V_{SS} - rV = 0.$$

$$\left.\frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}}\right|_{\mathcal{F}_T} = \exp\left(-\int_0^T \theta_t\,\mathrm{d}W_t - \frac{1}{2}\int_0^T \theta_t^2\,\mathrm{d}t\right).$$

Appendix D

Algorithm Blueprints

D.1 Walk-forward research

```
for decision_time in schedule:
    available = raw[raw.publication_time <= decision_time]
    train, validation = time_split(available, decision_time)
    transformer.fit(train.features)
    model = tune_only_on(train, validation, transformer)
    prediction = model.predict(features_at(decision_time))
    order = policy(prediction, positions, risk_limits)
    fill = execution_model(order, market_after(decision_time))
    ledger.apply(fill, costs=True, financing=True)
    audit.append(config_hash, data_hash, prediction, fill)
```

D.2 Hamilton filter

```
prob_filtered = initial_probability
loglik = 0.0
for observation in data:
    prob_predicted = transition.T @ prob_filtered
    density = emission_density(observation, parameters)
    normalizer = density @ prob_predicted
    prob_filtered = density * prob_predicted / normalizer
    loglik += log(normalizer)
```

D.3 Curve bootstrap

```
discounts = {}
for instrument in maturity_order:
    known_cashflows = present_value_with(discounts, instrument)
    discounts[instrument.maturity] = solve_last_discount(
        instrument.quote, known_cashflows, instrument.conventions
    )
validate_positive_discounts(discounts)
reprice_every_calibration_instrument(discounts)
```

D.4 Monte Carlo experiment

```
for step_count in grid_sizes:
    estimates = []
    for independent_seed in seeds:
        paths = simulate(model, step_count, paths_per_run,
   independent_seed)
        estimates.append(discounted_payoff(paths).mean())
    report(step_count, mean(estimates), standard_error(estimates)
   )
compare_grid_bias_and_sampling_error()
```

D.5 Model calibration

```
quotes = validate_and_clean(raw_quotes)
for start in multi_starts:
    fit = constrained_optimizer(
        objective=bid_ask_scaled_error,
        start=start,
        bounds=economic_bounds,
        analytic_gradient=gradient,
    )
    candidates.append(fit)
```

```
best = select_converged(candidates)
check_residuals_stability_withheld_hedges(best)
```

Appendix E

Selected Solutions and Proof Hints

E.1 Analysis and linear algebra

Convex inverse inequality. For positive X, apply Jensen to $f(x) = x^{-1}$: $(\mathbb{E}\,X)^{-1} \leq \mathbb{E}[X^{-1}]$. Multiply by $\mathbb{E}\,X > 0$.

Projection matrix. $P_X^\top = P_X$ because $X^\top X$ is symmetric. Multiplying P_X by itself cancels the middle $X^\top X$. A symmetric idempotent matrix has eigenvalues satisfying $\lambda^2 = \lambda$, hence 0 or 1.

Pseudoinverse. If $A = U_r \Sigma_r V_r^\top$, set $A^+ = V_r \Sigma_r^{-1} U_r^\top$. Every solution of $Ax = b$ has $x = A^+ b + z$ with $z \in \ker(A)$, orthogonal to $A^+ b$, so its norm is minimized at $z = 0$.

E.2 Probability

Gaussian conditional law. Partition the quadratic form in the joint normal density and complete the square in y. The remaining center is $\mu_Y + \Sigma_{YX}\Sigma_{XX}^{-1}(x - \mu_X)$ and the Schur complement is the conditional covariance.

Stochastic exponential. For $f(t, w) = \exp(\sigma w - \sigma^2 t/2)$, $f_t = -\sigma^2 f/2$, $f_w = \sigma f$, and $f_{ww} = \sigma^2 f$. Itô drift cancels, leaving $\mathrm{d}f = \sigma f\,\mathrm{d}W$. Positivity and expectation one, or an integrability argument, establish the martingale.

E.3 Econometrics

Within transformation. Average $y_{it} = \alpha_i + x_{it}^\mathsf{T}\beta + u_{it}$ over t and subtract:

$$y_{it} - \bar{y}_i = (x_{it} - \bar{x}_i)^\mathsf{T}\beta + (u_{it} - \bar{u}_i).$$

A time-invariant regressor subtracts to zero and is collinear with unit effects.

Overlapping returns. If $R_t^{(k)} = \sum_{j=1}^{k} r_{t+j}$ with iid r, adjacent observations share $k - h$ innovations at lag $h < k$, so $\mathrm{Cov}(R_t^{(k)}, R_{t+h}^{(k)}) = (k-h)\sigma^2$. It is zero at $h \geq k$.

E.4 Derivatives

Call convexity. For $K_1 < K_2 < K_3$ with K_2 a convex combination, a butterfly of calls has nonnegative terminal payoff. Its initial price must be nonnegative, which is the discrete convexity inequality. Take strike spacings to zero for $\partial_{KK} C \geq 0$.

Geometric Asian. For monitoring dates t_j, average log price is Gaussian because it is a linear combination of Brownian values. Its mean follows from each log-price mean and its variance from $\mathrm{Cov}(W_{t_i}, W_{t_j}) = \min(t_i, t_j)$. Exponentiation produces a lognormal geometric average, so the call is a Black-style lognormal expectation with adjusted mean and variance.

E.5 Portfolio and risk

Tangency portfolio. Write excess mean as $w^\mathsf{T}\mu_e = (\Sigma^{1/2}w)^\mathsf{T}(\Sigma^{-1/2}\mu_e)$. Cauchy–Schwarz bounds squared Sharpe by $\mu_e^\mathsf{T}\Sigma^{-1}\mu_e$, with equality when $\Sigma^{1/2}w$ is proportional to $\Sigma^{-1/2}\mu_e$.

Normal Expected Shortfall. For $L = \mu + \sigma Z$ and $z_\alpha = \Phi^{-1}(\alpha)$,

$$\mathbb{E}[L \mid L \geq \mu + \sigma z_\alpha] = \mu + \sigma \frac{\int_{z_\alpha}^{\infty} z\phi(z)\,\mathrm{d}z}{1-\alpha}.$$

Since $\phi'(z) = -z\phi(z)$, the numerator is $\phi(z_\alpha)$.

E.6 Numerical methods

Control variate.

$$\operatorname{Var}(X - c(Y - \mu_Y)) = \operatorname{Var}(X) - 2c \operatorname{Cov}(X, Y) + c^2 \operatorname{Var}(Y).$$

Differentiate in c to obtain $c^\star = \operatorname{Cov}(X, Y)/\operatorname{Var}(Y)$. Substitution gives $\operatorname{Var}(X)(1 - \rho^2)$.

Bisection. After n iterations, the containing interval length is $(b-a)/2^n$, so midpoint error is at most $(b-a)/2^{n+1}$. Choose $n \geq \lceil \log_2((b-a)/(2\varepsilon)) \rceil$.

E.7 What a complete solution contains

A complete quantitative solution does more than reach a formula. It states the domain, establishes existence and uniqueness when relevant, names the measure and information set, checks units, verifies limiting cases, quantifies numerical error, and explains the economic interpretation. An empirical solution additionally records data provenance, estimator uncertainty, point-in-time validation, costs, and failure modes.

Closing research checklist

A result is not ready because code ran. State the decision and estimand; fix the probability space and information set; enumerate economic and statistical assumptions; record source, vintage, timezone, units, and transformations; separate training, validation, and test periods; quantify sampling and model uncertainty; include costs and capacity; stress the tails and the implementation; preserve a complete audit trail; and explain what evidence would falsify the conclusion.

Bibliographic Guide

The manuscript is self-contained at the level of its main derivations, but mastery requires working with primary texts and research papers. The following guide names standard references by subject. Editions differ; consult the edition used and verify notation.

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Risk and market microstructure.

McNeil, Frey, and Embrechts, *Quantitative Risk Management*; Hasbrouck, *Empirical Market Microstructure*; Cartea, Jaimungal, and Penalva, *Algorithmic and High-Frequency Trading.*

For empirical data, use current methodology pages from the primary institutions listed in Chapter 36's data-source table and archive the exact metadata used in each project. For any model used in a consequential decision, consult current regulatory, legal, accounting, and market-convention sources rather than relying on a textbook alone.